


```

/* Private typedef -----*/

/* USER CODE BEGIN PTD */

/* USER CODE END PTD */

/* Private define -----*/

/* USER CODE BEGIN PD */

// TODO: Add values for below variables

#define NS      1024    // Number of samples in LUT

#define NS_WAVE  8192    // Number of samples in LUT

#define TIM2CLK 16000000 // STM Clock frequency: Hint You might want to check
the ioc file

#define F_SIGNAL 1    // Frequency of output analog signal

/* USER CODE END PD */

/* Private macro -----*/

/* USER CODE BEGIN PM */

/* USER CODE END PM */

/* Private variables -----*/

TIM_HandleTypeDef htim2;

TIM_HandleTypeDef htim3;

DMA_HandleTypeDef hdma_tim2_ch1;

uint32_t current_tick;

uint32_t last_button_press;

uint8_t current_waveform;

```

```

uint32_t *selected_lut;

char *waveform_name;

double frequency;

uint32_t ns_to_use;


/* USER CODE BEGIN PV */

// TODO: Add code for global variables, including LUTs

// Sin Wave LUT: 1024 samples, 4096 bytes (4.00KB)

uint32_t Sin_LUT[NS] = {

    2048, 2061, 2073, 2086, 2098, 2111, 2123, 2136, 2148, 2161, 2174, 2186, 2199, 2211,
    2224, 2236,

    2249, 2261, 2274, 2286, 2299, 2311, 2323, 2336, 2348, 2361, 2373, 2386, 2398, 2410,
    2423, 2435,

    2447, 2460, 2472, 2484, 2497, 2509, 2521, 2533, 2545, 2558, 2570, 2582, 2594, 2606,
    2618, 2630,

    2642, 2654, 2666, 2678, 2690, 2702, 2714, 2726, 2738, 2749, 2761, 2773, 2785, 2796,
    2808, 2820,

    2831, 2843, 2855, 2866, 2878, 2889, 2900, 2912, 2923, 2935, 2946, 2957, 2968, 2980,
    2991, 3002,

    3013, 3024, 3035, 3046, 3057, 3068, 3079, 3090, 3100, 3111, 3122, 3133, 3143, 3154,
    3164, 3175,

    3185, 3196, 3206, 3216, 3227, 3237, 3247, 3257, 3267, 3277, 3287, 3297, 3307, 3317,
    3327, 3337,

    3347, 3356, 3366, 3376, 3385, 3395, 3404, 3413, 3423, 3432, 3441, 3450, 3459, 3469,
    3478, 3487,

    3495, 3504, 3513, 3522, 3531, 3539, 3548, 3556, 3565, 3573, 3581, 3590, 3598, 3606,
    3614, 3622,

    3630, 3638, 3646, 3654, 3662, 3669, 3677, 3685, 3692, 3700, 3707, 3714, 3722, 3729,
    3736, 3743,

    3750, 3757, 3764, 3771, 3777, 3784, 3791, 3797, 3804, 3810, 3817, 3823, 3829, 3835,
    3841, 3847,

```

3853, 3859, 3865, 3871, 3876, 3882, 3888, 3893, 3898, 3904, 3909, 3914, 3919, 3924, 3929, 3934,

3939, 3944, 3949, 3953, 3958, 3962, 3967, 3971, 3975, 3980, 3984, 3988, 3992, 3996, 3999, 4003,

4007, 4010, 4014, 4017, 4021, 4024, 4027, 4031, 4034, 4037, 4040, 4042, 4045, 4048, 4051, 4053,

4056, 4058, 4060, 4063, 4065, 4067, 4069, 4071, 4073, 4075, 4076, 4078, 4080, 4081, 4083, 4084,

4085, 4086, 4087, 4088, 4089, 4090, 4091, 4092, 4093, 4093, 4094, 4094, 4094, 4095, 4095, 4095,

4095, 4095, 4095, 4095, 4094, 4094, 4094, 4093, 4093, 4092, 4091, 4090, 4089, 4088, 4087, 4086,

4085, 4084, 4083, 4081, 4080, 4078, 4076, 4075, 4073, 4071, 4069, 4067, 4065, 4063, 4060, 4058,

4056, 4053, 4051, 4048, 4045, 4042, 4040, 4037, 4034, 4031, 4027, 4024, 4021, 4017, 4014, 4010,

4007, 4003, 3999, 3996, 3992, 3988, 3984, 3980, 3975, 3971, 3967, 3962, 3958, 3953, 3949, 3944,

3939, 3934, 3929, 3924, 3919, 3914, 3909, 3904, 3898, 3893, 3888, 3882, 3876, 3871, 3865, 3859,

3853, 3847, 3841, 3835, 3829, 3823, 3817, 3810, 3804, 3797, 3791, 3784, 3777, 3771, 3764, 3757,

3750, 3743, 3736, 3729, 3722, 3714, 3707, 3700, 3692, 3685, 3677, 3669, 3662, 3654, 3646, 3638,

3630, 3622, 3614, 3606, 3598, 3590, 3581, 3573, 3565, 3556, 3548, 3539, 3531, 3522, 3513, 3504,

3495, 3487, 3478, 3469, 3459, 3450, 3441, 3432, 3423, 3413, 3404, 3395, 3385, 3376, 3366, 3356,

3347, 3337, 3327, 3317, 3307, 3297, 3287, 3277, 3267, 3257, 3247, 3237, 3227, 3216, 3206, 3196,

3185, 3175, 3164, 3154, 3143, 3133, 3122, 3111, 3100, 3090, 3079, 3068, 3057, 3046, 3035, 3024,

3013, 3002, 2991, 2980, 2968, 2957, 2946, 2935, 2923, 2912, 2900, 2889, 2878, 2866, 2855, 2843,

2831, 2820, 2808, 2796, 2785, 2773, 2761, 2749, 2738, 2726, 2714, 2702, 2690, 2678, 2666, 2654,

2642, 2630, 2618, 2606, 2594, 2582, 2570, 2558, 2545, 2533, 2521, 2509, 2497, 2484, 2472, 2460,

2447, 2435, 2423, 2410, 2398, 2386, 2373, 2361, 2348, 2336, 2323, 2311, 2299, 2286, 2274, 2261,

2249, 2236, 2224, 2211, 2199, 2186, 2174, 2161, 2148, 2136, 2123, 2111, 2098, 2086, 2073, 2061,

2048, 2035, 2023, 2010, 1998, 1985, 1973, 1960, 1948, 1935, 1922, 1910, 1897, 1885, 1872, 1860,

1847, 1835, 1822, 1810, 1797, 1785, 1773, 1760, 1748, 1735, 1723, 1710, 1698, 1686, 1673, 1661,

1649, 1636, 1624, 1612, 1599, 1587, 1575, 1563, 1551, 1538, 1526, 1514, 1502, 1490, 1478, 1466,

1454, 1442, 1430, 1418, 1406, 1394, 1382, 1370, 1358, 1347, 1335, 1323, 1311, 1300, 1288, 1276,

1265, 1253, 1241, 1230, 1218, 1207, 1196, 1184, 1173, 1161, 1150, 1139, 1128, 1116, 1105, 1094,

1083, 1072, 1061, 1050, 1039, 1028, 1017, 1006, 996, 985, 974, 963, 953, 942, 932, 921,

911, 900, 890, 880, 869, 859, 849, 839, 829, 819, 809, 799, 789, 779, 769, 759, 749, 740, 730, 720, 711, 701, 692, 683, 673, 664, 655, 646, 637, 627, 618, 609, 601, 592, 583, 574, 565, 557, 548, 540, 531, 523, 515, 506, 498, 490, 482, 474, 466, 458, 450, 442, 434, 427, 419, 411, 404, 396, 389, 382, 374, 367, 360, 353, 346, 339, 332, 325, 319, 312, 305, 299, 292, 286, 279, 273, 267, 261, 255, 249, 243, 237, 231, 225, 220, 214, 208, 203, 198, 192, 187, 182, 177, 172, 167, 162, 157, 152, 147, 143, 138, 134, 129, 125, 121, 116, 112, 108, 104, 100, 97, 93, 89, 86, 82, 79, 75, 72, 69, 65, 62, 59, 56, 54, 51, 48, 45, 43, 40, 38, 36, 33, 31, 29, 27, 25, 23, 21, 20, 18, 16, 15, 13, 12,

```

11, 10, 9, 8, 7, 6, 5, 4, 3, 3, 2, 2, 2, 1, 1, 1,
1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 5, 6, 7, 8, 9, 10,
11, 12, 13, 15, 16, 18, 20, 21, 23, 25, 27, 29, 31, 33, 36, 38,
40, 43, 45, 48, 51, 54, 56, 59, 62, 65, 69, 72, 75, 79, 82, 86,
89, 93, 97, 100, 104, 108, 112, 116, 121, 125, 129, 134, 138, 143, 147, 152,
157, 162, 167, 172, 177, 182, 187, 192, 198, 203, 208, 214, 220, 225, 231, 237,
243, 249, 255, 261, 267, 273, 279, 286, 292, 299, 305, 312, 319, 325, 332, 339,
346, 353, 360, 367, 374, 382, 389, 396, 404, 411, 419, 427, 434, 442, 450, 458,
466, 474, 482, 490, 498, 506, 515, 523, 531, 540, 548, 557, 565, 574, 583, 592,
601, 609, 618, 627, 637, 646, 655, 664, 673, 683, 692, 701, 711, 720, 730, 740,
749, 759, 769, 779, 789, 799, 809, 819, 829, 839, 849, 859, 869, 880, 890, 900,
911, 921, 932, 942, 953, 963, 974, 985, 996, 1006, 1017, 1028, 1039, 1050, 1061,
1072,
1083, 1094, 1105, 1116, 1128, 1139, 1150, 1161, 1173, 1184, 1196, 1207, 1218, 1230,
1241, 1253,
1265, 1276, 1288, 1300, 1311, 1323, 1335, 1347, 1358, 1370, 1382, 1394, 1406, 1418,
1430, 1442,
1454, 1466, 1478, 1490, 1502, 1514, 1526, 1538, 1551, 1563, 1575, 1587, 1599, 1612,
1624, 1636,
1649, 1661, 1673, 1686, 1698, 1710, 1723, 1735, 1748, 1760, 1773, 1785, 1797, 1810,
1822, 1835,
1847, 1860, 1872, 1885, 1897, 1910, 1922, 1935, 1948, 1960, 1973, 1985, 1998, 2010,
2023, 2035
};

```

```

// Saw Wave LUT: 1024 samples, 4096 bytes (4.00KB)

```

```

uint32_t Saw_LUT[NS] = {
    0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60,
    64, 68, 72, 76, 80, 84, 88, 92, 96, 100, 104, 108, 112, 116, 120, 124,
    128, 132, 136, 140, 144, 148, 152, 156, 160, 164, 168, 172, 176, 180, 184, 188,
    192, 196, 200, 204, 208, 212, 216, 220, 224, 228, 232, 236, 240, 244, 248, 252,

```

256, 260, 264, 268, 272, 276, 280, 284, 288, 292, 296, 300, 304, 308, 312, 316,
320, 324, 328, 332, 336, 340, 344, 348, 352, 356, 360, 364, 368, 372, 376, 380,
384, 388, 392, 396, 400, 404, 408, 412, 416, 420, 424, 428, 432, 436, 440, 444,
448, 452, 456, 460, 464, 468, 472, 476, 480, 484, 488, 492, 496, 500, 504, 508,
512, 516, 520, 524, 528, 532, 536, 540, 544, 548, 552, 556, 560, 564, 568, 572,
576, 580, 584, 588, 592, 596, 600, 604, 608, 612, 616, 620, 624, 628, 632, 636,
640, 644, 648, 652, 656, 660, 664, 668, 672, 676, 680, 684, 688, 692, 696, 700,
704, 708, 712, 716, 720, 724, 728, 732, 736, 740, 744, 748, 752, 756, 760, 764,
768, 772, 776, 780, 784, 788, 792, 796, 800, 804, 808, 812, 816, 820, 824, 828,
832, 836, 840, 844, 848, 852, 856, 860, 864, 868, 872, 876, 880, 884, 888, 892,
896, 900, 904, 908, 912, 916, 920, 924, 928, 932, 936, 940, 944, 948, 952, 956,
960, 964, 968, 972, 976, 980, 984, 988, 992, 996, 1000, 1004, 1008, 1012, 1016,
1020,

1024, 1028, 1032, 1036, 1040, 1044, 1048, 1052, 1056, 1060, 1064, 1068, 1072, 1076,
1080, 1084,

1088, 1092, 1096, 1100, 1104, 1108, 1112, 1116, 1120, 1124, 1128, 1132, 1136, 1140,
1144, 1148,

1152, 1156, 1160, 1164, 1168, 1172, 1176, 1180, 1184, 1188, 1192, 1196, 1200, 1204,
1208, 1212,

1216, 1220, 1224, 1228, 1232, 1236, 1240, 1244, 1248, 1252, 1256, 1260, 1264, 1268,
1272, 1276,

1280, 1284, 1288, 1292, 1296, 1300, 1304, 1308, 1312, 1316, 1320, 1324, 1328, 1332,
1336, 1340,

1344, 1348, 1352, 1356, 1360, 1364, 1368, 1372, 1376, 1380, 1384, 1388, 1392, 1396,
1400, 1404,

1408, 1412, 1416, 1420, 1424, 1428, 1432, 1436, 1440, 1444, 1448, 1452, 1456, 1460,
1464, 1468,

1472, 1476, 1480, 1484, 1488, 1492, 1496, 1500, 1504, 1508, 1512, 1516, 1520, 1524,
1528, 1532,

1536, 1540, 1544, 1548, 1552, 1556, 1560, 1564, 1568, 1572, 1576, 1580, 1584, 1588,
1592, 1596,

1600, 1604, 1608, 1612, 1616, 1620, 1624, 1628, 1632, 1636, 1640, 1644, 1648, 1652, 1656, 1660,

1664, 1668, 1672, 1676, 1680, 1684, 1688, 1692, 1696, 1700, 1704, 1708, 1712, 1716, 1720, 1724,

1728, 1732, 1736, 1740, 1744, 1748, 1752, 1756, 1760, 1764, 1768, 1772, 1776, 1780, 1784, 1788,

1792, 1796, 1800, 1804, 1808, 1812, 1816, 1820, 1824, 1828, 1832, 1836, 1840, 1844, 1848, 1852,

1856, 1860, 1864, 1868, 1872, 1876, 1880, 1884, 1888, 1892, 1896, 1900, 1904, 1908, 1912, 1916,

1920, 1924, 1928, 1932, 1936, 1940, 1944, 1948, 1952, 1956, 1960, 1964, 1968, 1972, 1976, 1980,

1984, 1988, 1992, 1996, 2000, 2004, 2008, 2012, 2016, 2020, 2024, 2028, 2032, 2036, 2040, 2044,

2048, 2052, 2056, 2060, 2064, 2068, 2072, 2076, 2080, 2084, 2088, 2092, 2096, 2100, 2104, 2108,

2112, 2116, 2120, 2124, 2128, 2132, 2136, 2140, 2144, 2148, 2152, 2156, 2160, 2164, 2168, 2172,

2176, 2180, 2184, 2188, 2192, 2196, 2200, 2204, 2208, 2212, 2216, 2220, 2224, 2228, 2232, 2236,

2240, 2244, 2248, 2252, 2256, 2260, 2264, 2268, 2272, 2276, 2280, 2284, 2288, 2292, 2296, 2300,

2304, 2308, 2312, 2316, 2320, 2324, 2328, 2332, 2336, 2340, 2344, 2348, 2352, 2356, 2360, 2364,

2368, 2372, 2376, 2380, 2384, 2388, 2392, 2396, 2400, 2404, 2408, 2412, 2416, 2420, 2424, 2428,

2432, 2436, 2440, 2444, 2448, 2452, 2456, 2460, 2464, 2468, 2472, 2476, 2480, 2484, 2488, 2492,

2496, 2500, 2504, 2508, 2512, 2516, 2520, 2524, 2528, 2532, 2536, 2540, 2544, 2548, 2552, 2556,

2560, 2564, 2568, 2572, 2576, 2580, 2584, 2588, 2592, 2596, 2600, 2604, 2608, 2612, 2616, 2620,

2624, 2628, 2632, 2636, 2640, 2644, 2648, 2652, 2656, 2660, 2664, 2668, 2672, 2676,
2680, 2684,

2688, 2692, 2696, 2700, 2704, 2708, 2712, 2716, 2720, 2724, 2728, 2732, 2736, 2740,
2744, 2748,

2752, 2756, 2760, 2764, 2768, 2772, 2776, 2780, 2784, 2788, 2792, 2796, 2800, 2804,
2808, 2812,

2816, 2820, 2824, 2828, 2832, 2836, 2840, 2844, 2848, 2852, 2856, 2860, 2864, 2868,
2872, 2876,

2880, 2884, 2888, 2892, 2896, 2900, 2904, 2908, 2912, 2916, 2920, 2924, 2928, 2932,
2936, 2940,

2944, 2948, 2952, 2956, 2960, 2964, 2968, 2972, 2976, 2980, 2984, 2988, 2992, 2996,
3000, 3004,

3008, 3012, 3016, 3020, 3024, 3028, 3032, 3036, 3040, 3044, 3048, 3052, 3056, 3060,
3064, 3068,

3072, 3076, 3080, 3084, 3088, 3092, 3096, 3100, 3104, 3108, 3112, 3116, 3120, 3124,
3128, 3132,

3136, 3140, 3144, 3148, 3152, 3156, 3160, 3164, 3168, 3172, 3176, 3180, 3184, 3188,
3192, 3196,

3200, 3204, 3208, 3212, 3216, 3220, 3224, 3228, 3232, 3236, 3240, 3244, 3248, 3252,
3256, 3260,

3264, 3268, 3272, 3276, 3280, 3284, 3288, 3292, 3296, 3300, 3304, 3308, 3312, 3316,
3320, 3324,

3328, 3332, 3336, 3340, 3344, 3348, 3352, 3356, 3360, 3364, 3368, 3372, 3376, 3380,
3384, 3388,

3392, 3396, 3400, 3404, 3408, 3412, 3416, 3420, 3424, 3428, 3432, 3436, 3440, 3444,
3448, 3452,

3456, 3460, 3464, 3468, 3472, 3476, 3480, 3484, 3488, 3492, 3496, 3500, 3504, 3508,
3512, 3516,

3520, 3524, 3528, 3532, 3536, 3540, 3544, 3548, 3552, 3556, 3560, 3564, 3568, 3572,
3576, 3580,

3584, 3588, 3592, 3596, 3600, 3604, 3608, 3612, 3616, 3620, 3624, 3628, 3632, 3636,
3640, 3644,

3648, 3652, 3656, 3660, 3664, 3668, 3672, 3676, 3680, 3684, 3688, 3692, 3696, 3700,
3704, 3708,

3712, 3716, 3720, 3724, 3728, 3732, 3736, 3740, 3744, 3748, 3752, 3756, 3760, 3764,
3768, 3772,

3776, 3780, 3784, 3788, 3792, 3796, 3800, 3804, 3808, 3812, 3816, 3820, 3824, 3828,
3832, 3836,

3840, 3844, 3848, 3852, 3856, 3860, 3864, 3868, 3872, 3876, 3880, 3884, 3888, 3892,
3896, 3900,

3904, 3908, 3912, 3916, 3920, 3924, 3928, 3932, 3936, 3940, 3944, 3948, 3952, 3956,
3960, 3964,

3968, 3972, 3976, 3980, 3984, 3988, 3992, 3996, 4000, 4004, 4008, 4012, 4016, 4020,
4024, 4028,

4032, 4036, 4040, 4044, 4048, 4052, 4056, 4060, 4064, 4068, 4072, 4076, 4080, 4084,
4088, 4092

};

// Triangle Wave LUT: 1024 samples, 4096 bytes (4.00KB)

uint32_t Triangle_LUT[NS] = {

0, 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120,
128, 136, 144, 152, 160, 168, 176, 184, 192, 200, 208, 216, 224, 232, 240, 248,
256, 264, 272, 280, 288, 296, 304, 312, 320, 328, 336, 344, 352, 360, 368, 376,
384, 392, 400, 408, 416, 424, 432, 440, 448, 456, 464, 472, 480, 488, 496, 504,
512, 520, 528, 536, 544, 552, 560, 568, 576, 584, 592, 600, 608, 616, 624, 632,
640, 648, 656, 664, 672, 680, 688, 696, 704, 712, 720, 728, 736, 744, 752, 760,
768, 776, 784, 792, 800, 808, 816, 824, 832, 840, 848, 856, 864, 872, 880, 888,
896, 904, 912, 920, 928, 936, 944, 952, 960, 968, 976, 984, 992, 1000, 1008,
1016,

1024, 1032, 1040, 1048, 1056, 1064, 1072, 1080, 1088, 1096, 1104, 1112, 1120, 1128,
1136, 1144,

1152, 1160, 1168, 1176, 1184, 1192, 1200, 1208, 1216, 1224, 1232, 1240, 1248, 1256,
1264, 1272,

1280, 1288, 1296, 1304, 1312, 1320, 1328, 1336, 1344, 1352, 1360, 1368, 1376, 1384,
1392, 1400,

1408, 1416, 1424, 1432, 1440, 1448, 1456, 1464, 1472, 1480, 1488, 1496, 1504, 1512,
1520, 1528,

1536, 1544, 1552, 1560, 1568, 1576, 1584, 1592, 1600, 1608, 1616, 1624, 1632, 1640,
1648, 1656,

1664, 1672, 1680, 1688, 1696, 1704, 1712, 1720, 1728, 1736, 1744, 1752, 1760, 1768,
1776, 1784,

1792, 1800, 1808, 1816, 1824, 1832, 1840, 1848, 1856, 1864, 1872, 1880, 1888, 1896,
1904, 1912,

1920, 1928, 1936, 1944, 1952, 1960, 1968, 1976, 1984, 1992, 2000, 2008, 2016, 2024,
2032, 2040,

2048, 2056, 2064, 2072, 2080, 2088, 2096, 2104, 2112, 2120, 2128, 2136, 2144, 2152,
2160, 2168,

2176, 2184, 2192, 2200, 2208, 2216, 2224, 2232, 2240, 2248, 2256, 2264, 2272, 2280,
2288, 2296,

2304, 2312, 2320, 2328, 2336, 2344, 2352, 2360, 2368, 2376, 2384, 2392, 2400, 2408,
2416, 2424,

2432, 2440, 2448, 2456, 2464, 2472, 2480, 2488, 2496, 2504, 2512, 2520, 2528, 2536,
2544, 2552,

2560, 2568, 2576, 2584, 2592, 2600, 2608, 2616, 2624, 2632, 2640, 2648, 2656, 2664,
2672, 2680,

2688, 2696, 2704, 2712, 2720, 2728, 2736, 2744, 2752, 2760, 2768, 2776, 2784, 2792,
2800, 2808,

2816, 2824, 2832, 2840, 2848, 2856, 2864, 2872, 2880, 2888, 2896, 2904, 2912, 2920,
2928, 2936,

2944, 2952, 2960, 2968, 2976, 2984, 2992, 3000, 3008, 3016, 3024, 3032, 3040, 3048,
3056, 3064,

3072, 3080, 3088, 3096, 3104, 3112, 3120, 3128, 3136, 3144, 3152, 3160, 3168, 3176,
3184, 3192,

3200, 3208, 3216, 3224, 3232, 3240, 3248, 3256, 3264, 3272, 3280, 3288, 3296, 3304,
3312, 3320,

3328, 3336, 3344, 3352, 3360, 3368, 3376, 3384, 3392, 3400, 3408, 3416, 3424, 3432, 3440, 3448,

3456, 3464, 3472, 3480, 3488, 3496, 3504, 3512, 3520, 3528, 3536, 3544, 3552, 3560, 3568, 3576,

3584, 3592, 3600, 3608, 3616, 3624, 3632, 3640, 3648, 3656, 3664, 3672, 3680, 3688, 3696, 3704,

3712, 3720, 3728, 3736, 3744, 3752, 3760, 3768, 3776, 3784, 3792, 3800, 3808, 3816, 3824, 3832,

3840, 3848, 3856, 3864, 3872, 3880, 3888, 3896, 3904, 3912, 3920, 3928, 3936, 3944, 3952, 3960,

3968, 3976, 3984, 3992, 4000, 4008, 4016, 4024, 4032, 4040, 4048, 4056, 4064, 4072, 4080, 4088,

4095, 4087, 4079, 4071, 4063, 4055, 4047, 4039, 4031, 4023, 4015, 4007, 3999, 3991, 3983, 3975,

3967, 3959, 3951, 3943, 3935, 3927, 3919, 3911, 3903, 3895, 3887, 3879, 3871, 3863, 3855, 3847,

3839, 3831, 3823, 3815, 3807, 3799, 3791, 3783, 3775, 3767, 3759, 3751, 3743, 3735, 3727, 3719,

3711, 3703, 3695, 3687, 3679, 3671, 3663, 3655, 3647, 3639, 3631, 3623, 3615, 3607, 3599, 3591,

3583, 3575, 3567, 3559, 3551, 3543, 3535, 3527, 3519, 3511, 3503, 3495, 3487, 3479, 3471, 3463,

3455, 3447, 3439, 3431, 3423, 3415, 3407, 3399, 3391, 3383, 3375, 3367, 3359, 3351, 3343, 3335,

3327, 3319, 3311, 3303, 3295, 3287, 3279, 3271, 3263, 3255, 3247, 3239, 3231, 3223, 3215, 3207,

3199, 3191, 3183, 3175, 3167, 3159, 3151, 3143, 3135, 3127, 3119, 3111, 3103, 3095, 3087, 3079,

3071, 3063, 3055, 3047, 3039, 3031, 3023, 3015, 3007, 2999, 2991, 2983, 2975, 2967, 2959, 2951,

2943, 2935, 2927, 2919, 2911, 2903, 2895, 2887, 2879, 2871, 2863, 2855, 2847, 2839, 2831, 2823,

2815, 2807, 2799, 2791, 2783, 2775, 2767, 2759, 2751, 2743, 2735, 2727, 2719, 2711, 2703, 2695,

2687, 2679, 2671, 2663, 2655, 2647, 2639, 2631, 2623, 2615, 2607, 2599, 2591, 2583, 2575, 2567,

2559, 2551, 2543, 2535, 2527, 2519, 2511, 2503, 2495, 2487, 2479, 2471, 2463, 2455, 2447, 2439,

2431, 2423, 2415, 2407, 2399, 2391, 2383, 2375, 2367, 2359, 2351, 2343, 2335, 2327, 2319, 2311,

2303, 2295, 2287, 2279, 2271, 2263, 2255, 2247, 2239, 2231, 2223, 2215, 2207, 2199, 2191, 2183,

2175, 2167, 2159, 2151, 2143, 2135, 2127, 2119, 2111, 2103, 2095, 2087, 2079, 2071, 2063, 2055,

2047, 2039, 2031, 2023, 2015, 2007, 1999, 1991, 1983, 1975, 1967, 1959, 1951, 1943, 1935, 1927,

1919, 1911, 1903, 1895, 1887, 1879, 1871, 1863, 1855, 1847, 1839, 1831, 1823, 1815, 1807, 1799,

1791, 1783, 1775, 1767, 1759, 1751, 1743, 1735, 1727, 1719, 1711, 1703, 1695, 1687, 1679, 1671,

1663, 1655, 1647, 1639, 1631, 1623, 1615, 1607, 1599, 1591, 1583, 1575, 1567, 1559, 1551, 1543,

1535, 1527, 1519, 1511, 1503, 1495, 1487, 1479, 1471, 1463, 1455, 1447, 1439, 1431, 1423, 1415,

1407, 1399, 1391, 1383, 1375, 1367, 1359, 1351, 1343, 1335, 1327, 1319, 1311, 1303, 1295, 1287,

1279, 1271, 1263, 1255, 1247, 1239, 1231, 1223, 1215, 1207, 1199, 1191, 1183, 1175, 1167, 1159,

1151, 1143, 1135, 1127, 1119, 1111, 1103, 1095, 1087, 1079, 1071, 1063, 1055, 1047, 1039, 1031,

1023, 1015, 1007, 999, 991, 983, 975, 967, 959, 951, 943, 935, 927, 919, 911, 903,

895, 887, 879, 871, 863, 855, 847, 839, 831, 823, 815, 807, 799, 791, 783, 775,

767, 759, 751, 743, 735, 727, 719, 711, 703, 695, 687, 679, 671, 663, 655, 647,

639, 631, 623, 615, 607, 599, 591, 583, 575, 567, 559, 551, 543, 535, 527, 519,

```

511, 503, 495, 487, 479, 471, 463, 455, 447, 439, 431, 423, 415, 407, 399, 391,
383, 375, 367, 359, 351, 343, 335, 327, 319, 311, 303, 295, 287, 279, 271, 263,
255, 247, 239, 231, 223, 215, 207, 199, 191, 183, 175, 167, 159, 151, 143, 135,
127, 119, 111, 103, 95, 87, 79, 71, 63, 55, 47, 39, 31, 23, 15, 7
};

// Piano: 8192 samples, 32,768 bytes (32.00KB)

uint32_t Piano_LUT[NS_WAVE] = {
    1900, 1986, 2032, 2240, 2341, 2177, 1941, 1808, 2042, 1947, 2090, 2239, 2201, 1960,
    1820, 2092,
    1884, 2017, 2219, 2080, 1866, 2065, 2136, 2175, 2168, 1932, 2013, 2011, 2267, 2008,
    1958, 2069,
    2225, 1944, 2070, 2084, 2081, 2075, 2069, 2063, 2055, 2048, 2039, 2030, 2022, 2012,
    2003, 1995,
    1985, 1977, 1965, 1945, 1915, 1872, 1815, 1744, 1661, 1563, 1455, 1339, 1216, 1091,
    968, 850,
    740, 642, 566, 562, 590, 620, 675, 735, 808, 882, 968, 1065, 1168, 1292, 1356,
    1413,
    1499, 1571, 1635, 1717, 1834, 1943, 1982, 2015, 2043, 2011, 2012, 1971, 1853, 1872,
    1945, 2033,
    2095, 2104, 2121, 2006, 1873, 1819, 1913, 2115, 1899, 2304, 2023, 1678, 1582, 1727,
    2077, 1892,
    2240, 2507, 2316, 2264, 1931, 1603, 1567, 1584, 1610, 1709, 1770, 1867, 1972, 2025,
    2101, 2289,
    2649, 2905, 2891, 2738, 2700, 2766, 2843, 2791, 2540, 2246, 2048, 1911, 1866, 1789,
    1763, 1759,
    1778, 1791, 1777, 1731, 1636, 1575, 1544, 1558, 1680, 1896, 2153, 2353, 2503, 2611,
    2699, 2783,
    2866, 2935, 2935, 2932, 2891, 2831, 2774, 2740, 2712, 2705, 2712, 2709, 2702, 2696,
    2685, 2668,
    2644, 2611, 2571, 2529, 2489, 2449, 2415, 2388, 2373, 2367, 2367, 2377, 2399, 2430,
    2464, 2496,

```

2533, 2576, 2627, 2674, 2713, 2741, 2777, 2817, 2850, 2882, 2915, 2950, 2979, 3012, 3050, 3094,

3145, 3209, 3273, 3332, 3382, 3420, 3451, 3479, 3503, 3520, 3532, 3538, 3538, 3535, 3527, 3515,

3499, 3476, 3440, 3397, 3346, 3295, 3243, 3183, 3119, 3046, 2971, 2893, 2816, 2741, 2679, 2625,

2584, 2546, 2505, 2466, 2430, 2397, 2371, 2346, 2321, 2295, 2268, 2240, 2210, 2176, 2140, 2100,

2060, 2013, 1965, 1913, 1861, 1806, 1750, 1691, 1631, 1573, 1517, 1463, 1407, 1349, 1290, 1230,

1176, 1124, 1079, 1034, 993, 953, 916, 881, 851, 825, 803, 784, 766, 746, 726, 704,

682, 660, 639, 620, 606, 595, 586, 575, 568, 565, 578, 596, 616, 630, 641, 659,

686, 722, 762, 806, 852, 896, 941, 988, 1037, 1088, 1148, 1216, 1273, 1320, 1360, 1393,

1423, 1454, 1488, 1524, 1563, 1602, 1635, 1663, 1689, 1716, 1747, 1786, 1828, 1865, 1892, 1911,

1931, 1956, 1983, 2004, 2021, 2034, 2023, 2013, 2007, 2005, 2012, 2037, 2098, 2187, 2292, 2418,

2472, 2464, 2400, 2254, 2071, 1913, 1721, 1553, 1425, 1318, 1245, 1229, 1238, 1299, 1379, 1496,

1662, 1842, 2048, 2258, 2429, 2576, 2648, 2700, 2701, 2666, 2582, 2491, 2391, 2283, 2161, 2045,

1916, 1782, 1684, 1612, 1546, 1495, 1488, 1490, 1500, 1533, 1603, 1694, 1807, 1953, 2089, 2222,

2374, 2524, 2655, 2771, 2865, 2947, 2997, 3025, 3034, 3020, 2984, 2908, 2812, 2689, 2547, 2397,

2241, 2084, 1928, 1794, 1693, 1614, 1556, 1516, 1524, 1549, 1576, 1626, 1698, 1772, 1843, 1915,

1987, 2069, 2168, 2262, 2341, 2410, 2472, 2524, 2569, 2615, 2655, 2685, 2706, 2714, 2720, 2726,

2731, 2734, 2739, 2746, 2753, 2760, 2766, 2771, 2776, 2783, 2789, 2794, 2798, 2803, 2813, 2826,

2837, 2849, 2864, 2882, 2904, 2933, 2964, 2993, 3015, 3037, 3062, 3090, 3115, 3135, 3150, 3163,

3176, 3186, 3190, 3191, 3191, 3186, 3178, 3167, 3151, 3131, 3109, 3086, 3062, 3038, 3015, 2991,

2963, 2932, 2898, 2859, 2815, 2770, 2724, 2679, 2631, 2583, 2534, 2486, 2437, 2385, 2332, 2277,

2219, 2160, 2099, 2038, 1978, 1915, 1851, 1788, 1730, 1674, 1620, 1567, 1516, 1464, 1413, 1364,

1316, 1271, 1228, 1189, 1152, 1117, 1085, 1056, 1030, 1008, 989, 978, 971, 966, 960, 958,

962, 974, 991, 1009, 1027, 1045, 1066, 1090, 1116, 1145, 1174, 1203, 1231, 1259, 1287, 1314,

1336, 1353, 1364, 1371, 1373, 1370, 1362, 1353, 1345, 1340, 1337, 1331, 1320, 1306, 1290, 1272,

1255, 1241, 1232, 1227, 1224, 1224, 1228, 1237, 1254, 1281, 1319, 1366, 1428, 1501, 1590, 1695,

1807, 1916, 2026, 2129, 2222, 2299, 2364, 2424, 2471, 2502, 2518, 2521, 2504, 2465, 2411, 2340,

2252, 2146, 2018, 1881, 1739, 1592, 1444, 1323, 1237, 1171, 1129, 1125, 1153, 1205, 1289, 1399,

1540, 1698, 1853, 2010, 2165, 2306, 2422, 2524, 2600, 2640, 2653, 2638, 2604, 2540, 2434, 2288,

2109, 1926, 1752, 1624, 1560, 1573, 1652, 1802, 2004, 2249, 2475, 2587, 2574, 2417, 2184, 1983,

1931, 1985, 2056, 2120, 2158, 2174, 2192, 2210, 2228, 2241, 2252, 2265, 2280, 2294, 2308, 2322,

2340, 2360, 2383, 2410, 2438, 2467, 2497, 2528, 2562, 2600, 2644, 2689, 2736, 2782, 2829, 2878,

2929, 2980, 3029, 3075, 3118, 3158, 3195, 3230, 3262, 3291, 3317, 3340, 3360, 3378, 3392, 3402,

3409, 3412, 3410, 3404, 3393, 3380, 3362, 3341, 3318, 3293, 3263, 3231, 3196, 3156, 3115, 3071,

3024, 2975, 2924, 2869, 2813, 2755, 2696, 2636, 2575, 2513, 2450, 2386, 2320, 2253, 2185, 2115,

2045, 1974, 1904, 1834, 1765, 1697, 1629, 1563, 1499, 1437, 1377, 1320, 1267, 1216, 1167, 1121,

1078, 1039, 1003, 973, 946, 923, 903, 887, 875, 869, 870, 877, 892, 913, 938, 967,

1001, 1042, 1090, 1146, 1207, 1271, 1336, 1405, 1477, 1550, 1619, 1682, 1739, 1786, 1822, 1847,

1865, 1877, 1879, 1873, 1858, 1835, 1802, 1760, 1715, 1667, 1619, 1572, 1533, 1506, 1495, 1534,

1652, 1844, 2069, 2261, 2412, 2507, 2550, 2535, 2467, 2336, 2157, 1942, 1722, 1548, 1432, 1404,

1475, 1637, 1862, 2115, 2308, 2386, 2343, 2180, 1985, 2006, 2344, 2268, 2090, 2088, 2275, 2455,

2471, 2420, 2373, 2324, 2275, 2231, 2193, 2160, 2133, 2110, 2091, 2078, 2071, 2071, 2078, 2091,

2109, 2130, 2154, 2181, 2213, 2251, 2294, 2343, 2396, 2452, 2513, 2577, 2646, 2716, 2786, 2854,

2921, 2986, 3050, 3110, 3168, 3221, 3270, 3313, 3351, 3385, 3416, 3443, 3465, 3482, 3496, 3505,

3509, 3511, 3510, 3505, 3497, 3485, 3469, 3452, 3429, 3405, 3377, 3347, 3315, 3280, 3244, 3206,

3165, 3123, 3076, 3029, 2978, 2926, 2871, 2816, 2758, 2699, 2639, 2578, 2515, 2451, 2387, 2320,

2253, 2185, 2117, 2048, 1980, 1912, 1844, 1777, 1711, 1646, 1582, 1520, 1459, 1401, 1344, 1289,

1235, 1183, 1133, 1084, 1037, 992, 949, 909, 870, 835, 802, 772, 745, 720, 699, 680,

664, 653, 643, 639, 637, 639, 645, 655, 669, 686, 708, 736, 768, 805, 848, 896, 947, 1001, 1058, 1117, 1180, 1247, 1315, 1385, 1454, 1520, 1583, 1642, 1695, 1745, 1791, 1835,

1875, 1909, 1936, 1957, 1972, 1980, 1981, 1976, 1966, 1951, 1932, 1908, 1880, 1848, 1814, 1777,

1738, 1716, 1745, 1935, 2043, 1947, 1716, 1670, 1995, 2205, 2106, 1875, 1742, 1802, 2047, 2352,

2553, 2562, 2412, 2156, 1905, 1751, 1697, 1740, 1876, 2117, 2388, 2531, 2538, 2497, 2435, 2372,

2314, 2261, 2215, 2177, 2150, 2132, 2125, 2123, 2129, 2145, 2174, 2217, 2270, 2330, 2395, 2467,

2544, 2624, 2703, 2781, 2856, 2927, 2991, 3046, 3094, 3135, 3168, 3195, 3215, 3228, 3234, 3233,

3227, 3214, 3196, 3172, 3143, 3109, 3069, 3026, 2978, 2925, 2869, 2807, 2743, 2675, 2606, 2534,

2460, 2385, 2309, 2231, 2153, 2075, 1998, 1921, 1846, 1771, 1699, 1628, 1558, 1491, 1425, 1361,

1300, 1241, 1186, 1135, 1089, 1047, 1009, 974, 942, 915, 893, 876, 864, 858, 858, 862,

871, 886, 908, 936, 971, 1010, 1055, 1105, 1161, 1223, 1289, 1356, 1420, 1482, 1542, 1600,

1656, 1709, 1758, 1800, 1833, 1862, 1886, 1906, 1925, 1943, 1962, 1980, 1998, 2017, 2041, 2082,

2138, 2195, 2130, 1907, 1685, 1544, 1520, 1624, 1829, 2065, 2277, 2440, 2539, 2576, 2531, 2423,

2258, 2061, 1862, 1692, 1571, 1495, 1461, 1468, 1514, 1598, 1714, 1865, 2046, 2241, 2426, 2587,

2711, 2785, 2812, 2791, 2721, 2600, 2434, 2244, 2056, 1896, 1777, 1705, 1690, 1723, 1801, 1907,

2040, 2189, 2326, 2427, 2489, 2516, 2519, 2509, 2492, 2475, 2463, 2456, 2454, 2459, 2470, 2489,

2523, 2566, 2613, 2663, 2714, 2762, 2805, 2842, 2871, 2892, 2903, 2906, 2900, 2883, 2856, 2822,

2778, 2725, 2664, 2596, 2521, 2440, 2354, 2264, 2171, 2077, 1983, 1891, 1801, 1716, 1634, 1557,

1485, 1421, 1363, 1311, 1267, 1231, 1207, 1190, 1181, 1181, 1194, 1215, 1244, 1281, 1326, 1375,

1424, 1472, 1516, 1551, 1580, 1602, 1620, 1631, 1636, 1634, 1630, 1628, 1633, 1652, 1693, 1765,

1870, 2000, 2125, 2239, 2321, 2360, 2350, 2291, 2181, 2024, 1831, 1638, 1474, 1355, 1295, 1292,

1342, 1443, 1589, 1768, 1960, 2139, 2297, 2425, 2518, 2576, 2595, 2573, 2508, 2396, 2243, 2059,

1866, 1694, 1561, 1482, 1462, 1499, 1587, 1727, 1908, 2118, 2325, 2506, 2623, 2667, 2631, 2523,

2344, 2125, 1937, 1844, 1856, 1924, 2005, 2065, 2109, 2143, 2170, 2195, 2220, 2247, 2277, 2314,

2359, 2410, 2468, 2531, 2595, 2659, 2725, 2789, 2847, 2902, 2950, 2989, 3020, 3040, 3051, 3051,

3042, 3024, 2998, 2964, 2922, 2873, 2815, 2752, 2683, 2608, 2526, 2439, 2348, 2255, 2159, 2063,

1966, 1869, 1773, 1680, 1592, 1510, 1434, 1367, 1308, 1256, 1215, 1184, 1163, 1152, 1155, 1172,

1208, 1259, 1326, 1409, 1503, 1602, 1701, 1793, 1874, 1935, 1972, 1982, 1969, 1934, 1878, 1804,

1719, 1640, 1597, 1672, 1922, 2168, 2331, 2391, 2343, 2191, 1961, 1713, 1531, 1479, 1570, 1780,

2055, 2299, 2425, 2407, 2242, 1970, 1790, 1944, 2307, 2253, 2101, 2249, 2222, 2175, 2135, 2102,

2078, 2065, 2061, 2066, 2078, 2099, 2130, 2168, 2216, 2272, 2337, 2408, 2488, 2571, 2656, 2739,

2817, 2891, 2959, 3021, 3075, 3121, 3160, 3191, 3214, 3230, 3238, 3239, 3233, 3222, 3203, 3179,

3150, 3115, 3075, 3028, 2977, 2921, 2861, 2796, 2726, 2653, 2576, 2496, 2414, 2331, 2247, 2161,

2076, 1991, 1906, 1822, 1738, 1655, 1575, 1497, 1423, 1353, 1287, 1226, 1170, 1119, 1073, 1033,

998, 970, 948, 931, 922, 920, 927, 942, 965, 997, 1040, 1092, 1155, 1228, 1311, 1402,

1500, 1600, 1698, 1790, 1876, 1953, 2017, 2066, 2100, 2120, 2128, 2122, 2101, 2064, 2011, 1944,

1860, 1758, 1671, 1903, 2083, 1933, 1751, 2140, 1982, 1802, 2058, 2407, 2377, 2060, 1834, 1846,

2086, 2400, 2432, 2355, 2271, 2195, 2137, 2096, 2075, 2072, 2090, 2125, 2181, 2258, 2350, 2449,

2550, 2646, 2734, 2813, 2881, 2938, 2982, 3011, 3027, 3029, 3021, 3003, 2974, 2937, 2891, 2835,

2771, 2699, 2621, 2538, 2449, 2356, 2260, 2162, 2062, 1962, 1863, 1766, 1671, 1581, 1495, 1414,

1340, 1273, 1212, 1158, 1113, 1078, 1051, 1034, 1027, 1030, 1043, 1070, 1107, 1159, 1225, 1306,

1399, 1497, 1600, 1702, 1800, 1888, 1964, 2025, 2071, 2100, 2115, 2115, 2102, 2076, 2039, 2002,

1994, 1837, 1662, 1757, 2048, 2267, 2367, 2320, 2118, 1876, 1687, 1582, 1555, 1602, 1716, 1892,

2104, 2323, 2517, 2650, 2714, 2709, 2630, 2483, 2283, 2065, 1877, 1748, 1687, 1693, 1755, 1866,

2026, 2218, 2386, 2480, 2509, 2501, 2474, 2437, 2402, 2375, 2367, 2381, 2418, 2469, 2526, 2584,

2637, 2682, 2710, 2720, 2706, 2673, 2621, 2550, 2464, 2365, 2255, 2137, 2015, 1892, 1774, 1665,

1568, 1486, 1422, 1376, 1350, 1346, 1365, 1406, 1470, 1546, 1625, 1698, 1756, 1794, 1817, 1829,

1840, 1879, 1971, 2098, 2185, 2185, 2070, 1851, 1637, 1490, 1438, 1502, 1670, 1891, 2110, 2292,

2422, 2498, 2508, 2449, 2320, 2136, 1921, 1721, 1580, 1512, 1506, 1558, 1669, 1832, 2033, 2250,

2448, 2595, 2667, 2661, 2582, 2426, 2212, 1989, 1834, 1777, 1798, 1877, 1989, 2107, 2197, 2263,

2316, 2358, 2392, 2425, 2462, 2508, 2558, 2611, 2660, 2704, 2741, 2764, 2774, 2769, 2751, 2721,

2675, 2613, 2537, 2449, 2349, 2240, 2125, 2007, 1886, 1768, 1656, 1556, 1474, 1411, 1374, 1368,

1395, 1462, 1562, 1686, 1819, 1932, 1996, 2000, 1951, 1862, 1751, 1702, 1957, 2206, 2320, 2266,

2071, 1797, 1574, 1517, 1653, 1910, 2182, 2349, 2376, 2253, 1999, 1747, 1706, 1887, 2211, 2423,

2337, 2094, 2086, 2113, 2111, 2110, 2114, 2126, 2149, 2183, 2232, 2294, 2371, 2463, 2559, 2648,

2728, 2798, 2855, 2900, 2933, 2953, 2960, 2955, 2938, 2909, 2869, 2819, 2758, 2689, 2611, 2525,

2431, 2331, 2228, 2121, 2013, 1904, 1797, 1694, 1596, 1503, 1418, 1340, 1274, 1219, 1178, 1150,

1139, 1144, 1166, 1209, 1273, 1359, 1467, 1592, 1730, 1868, 1995, 2103, 2181, 2230, 2247, 2232,

2184, 2102, 1979, 1804, 1654, 1897, 2127, 1998, 1811, 2097, 1927, 2329, 1954, 2111, 2317, 2228,

2157, 2108, 2084, 2088, 2128, 2202, 2300, 2415, 2530, 2633, 2717, 2784, 2834, 2865, 2878, 2873,

2852, 2814, 2762, 2698, 2620, 2532, 2435, 2329, 2217, 2100, 1980, 1861, 1743, 1631, 1526, 1430,

1345, 1272, 1214, 1168, 1137, 1122, 1124, 1146, 1188, 1249, 1332, 1435, 1555, 1688, 1823, 1950,

2059, 2145, 2204, 2238, 2244, 2219, 2164, 2068, 1904, 1880, 1865, 2140, 2204, 1881, 1701, 1688,

1826, 2069, 2328, 2519, 2600, 2534, 2343, 2088, 1862, 1730, 1700, 1758, 1906, 2128, 2369, 2488,

2493, 2430, 2347, 2277, 2246, 2259, 2313, 2393, 2477, 2547, 2591, 2600, 2571, 2510, 2421, 2305,

2167, 2019, 1869, 1727, 1598, 1491, 1411, 1364, 1358, 1392, 1464, 1576, 1712, 1836, 1929, 1987,

2009, 2010, 1997, 2045, 2009, 1766, 1613, 1670, 1924, 2179, 2354, 2424, 2368, 2186, 1941, 1737,

1604, 1552, 1577, 1670, 1822, 2024, 2249, 2452, 2600, 2679, 2688, 2621, 2484, 2290, 2062, 1865,

1743, 1693, 1713, 1795, 1924, 2088, 2247, 2357, 2418, 2449, 2458, 2464, 2476, 2491, 2506, 2509,

2491, 2449, 2375, 2267, 2130, 1974, 1811, 1663, 1560, 1534, 1609, 1762, 1925, 1990, 1953, 1830,

1859, 2157, 2248, 2106, 1807, 1591, 1598, 1832, 2137, 2342, 2380, 2232, 1960, 1748, 1725, 1872,

2137, 2411, 2519, 2397, 2117, 1921, 1918, 1994, 2068, 2126, 2181, 2236, 2299, 2378, 2467, 2556,

2632, 2688, 2726, 2741, 2735, 2710, 2666, 2604, 2524, 2427, 2316, 2196, 2067, 1935, 1805, 1679,

1564, 1464, 1384, 1330, 1305, 1316, 1366, 1459, 1592, 1756, 1925, 2075, 2182, 2239, 2244, 2190,

2072, 1881, 1661, 1814, 2126, 2199, 1977, 1714, 2085, 2002, 2193, 2103, 2167, 2177, 2105, 2071,

2090, 2159, 2279, 2420, 2548, 2646, 2712, 2746, 2750, 2729, 2683, 2616, 2530, 2426, 2307, 2175,

2036, 1893, 1753, 1623, 1507, 1409, 1329, 1271, 1237, 1230, 1252, 1305, 1392, 1512, 1659, 1823,

1986, 2130, 2241, 2314, 2344, 2326, 2258, 2131, 1895, 1776, 2029, 2046, 1843, 1777, 1940, 2229,

2466, 2525, 2374, 2085, 1852, 1764, 1831, 2047, 2347, 2423, 2289, 2162, 2112, 2175, 2313, 2436,

2510, 2523, 2467, 2347, 2181, 1988, 1796, 1628, 1504, 1440, 1447, 1529, 1682, 1877, 2040, 2124,

2123, 2033, 1943, 1810, 1779, 2067, 2282, 2308, 2139, 1866, 1675, 1586, 1582, 1655, 1804, 2014,

2244, 2442, 2587, 2668, 2671, 2596, 2447, 2234, 2003, 1817, 1705, 1674, 1717, 1828, 2005, 2226,

2391, 2443, 2407, 2337, 2302, 2313, 2308, 2242, 2067, 1809, 1671, 1846, 2023, 1896, 2098, 2115,

1759, 1745, 2110, 2287, 2120, 1808, 1771, 1952, 2245, 2456, 2442, 2231, 1971, 1856, 1851, 1917,

2034, 2172, 2317, 2451, 2550, 2590, 2578, 2523, 2426, 2299, 2156, 2007, 1862, 1737, 1647, 1634,

1763, 1997, 2102, 1989, 1940, 1995, 2027, 2162, 2000, 1988, 2098, 2257, 2422, 2518, 2529, 2476,

2377, 2244, 2092, 1931, 1776, 1648, 1570, 1563, 1663, 1874, 2085, 2204, 2199, 2005, 1753, 2056,

2099, 1868, 2193, 2147, 2087, 2220, 2235, 2317, 2336, 2265, 2118, 1935, 1747, 1627, 1647, 1833,

2053, 2160, 2112, 1840, 1806, 2122, 2125, 1806, 1886, 2163, 2387, 2378, 2104, 1882, 1869, 2023,

2268, 2405, 2367, 2215, 2095, 2010, 1854, 1870, 2069, 1838, 2161, 1904, 2151, 2390, 2240, 1940,

1832, 1872, 2005, 2186, 2377, 2501, 2531, 2461, 2326, 2160, 1997, 1863, 1793, 1922, 1985, 1848,

2027, 2100, 2159, 1954, 2060, 2262, 2434, 2473, 2393, 2247, 2063, 1877, 1732, 1687, 1852, 2111,

2121, 1808, 2072, 1946, 2172, 2141, 2208, 2314, 2223, 2006, 1778, 1735, 2002, 2151, 1991, 1775,

2083, 1977, 2059, 2315, 1990, 1921, 2223, 2272, 2096, 1969, 1895, 2074, 1892, 2071, 2119, 2236,

1893, 1926, 2150, 2372, 2398, 2217, 1992, 1879, 1980, 2003, 2080, 1979, 2009, 2266, 2407, 2309,

2077, 1856, 1792, 2094, 1975, 1977, 2109, 2086, 2272, 2248, 2000, 1795, 2050, 2118, 1808, 1956,

2098, 2176, 1964, 2198, 2168, 1963, 2096, 1832, 2157, 2173, 1929, 1999, 2271, 2341, 2107, 1956,

2006, 2058, 1930, 2144, 2370, 2289, 2026, 1859, 2036, 2119, 2092, 2320, 2136, 1880, 2049, 1858,

2044, 2016, 2199, 2063, 1949, 1880, 2103, 2025, 2122, 2186, 1977, 1946, 2067, 2075, 2320, 2073,

1904, 1972, 2062, 2281, 1991, 1959, 1989, 2123, 2138, 2143, 2143, 2140, 2137, 2131, 2124, 2119,

2112, 2106, 2098, 2094, 2095, 2086, 2065, 2037, 2001, 1958, 1900, 1836, 1762, 1681, 1592, 1496,

1395, 1289, 1181, 1070, 961, 855, 761, 703, 687, 664, 650, 646, 650, 662, 682, 709,

742, 786, 831, 885, 941, 1003, 1066, 1131, 1197, 1267, 1345, 1412, 1492, 1592, 1659, 1687,

1732, 1790, 1854, 1909, 1950, 2000, 2048, 2102, 2202, 2284, 2311, 2336, 2332, 2369, 2352, 2289,

2324, 2198, 2110, 2189, 2269, 2035, 2020, 2201, 2026, 2068, 1802, 1972, 2335, 2364, 2214, 2431,

2672, 2719, 2566, 2539, 2505, 2324, 2199, 2212, 2226, 2229, 2219, 2176, 2090, 2074, 2157, 2287,

2369, 2419, 2625, 2923, 2918, 2764, 2559, 2360, 2204, 2193, 2258, 2313, 2340, 2331, 2242, 2131,

1994, 1885, 1773, 1680, 1553, 1459, 1358, 1279, 1228, 1195, 1222, 1292, 1406, 1430, 1406, 1407,

1472, 1623, 1799, 1890, 1916, 1945, 1963, 1985, 2031, 2110, 2153, 2110, 1983, 1819, 1547, 1477,

1564, 1675, 1782, 1897, 1971, 2012, 2041, 2102, 2143, 2179, 2258, 2371, 2508, 2649, 2777, 2882,

2970, 2987, 2975, 2932, 2853, 2786, 2710, 2645, 2595, 2515, 2425, 2332, 2227, 2100, 1982, 1879,

1784, 1715, 1664, 1623, 1587, 1557, 1539, 1528, 1507, 1486, 1478, 1510, 1561, 1597, 1660, 1737,

1828, 1917, 1998, 2087, 2179, 2277, 2378, 2453, 2511, 2568, 2621, 2669, 2715, 2755, 2784, 2808,

2822, 2833, 2849, 2862, 2875, 2894, 2921, 2949, 2968, 2978, 2977, 2978, 2988, 2995, 2995, 2997,

3004, 3000, 2988, 2983, 2973, 2950, 2915, 2881, 2845, 2807, 2772, 2735, 2699, 2668, 2642, 2621,

2603, 2584, 2565, 2550, 2539, 2526, 2507, 2487, 2468, 2447, 2421, 2391, 2361, 2330, 2298, 2263,

2227, 2193, 2159, 2121, 2081, 2039, 1999, 1961, 1921, 1878, 1833, 1788, 1743, 1698, 1652, 1608,

1567, 1530, 1493, 1455, 1416, 1377, 1339, 1299, 1260, 1221, 1182, 1144, 1108, 1074, 1044, 1017,

993, 970, 947, 923, 898, 874, 853, 830, 806, 783, 767, 759, 760, 766, 775, 784, 793, 802, 811, 822, 836, 857, 881, 905, 929, 952, 977, 1003, 1035, 1072, 1113, 1158,

1207, 1256, 1305, 1353, 1403, 1456, 1513, 1576, 1643, 1712, 1782, 1852, 1920, 1986, 2048, 2107,

2165, 2226, 2290, 2354, 2413, 2465, 2513, 2561, 2610, 2653, 2689, 2719, 2746, 2767, 2780, 2788,

2790, 2786, 2780, 2774, 2760, 2733, 2695, 2648, 2597, 2541, 2473, 2389, 2298, 2206, 2106, 1988,

1859, 1741, 1641, 1546, 1450, 1360, 1285, 1227, 1181, 1152, 1140, 1138, 1156, 1189, 1237, 1299,

1376, 1461, 1559, 1662, 1762, 1865, 1976, 2089, 2196, 2299, 2391, 2476, 2536, 2585, 2621, 2651,

2660, 2628, 2547, 2409, 2210, 2012, 1868, 1810, 1780, 1797, 1832, 1878, 1964, 2064, 2163, 2272,

2401, 2527, 2626, 2710, 2753, 2763, 2733, 2677, 2587, 2467, 2357, 2249, 2152, 2064, 1999, 1949,

1922, 1979, 2035, 2044, 2002, 1928, 1841, 1736, 1628, 1525, 1446, 1401, 1398, 1421, 1483, 1564,

1683, 1802, 1925, 2051, 2162, 2233, 2245, 2192, 2095, 2109, 2195, 2350, 2468, 2527, 2487, 2365,

2204, 2015, 1868, 1765, 1689, 1650, 1631, 1621, 1619, 1626, 1644, 1669, 1700, 1735, 1774, 1815,

1860, 1908, 1959, 2011, 2065, 2120, 2176, 2233, 2293, 2354, 2415, 2477, 2540, 2603, 2665, 2725,

2785, 2844, 2903, 2960, 3015, 3066, 3113, 3155, 3192, 3226, 3258, 3286, 3311, 3332, 3350, 3364,

3374, 3381, 3383, 3381, 3375, 3365, 3352, 3336, 3316, 3293, 3267, 3238, 3208, 3176, 3142, 3106,

3069, 3030, 2990, 2948, 2906, 2862, 2817, 2772, 2725, 2678, 2630, 2581, 2532, 2482, 2431, 2380,

2329, 2277, 2226, 2174, 2123, 2073, 2024, 1975, 1926, 1877, 1828, 1778, 1728, 1678, 1628, 1578,

1530, 1483, 1438, 1395, 1353, 1313, 1276, 1239, 1202, 1167, 1131, 1096, 1062, 1031, 1004, 979,

958, 938, 922, 908, 896, 888, 885, 887, 895, 909, 929, 954, 983, 1018, 1056, 1099,

1145, 1195, 1246, 1300, 1359, 1423, 1489, 1554, 1616, 1673, 1727, 1778, 1826, 1872, 1916, 1956,

1995, 2030, 2058, 2081, 2099, 2109, 2103, 2082, 2027, 1982, 1993, 1800, 1682, 1676, 1714, 1777,

1851, 1949, 2053, 2157, 2265, 2374, 2483, 2591, 2684, 2752, 2790, 2804, 2797, 2761, 2687, 2589,

2475, 2348, 2222, 2108, 2005, 1920, 1854, 1805, 1782, 1785, 1810, 1863, 1944, 2050, 2204, 2369,

2518, 2632, 2696, 2724, 2724, 2704, 2668, 2616, 2552, 2476, 2391, 2296, 2197, 2098, 1998, 1899,

1798, 1699, 1607, 1522, 1444, 1374, 1316, 1272, 1239, 1219, 1216, 1229, 1252, 1292, 1358, 1446,

1534, 1626, 1728, 1827, 1917, 1993, 2053, 2097, 2124, 2129, 2101, 2033, 1888, 1721, 1687, 1746,

1862, 2030, 2206, 2381, 2509, 2595, 2625, 2602, 2533, 2420, 2268, 2096, 1940, 1799, 1688, 1625,

1612, 1640, 1719, 1820, 1937, 2046, 2149, 2236, 2306, 2368, 2426, 2476, 2516, 2549, 2578, 2605,

2630, 2653, 2672, 2685, 2697, 2710, 2721, 2729, 2735, 2743, 2750, 2754, 2753, 2748, 2739, 2727,

2711, 2692, 2671, 2648, 2621, 2591, 2560, 2529, 2496, 2461, 2423, 2383, 2340, 2296, 2249, 2201,

2151, 2100, 2048, 1994, 1940, 1887, 1834, 1781, 1727, 1671, 1614, 1557, 1501, 1446, 1393, 1342,

1293, 1245, 1199, 1156, 1116, 1079, 1046, 1017, 995, 979, 969, 965, 965, 970, 983, 1002,

1030, 1065, 1104, 1148, 1194, 1243, 1296, 1354, 1416, 1479, 1545, 1614, 1685, 1757, 1832, 1906,

1979, 2050, 2119, 2186, 2251, 2314, 2373, 2427, 2476, 2519, 2553, 2575, 2585, 2584, 2569, 2535,

2478, 2399, 2298, 2173, 2033, 1885, 1743, 1617, 1512, 1438, 1394, 1374, 1383, 1426, 1500, 1604,

1728, 1861, 2005, 2146, 2268, 2365, 2439, 2484, 2484, 2393, 2182, 2008, 1951, 1992, 2133, 2323,

2446, 2423, 2286, 2152, 2099, 2137, 2153, 2128, 2079, 2006, 1916, 1814, 1707, 1608, 1514, 1432,

1371, 1335, 1330, 1354, 1398, 1462, 1550, 1659, 1775, 1891, 2010, 2128, 2229, 2306, 2350, 2340,

2210, 1898, 1899, 2230, 2349, 2136, 1900, 1797, 1753, 1734, 1734, 1749, 1774, 1807, 1845, 1888,

1938, 1994, 2053, 2115, 2180, 2246, 2316, 2387, 2460, 2534, 2607, 2679, 2750, 2819, 2884, 2944,

2998, 3047, 3090, 3126, 3156, 3180, 3197, 3207, 3211, 3208, 3200, 3187, 3168, 3144, 3115, 3082,

3046, 3007, 2965, 2920, 2873, 2823, 2771, 2717, 2662, 2605, 2547, 2488, 2428, 2368, 2308, 2247,

2187, 2127, 2067, 2007, 1948, 1888, 1829, 1771, 1713, 1657, 1603, 1550, 1499, 1450, 1403, 1359,

1316, 1276, 1238, 1204, 1174, 1150, 1130, 1116, 1107, 1107, 1114, 1131, 1157, 1192, 1236, 1287,

1348, 1415, 1487, 1559, 1629, 1697, 1762, 1823, 1879, 1927, 1969, 2005, 2036, 2056, 2061, 2039,

2085, 1988, 1752, 1683, 1724, 1823, 1956, 2117, 2288, 2441, 2555, 2612, 2597, 2503, 2344, 2148,

1976, 1854, 1774, 1731, 1718, 1733, 1773, 1834, 1916, 2020, 2140, 2270, 2404, 2528, 2637, 2727,

2791, 2827, 2844, 2841, 2817, 2776, 2719, 2649, 2570, 2483, 2390, 2294, 2194, 2095, 1997, 1902,

1812, 1727, 1648, 1576, 1510, 1454, 1411, 1386, 1377, 1388, 1421, 1477, 1555, 1648, 1746, 1829,

1892, 1939, 1960, 1950, 1906, 1822, 1733, 1718, 1790, 1911, 2074, 2258, 2429, 2543, 2586, 2563,

2478, 2343, 2175, 2000, 1836, 1715, 1650, 1660, 1781, 1988, 2174, 2286, 2338, 2345, 2334, 2328,

2336, 2356, 2389, 2440, 2491, 2538, 2568, 2575, 2563, 2534, 2495, 2445, 2387, 2323, 2254, 2181,

2104, 2022, 1936, 1850, 1764, 1679, 1596, 1516, 1440, 1372, 1314, 1266, 1228, 1203, 1190, 1195,

1214, 1251, 1304, 1369, 1443, 1528, 1623, 1724, 1828, 1934, 2038, 2136, 2226, 2306, 2374, 2429,

2472, 2499, 2506, 2488, 2442, 2365, 2260, 2125, 1967, 1806, 1659, 1543, 1465, 1428, 1429, 1470,

1548, 1658, 1793, 1934, 2075, 2209, 2322, 2415, 2475, 2483, 2415, 2234, 2070, 2015, 2114, 2274,

2190, 2215, 2221, 2184, 2124, 2041, 1942, 1838, 1727, 1613, 1515, 1440, 1394, 1383, 1404, 1458,

1536, 1634, 1749, 1876, 2006, 2128, 2236, 2319, 2361, 2319, 2043, 1879, 2215, 2157, 1873, 1794,

1776, 1784, 1813, 1857, 1913, 1979, 2050, 2130, 2212, 2296, 2380, 2463, 2545, 2626, 2702, 2773,

2838, 2895, 2942, 2977, 3000, 3010, 3008, 2996, 2973, 2941, 2901, 2854, 2799, 2740, 2675, 2607,

2538, 2467, 2394, 2320, 2245, 2169, 2093, 2017, 1942, 1869, 1797, 1728, 1663, 1603, 1549, 1502,

1461, 1426, 1401, 1389, 1394, 1418, 1462, 1528, 1609, 1699, 1775, 1834, 1880, 1909, 1919, 1901,

1861, 2135, 2111, 1819, 1703, 1717, 1811, 1951, 2120, 2295, 2450, 2562, 2620, 2621, 2568, 2457,

2294, 2109, 1938, 1800, 1700, 1640, 1615, 1624, 1662, 1725, 1810, 1919, 2047, 2187, 2328, 2460,

2576, 2670, 2737, 2775, 2786, 2770, 2728, 2663, 2578, 2480, 2374, 2264, 2154, 2044, 1938, 1837,

1744, 1663, 1595, 1545, 1513, 1505, 1529, 1593, 1684, 1783, 1863, 1909, 1915, 1874, 1804, 1755,

1817, 1999, 2233, 2413, 2450, 2332, 2104, 1867, 1725, 1729, 1891, 2152, 2329, 2368, 2309, 2242,

2211, 2227, 2293, 2392, 2470, 2496, 2478, 2437, 2379, 2306, 2224, 2133, 2036, 1932, 1824, 1717,

1614, 1520, 1438, 1372, 1325, 1298, 1295, 1314, 1360, 1429, 1517, 1621, 1737, 1857, 1979, 2098,

2205, 2302, 2384, 2447, 2485, 2495, 2471, 2407, 2296, 2143, 1962, 1778, 1627, 1534, 1514, 1571,

1698, 1887, 2076, 2224, 2294, 2203, 2012, 2221, 2283, 2171, 2160, 2249, 2360, 2385, 2360, 2313,

2254, 2184, 2104, 2017, 1922, 1824, 1724, 1625, 1532, 1449, 1378, 1318, 1274, 1248, 1240, 1252,

1284, 1335, 1402, 1482, 1574, 1673, 1775, 1877, 1977, 2073, 2164, 2251, 2330, 2395, 2438, 2445,

2384, 2200, 1936, 1936, 2076, 1821, 1800, 1846, 1919, 2010, 2111, 2220, 2333, 2443, 2547, 2635,

2702, 2746, 2769, 2765, 2736, 2684, 2610, 2519, 2416, 2309, 2202, 2099, 2001, 1910, 1831, 1765,

1721, 1710, 1748, 1831, 1872, 1838, 1799, 1933, 2133, 1954, 1755, 1767, 1862, 2012, 2189, 2366,

2505, 2581, 2597, 2555, 2450, 2299, 2118, 1937, 1786, 1683, 1627, 1612, 1638, 1701, 1798, 1926,

2076, 2235, 2384, 2514, 2611, 2667, 2685, 2662, 2603, 2512, 2396, 2265, 2129, 1999, 1880, 1776,

1696, 1645, 1634, 1678, 1773, 1867, 1909, 1888, 1807, 1799, 1949, 2202, 2389, 2428, 2335, 2141,

1900, 1716, 1635, 1654, 1774, 1969, 2186, 2357, 2441, 2406, 2279, 2171, 2129, 2150, 2242, 2377,

2453, 2457, 2416, 2342, 2250, 2143, 2025, 1903, 1781, 1665, 1559, 1472, 1410, 1379, 1385, 1429,

1514, 1630, 1763, 1905, 2046, 2171, 2271, 2336, 2357, 2316, 2180, 1927, 1706, 1671, 1850, 2128,

2261, 2024, 2086, 2082, 2141, 2247, 2382, 2491, 2551, 2569, 2553, 2514, 2460, 2397, 2325, 2245,

2159, 2068, 1974, 1879, 1785, 1693, 1603, 1519, 1440, 1371, 1314, 1272, 1249, 1246, 1263, 1300,

1356, 1429, 1518, 1616, 1721, 1827, 1929, 2025, 2110, 2184, 2247, 2291, 2303, 2229, 1912, 2192,

1888, 1921, 2045, 2206, 2371, 2499, 2559, 2533, 2434, 2289, 2124, 1978, 1890, 1924, 1960, 1827,

2060, 2024, 1772, 1735, 1821, 1975, 2169, 2357, 2486, 2522, 2466, 2307, 2092, 1894, 1770, 1725,

1749, 1836, 1972, 2141, 2314, 2454, 2550, 2585, 2559, 2485, 2371, 2239, 2107, 1984, 1877, 1790,

1726, 1701, 1716, 1756, 1793, 1814, 1840, 1914, 2074, 2298, 2408, 2381, 2265, 2085, 1882, 1700,

1597, 1592, 1687, 1860, 2062, 2237, 2361, 2365, 2200, 2079, 2088, 2192, 2361, 2448, 2424, 2336,

2213, 2062, 1900, 1744, 1621, 1551, 1550, 1641, 1829, 2034, 2178, 2241, 2203, 2003, 1726, 1751,

2015, 2244, 2141, 2028, 1982, 2056, 2177, 2335, 2481, 2568, 2592, 2568, 2509, 2431, 2340, 2240,

2134, 2023, 1910, 1799, 1694, 1600, 1520, 1457, 1415, 1398, 1414, 1465, 1548, 1657, 1780, 1903,

2016, 2110, 2186, 2239, 2203, 1927, 2061, 2211, 2315, 2335, 2239, 2120, 2003, 1836, 2026, 1883,

2143, 1966, 2070, 2203, 2324, 2410, 2441, 2417, 2350, 2262, 2159, 2047, 1936, 1829, 1730, 1646,

1586, 1558, 1569, 1621, 1698, 1789, 1892, 2007, 2127, 2245, 2287, 2060, 1839, 2000, 2184, 2241,

2218, 2160, 1916, 1912, 1902, 2020, 1974, 2068, 2217, 2366, 2435, 2396, 2303, 2172, 2030, 1896,

1771, 1675, 1646, 1702, 1820, 1962, 2104, 2212, 1954, 1890, 2100, 2261, 2255, 2160, 2133, 1879,

1884, 1998, 1959, 2110, 2250, 2369, 2388, 2322, 2209, 2086, 1960, 1839, 1731, 1654, 1631, 1671,

1776, 1904, 2035, 2182, 2044, 2008, 2238, 2118, 2142, 1894, 1852, 1973, 2066, 2187, 2325, 2377,

2312, 2188, 2044, 1900, 1778, 1691, 1668, 1751, 1894, 2055, 2182, 1917, 2035, 2238, 2109, 2183,

1969, 1825, 1807, 1919, 2114, 2118, 2271, 2336, 2243, 2088, 1921, 1777, 1732, 1858, 2057, 2119,

1897, 2164, 2150, 2176, 1979, 1844, 1988, 2075, 2218, 2304, 2226, 2068, 1898, 1771, 1837, 2044,

1994, 2136, 2039, 2241, 2067, 1869, 1789, 1885, 2049, 2138, 2235, 2124, 1919, 1837, 2071, 1889,

2161, 2085, 2220, 2052, 1849, 1816, 2000, 1989, 2163, 2231, 2070, 1895, 1960, 2078, 2227, 2066,

1855, 1895, 2119, 2189, 2116, 1897, 1941, 2081, 2066, 1856, 1851, 2093, 2049, 2148, 1938, 2131,

2153, 2084, 1865, 1855, 2101, 1955, 2210, 2103, 2018, 2072, 2066, 2061, 2055, 2051, 2049, 2041,

2010, 1941, 1823, 1664, 1482, 1300, 1139, 1039, 1049, 1091, 1166, 1268, 1397, 1540, 1692, 1850,

2016, 2203, 2256, 2354, 2396, 2440, 2502, 2620, 2691, 2621, 2474, 2362, 2079, 2004, 2088, 1875,

1627, 1473, 1565, 1702, 1938, 2165, 2179, 2494, 2336, 1791, 1965, 1672, 1642, 1752, 1739, 2029,

2281, 2344, 2634, 2591, 2143, 2066, 2146, 2176, 2191, 2152, 2089, 2049, 1989, 1959, 1958, 1970,

2005, 2058, 2128, 2207, 2273, 2315, 2345, 2367, 2385, 2407, 2448, 2510, 2597, 2697, 2813, 2918,

3020, 3099, 3137, 3172, 3201, 3206, 3197, 3201, 3232, 3282, 3340, 3379, 3390, 3385, 3365, 3326,

3272, 3214, 3148, 3070, 2974, 2871, 2770, 2669, 2558, 2440, 2327, 2240, 2173, 2118, 2069, 2035,

2011, 1988, 1964, 1935, 1902, 1864, 1817, 1761, 1700, 1638, 1576, 1510, 1439, 1365, 1299, 1243,

1198, 1167, 1147, 1128, 1104, 1077, 1057, 1041, 1018, 995, 1000, 1018, 1029, 1057, 1094, 1162,

1261, 1390, 1538, 1651, 1743, 1822, 1893, 1987, 2114, 2229, 2340, 2405, 2412, 2345, 2255, 2166,

2101, 2075, 2082, 1952, 1688, 1628, 1850, 2206, 2411, 2241, 2082, 1977, 1757, 1628, 1656, 1830,

2102, 2415, 2531, 2420, 2337, 2302, 2304, 2266, 2230, 2227, 2206, 2182, 2163, 2154, 2161, 2171,

2181, 2189, 2202, 2228, 2265, 2315, 2377, 2445, 2513, 2584, 2656, 2724, 2794, 2859, 2920, 2981,

3044, 3102, 3145, 3184, 3215, 3233, 3238, 3241, 3242, 3235, 3218, 3193, 3163, 3129, 3091, 3049,

3007, 2967, 2930, 2892, 2852, 2809, 2763, 2714, 2663, 2611, 2560, 2508, 2455, 2402, 2350, 2297,

2243, 2188, 2131, 2072, 2011, 1949, 1885, 1820, 1757, 1697, 1639, 1581, 1522, 1460, 1399, 1340,

1281, 1225, 1171, 1121, 1076, 1032, 990, 954, 924, 898, 873, 853, 842, 842, 847, 858,

877, 905, 943, 990, 1049, 1117, 1191, 1264, 1335, 1410, 1484, 1556, 1636, 1718, 1796, 1863,

1918, 1972, 2027, 2072, 2108, 2138, 2167, 2193, 2219, 2241, 2259, 2272, 2262, 2180, 1977, 1741,

1921, 1995, 1968, 1893, 1700, 1610, 1665, 1890, 2126, 2312, 2371, 2275, 2132, 2037, 1982, 1939,

1923, 1936, 1969, 2014, 2079, 2152, 2237, 2332, 2426, 2520, 2617, 2708, 2796, 2875, 2933, 2972,

2994, 3007, 3015, 3016, 3005, 2984, 2953, 2918, 2876, 2825, 2768, 2705, 2637, 2565, 2493, 2420,

2344, 2264, 2181, 2095, 2007, 1918, 1830, 1746, 1667, 1592, 1520, 1452, 1388, 1328, 1276, 1233,

1198, 1170, 1149, 1136, 1135, 1148, 1172, 1211, 1271, 1346, 1428, 1520, 1633, 1752, 1868, 1976,

2073, 2163, 2241, 2303, 2344, 2367, 2368, 2313, 2152, 1878, 1871, 2125, 2382, 2342, 2076, 1802,

1686, 1794, 2074, 2338, 2343, 2249, 2148, 2100, 2069, 2057, 2070, 2095, 2130, 2180, 2244, 2316,

2395, 2485, 2577, 2658, 2735, 2811, 2879, 2929, 2963, 2984, 2992, 2985, 2969, 2946, 2917, 2882,

2841, 2796, 2744, 2689, 2631, 2570, 2506, 2439, 2370, 2299, 2228, 2156, 2082, 2006, 1928, 1849,

1771, 1696, 1625, 1558, 1494, 1430, 1368, 1307, 1251, 1202, 1157, 1118, 1085, 1059, 1042, 1036,

1040, 1053, 1074, 1106, 1152, 1210, 1273, 1340, 1415, 1504, 1602, 1700, 1793, 1882, 1963, 2035,

2100, 2161, 2213, 2254, 2282, 2292, 2270, 2214, 2084, 1805, 1959, 2192, 2029, 1751, 1699, 1797,

2005, 2254, 2458, 2531, 2462, 2256, 2069, 1957, 1889, 1862, 1867, 1894, 1956, 2046, 2157, 2296,

2432, 2565, 2681, 2758, 2808, 2839, 2846, 2832, 2804, 2765, 2717, 2658, 2593, 2521, 2443, 2360,

2275, 2187, 2097, 2004, 1910, 1816, 1724, 1638, 1557, 1480, 1408, 1342, 1282, 1231, 1186, 1151,

1127, 1117, 1120, 1136, 1166, 1215, 1281, 1357, 1440, 1534, 1639, 1747, 1850, 1946, 2032, 2113,

2185, 2244, 2285, 2305, 2293, 2231, 2068, 1838, 2161, 2270, 1957, 1733, 1752, 1973, 2265, 2420,

2390, 2294, 2213, 2152, 2120, 2107, 2104, 2115, 2138, 2172, 2212, 2261, 2320, 2387, 2461, 2542,

2627, 2707, 2780, 2844, 2901, 2948, 2981, 3001, 3010, 3012, 3004, 2988, 2965, 2935, 2900, 2859,

2814, 2764, 2711, 2655, 2595, 2534, 2470, 2405, 2338, 2269, 2197, 2125, 2052, 1980, 1907, 1834,

1762, 1690, 1620, 1554, 1490, 1430, 1372, 1318, 1267, 1219, 1174, 1134, 1100, 1070, 1047, 1032,

1024, 1025, 1035, 1054, 1084, 1124, 1178, 1242, 1315, 1391, 1473, 1560, 1651, 1742, 1829, 1910,

1984, 2050, 2109, 2161, 2203, 2235, 2252, 2255, 2232, 2166, 2041, 1789, 1813, 2115, 2187, 2050,

1772, 1638, 1698, 1895, 2165, 2378, 2433, 2299, 2118, 2012, 1964, 1948, 1963, 2006, 2064, 2136,

2224, 2328, 2445, 2560, 2665, 2753, 2816, 2854, 2872, 2871, 2850, 2812, 2762, 2700, 2629, 2551,

2466, 2376, 2281, 2183, 2085, 1985, 1888, 1795, 1709, 1629, 1556, 1488, 1428, 1381, 1347, 1329,

1332, 1359, 1416, 1507, 1622, 1756, 1889, 2003, 2092, 2151, 2177, 2140, 1979, 1863, 2194, 2125,

1805, 1678, 1789, 2057, 2321, 2399, 2283, 2146, 2071, 2041, 2039, 2052, 2083, 2137, 2204, 2281,

2370, 2466, 2564, 2659, 2742, 2807, 2854, 2880, 2888, 2880, 2861, 2829, 2787, 2734, 2674, 2608,

2537, 2461, 2381, 2297, 2210, 2121, 2032, 1943, 1855, 1770, 1687, 1608, 1533, 1466, 1403, 1346,

1297, 1260, 1236, 1227, 1234, 1258, 1301, 1364, 1449, 1554, 1668, 1790, 1912, 2025, 2126, 2214,

2285, 2336, 2355, 2323, 2205, 1911, 1825, 2046, 1972, 1740, 1847, 2188, 2242, 2041, 1975, 1976,

2015, 2083, 2177, 2286, 2410, 2535, 2644, 2717, 2758, 2771, 2757, 2723, 2670, 2600, 2518, 2428,

2331, 2229, 2125, 2018, 1914, 1813, 1717, 1629, 1553, 1488, 1434, 1396, 1380, 1394, 1449, 1544,

1675, 1816, 1958, 2079, 2165, 2205, 2161, 1928, 2178, 2325, 2146, 1832, 1672, 1770, 2090, 2268,

2128, 2011, 1988, 2018, 2087, 2188, 2307, 2440, 2567, 2661, 2711, 2730, 2719, 2685, 2635, 2570,

2493, 2406, 2313, 2215, 2113, 2010, 1906, 1805, 1707, 1614, 1532, 1462, 1405, 1365, 1345, 1352,

1393, 1471, 1583, 1716, 1851, 1980, 2088, 2171, 2217, 2206, 2055, 2084, 1865, 1924, 2291, 2306,

2001, 1879, 1870, 1917, 2008, 2148, 2316, 2477, 2597, 2661, 2683, 2671, 2632, 2574, 2498, 2406,

2302, 2190, 2074, 1957, 1846, 1742, 1645, 1556, 1479, 1417, 1373, 1355, 1366, 1414, 1498, 1613,

1753, 1889, 2012, 2116, 2192, 2220, 2154, 1901, 2252, 1940, 1782, 2054, 2305, 2261, 2172, 2142,

2149, 2191, 2258, 2343, 2440, 2537, 2616, 2660, 2668, 2650, 2612, 2558, 2490, 2410, 2320, 2225,

2128, 2030, 1931, 1832, 1737, 1645, 1559, 1484, 1424, 1379, 1355, 1356, 1388, 1457, 1554, 1667,

1788, 1899, 1994, 2069, 2112, 2106, 2010, 2089, 1992, 1816, 2176, 2271, 2008, 1936, 1950, 2008,

2096, 2209, 2343, 2475, 2588, 2667, 2711, 2716, 2692, 2643, 2574, 2490, 2396, 2296, 2191, 2081,

1971, 1864, 1763, 1669, 1587, 1516, 1462, 1430, 1430, 1472, 1556, 1682, 1825, 1945, 2032, 2077,

2060, 1930, 2102, 1775, 2045, 2243, 2071, 2038, 2073, 2147, 2249, 2373, 2499, 2596, 2645, 2644,

2609, 2548, 2464, 2360, 2240, 2115, 1988, 1865, 1749, 1645, 1564, 1515, 1509, 1568, 1712, 1910,

2085, 2199, 2214, 1978, 2096, 1900, 1871, 2219, 2030, 1991, 2057, 2186, 2349, 2490, 2570, 2586,

2550, 2475, 2375, 2261, 2140, 2013, 1883, 1759, 1651, 1566, 1511, 1498, 1541, 1648, 1802, 1956,

2074, 2130, 2082, 2228, 1956, 1753, 2026, 2219, 2019, 2009, 2091, 2231, 2386, 2514, 2576, 2572,

2525, 2442, 2334, 2209, 2074, 1935, 1801, 1682, 1587, 1526, 1511, 1555, 1676, 1847, 2008, 2117,

2130, 2019, 2137, 1910, 1970, 2116, 2306, 2467, 2546, 2545, 2489, 2394, 2272, 2131, 1985, 1843,

1716, 1614, 1546, 1530, 1583, 1716, 1897, 2045, 2120, 2046, 2085, 1823, 2127, 2298, 2085, 2042,

2138, 2317, 2455, 2469, 2397, 2271, 2116, 1952, 1794, 1663, 1585, 1590, 1690, 1870, 2022, 2076,

1944, 1866, 2194, 2009, 1932, 1970, 2071, 2212, 2367, 2505, 2599, 2634, 2620, 2565, 2485, 2384,

2269, 2143, 2014, 1889, 1772, 1667, 1582, 1525, 1510, 1547, 1648, 1798, 1946, 2047, 2083, 1992,

1945, 2148, 2018, 2097, 2278, 2443, 2476, 2392, 2241, 2047, 1858, 1724, 1696, 1890, 2119, 1986,

2063, 1951, 2172, 2011, 2092, 2297, 2446, 2451, 2348, 2198, 2025, 1854, 1708, 1630, 1669, 1826,

2032, 2130, 2143, 1899, 1812, 2139, 2080, 1984, 2115, 2327, 2465, 2457, 2359, 2209, 2028, 1849,

1705, 1634, 1670, 1850, 2059, 2112, 1972, 1997, 2053, 2299, 2418, 2334, 2152, 1946, 1772, 1701,

1782, 1981, 2067, 1962, 2231, 2094, 2266, 2181, 1980, 1804, 1747, 1851, 2027, 2028, 2086, 2123,

2279, 2382, 2320, 2185, 2027, 1880, 1765, 1737, 1837, 2012, 2122, 2087, 2238, 2258, 2084, 1902,

1813, 1977, 1960, 2050, 2216, 2305, 2164, 1977, 1804, 1809, 2016, 2212, 1920, 2051, 2214, 2280,

2116, 1932, 1812, 1912, 2156, 2024, 2227, 2183, 1996, 1835, 1825, 2017, 2194, 1963, 2001, 2253,

2101, 1889, 1855, 2080, 2106, 2186, 2258, 2075, 1892, 1803, 1921, 2155, 2046, 2237, 2211, 2021,

1841, 1851, 2097, 2033, 2203, 2166, 1924, 1921, 2090, 2059, 2235, 2131, 1910, 1855, 2082, 2071,

2229, 2110, 1892, 1838, 2081, 2128, 2228, 2057, 1845, 2011, 2108, 2153, 1908, 1904, 2103, 2092,

2216, 2166, 1938, 1809, 2033, 2006, 2155, 1893, 1990, 2172, 1885, 2090, 2027, 2105, 2063, 1940,

2075, 1981, 2025, 2004, 2086, 1916, 1999, 2150, 1896, 1969, 2108, 1954, 2100, 2106, 2103, 2101,

2099, 2098, 2094, 2092, 2090, 2089, 2087, 2085, 2082, 2080, 2078, 2076, 2074, 2073, 2073, 2069,

2067, 2067, 2066, 2065, 2062, 2061, 2060, 2052, 2045, 2037, 2025, 2011, 1992, 1971, 1949, 1918,

1886, 1850, 1811, 1770, 1722, 1672, 1618, 1561, 1502, 1438, 1373, 1306, 1235, 1162, 1086, 1009,

931, 853, 773, 693, 614, 535, 456, 380, 311, 248, 212, 186, 174, 158, 140, 128, 118, 115, 111, 107, 112, 118, 125, 134, 148, 164, 183, 204, 226, 254, 283, 312, 343, 376, 411, 447, 488, 530, 573, 617, 662, 708, 756, 805, 854, 908, 961, 1014,

1067, 1121, 1172, 1222, 1269, 1316, 1368, 1422, 1478, 1532, 1584, 1631, 1677, 1731, 1790, 1865,

1939, 2008, 2047, 2064, 2070, 2072, 2104, 2136, 2163, 2195, 2234, 2276, 2321, 2337, 2348, 2357,

2369, 2389, 2407, 2424, 2438, 2450, 2460, 2489, 2544, 2611, 2671, 2702, 2735, 2759, 2767, 2728,

2700, 2702, 2714, 2666, 2541, 2458, 2415, 2210, 1969, 1834, 1858, 1925, 1997, 1977, 1887, 1639,

1343, 1147, 1116, 1156, 1222, 1425, 1628, 1753, 1850, 1930, 1971, 1977, 2076, 2165, 2160, 2359,

2611, 2722, 2693, 2640, 2626, 2639, 2633, 2615, 2657, 2773, 2920, 2999, 3006, 2963, 2870, 2692,

2469, 2216, 2090, 2070, 2010, 1961, 1971, 1972, 1862, 1679, 1450, 1306, 1237, 1266, 1379, 1514,

1625, 1763, 1911, 1854, 1650, 1508, 1434, 1580, 1917, 2253, 2461, 2578, 2644, 2611, 2556, 2554,

2672, 2721, 2673, 2396, 2159, 2095, 2038, 1976, 1939, 1873, 1770, 1683, 1601, 1588, 1627, 1670,

1729, 1777, 1835, 1912, 1966, 2005, 2023, 2032, 2047, 2070, 2120, 2197, 2321, 2471, 2650, 2834,

2984, 3125, 3222, 3263, 3265, 3223, 3152, 3080, 3018, 2963, 2918, 2891, 2900, 2912, 2937, 2977,

3016, 3045, 3064, 3068, 3060, 3031, 2973, 2898, 2828, 2754, 2665, 2573, 2494, 2413, 2319, 2240,

2182, 2128, 2064, 1996, 1930, 1868, 1801, 1731, 1657, 1577, 1504, 1441, 1380, 1324, 1275, 1225,

1189, 1162, 1148, 1144, 1147, 1154, 1165, 1181, 1199, 1220, 1245, 1270, 1293, 1315, 1337, 1357,

1376, 1394, 1409, 1420, 1431, 1441, 1456, 1477, 1499, 1523, 1546, 1568, 1589, 1607, 1627, 1652,

1678, 1709, 1741, 1769, 1798, 1834, 1870, 1911, 1953, 1997, 2042, 2091, 2143, 2198, 2256, 2316,

2379, 2442, 2507, 2572, 2632, 2691, 2736, 2780, 2822, 2863, 2907, 2951, 2996, 3042, 3088, 3135,

3179, 3220, 3260, 3291, 3322, 3346, 3370, 3387, 3405, 3421, 3437, 3458, 3482, 3507, 3533, 3560,

3589, 3618, 3654, 3689, 3729, 3768, 3807, 3847, 3880, 3909, 3934, 3955, 3975, 3995, 4014, 4035,

4055, 4071, 4085, 4095, 4095, 4095, 4090, 4084, 4077, 4070, 4062, 4052, 4041, 4027, 4010, 3993,

3970, 3947, 3922, 3893, 3864, 3830, 3794, 3758, 3718, 3677, 3636, 3594, 3551, 3511, 3473, 3435,

3398, 3361, 3325, 3285, 3244, 3203, 3162, 3120, 3078, 3035, 2993, 2950, 2910, 2874, 2837, 2802,

2771, 2740, 2709, 2680, 2652, 2624, 2598, 2573, 2547, 2522, 2497, 2472, 2447, 2427, 2406, 2385,

2366, 2347, 2329, 2310, 2290, 2270, 2250, 2229, 2208, 2187, 2165, 2144, 2122, 2100, 2075, 2048,

2022, 1994, 1965, 1935, 1905, 1876, 1848, 1819, 1791, 1762, 1733, 1704, 1676, 1648, 1620, 1592,

1564, 1535, 1506, 1475, 1443, 1412, 1380, 1348, 1317, 1285, 1255, 1225, 1195, 1166, 1137, 1108,

1078, 1048, 1017, 988, 959, 931, 903, 875, 847, 819, 792, 764, 737, 715, 694, 672,

655, 637, 620, 601, 581, 562, 541, 521, 501, 482, 463, 446, 429, 413, 399, 384, 370, 355, 340, 327, 315, 305, 295, 285, 275, 266, 256, 247, 241, 235, 236, 237, 242, 247, 250, 252, 253, 256, 265, 279, 301, 329, 363, 402, 445, 486, 527, 565, 602, 638, 673, 707, 741, 777, 813, 848, 885, 922, 961, 1001, 1043, 1086, 1130, 1175,

1220, 1262, 1305, 1344, 1384, 1426, 1470, 1517, 1568, 1620, 1671, 1723, 1771, 1820, 1870, 1916,

1959, 1997, 2033, 2069, 2106, 2142, 2176, 2206, 2231, 2254, 2278, 2304, 2331, 2351, 2367, 2376,

2384, 2392, 2398, 2406, 2413, 2418, 2422, 2432, 2447, 2461, 2466, 2461, 2452, 2444, 2431, 2408,

2361, 2286, 2162, 1965, 1762, 1562, 1404, 1363, 1367, 1404, 1478, 1562, 1636, 1749, 1881, 1999,

2095, 2200, 2336, 2440, 2515, 2598, 2648, 2658, 2639, 2575, 2504, 2393, 2264, 2129, 1968, 1864,

1780, 1704, 1658, 1608, 1582, 1541, 1534, 1546, 1563, 1621, 1722, 1864, 2011, 2158, 2311, 2453,

2575, 2679, 2764, 2838, 2885, 2906, 2909, 2891, 2857, 2812, 2758, 2685, 2602, 2511, 2423, 2307,

2182, 2060, 1938, 1823, 1719, 1633, 1553, 1478, 1425, 1391, 1383, 1395, 1415, 1444, 1494, 1556,

1628, 1707, 1784, 1863, 1958, 2059, 2154, 2240, 2314, 2390, 2469, 2547, 2617, 2684, 2748, 2804,

2856, 2894, 2917, 2934, 2947, 2938, 2899, 2848, 2778, 2695, 2614, 2517, 2408, 2293, 2179, 2066,

1962, 1879, 1814, 1756, 1704, 1655, 1608, 1564, 1527, 1501, 1484, 1473, 1467, 1465, 1465, 1468,

1474, 1482, 1493, 1505, 1519, 1535, 1553, 1573, 1595, 1620, 1646, 1674, 1703, 1733, 1764, 1795,

1826, 1856, 1886, 1917, 1949, 1982, 2018, 2054, 2094, 2134, 2176, 2218, 2262, 2306, 2351, 2396,

2442, 2487, 2531, 2576, 2620, 2666, 2712, 2758, 2805, 2852, 2899, 2945, 2990, 3035, 3080, 3124,

3168, 3212, 3255, 3298, 3341, 3382, 3422, 3459, 3496, 3530, 3563, 3592, 3620, 3647, 3672, 3697,

3721, 3744, 3767, 3789, 3810, 3829, 3847, 3861, 3875, 3883, 3891, 3896, 3900, 3902, 3904, 3904,

3903, 3901, 3897, 3892, 3884, 3876, 3865, 3854, 3842, 3828, 3814, 3799, 3784, 3768, 3752, 3735,

3718, 3701, 3684, 3666, 3648, 3629, 3610, 3590, 3569, 3549, 3527, 3504, 3481, 3457, 3432, 3406,

3379, 3352, 3324, 3295, 3267, 3237, 3208, 3179, 3149, 3120, 3091, 3061, 3031, 3001, 2970, 2940,

2909, 2878, 2847, 2816, 2785, 2754, 2722, 2691, 2659, 2628, 2596, 2564, 2531, 2499, 2466, 2433,

2401, 2368, 2335, 2302, 2269, 2236, 2203, 2170, 2137, 2104, 2072, 2039, 2006, 1973, 1940, 1908,

1875, 1843, 1812, 1780, 1749, 1718, 1687, 1656, 1624, 1592, 1560, 1527, 1495, 1462, 1429, 1396,

1364, 1331, 1299, 1268, 1236, 1206, 1176, 1147, 1118, 1089, 1061, 1034, 1007, 980, 954, 928,

902, 877, 852, 828, 805, 782, 759, 737, 714, 691, 668, 646, 623, 600, 579, 557,

537, 518, 500, 484, 469, 455, 443, 431, 422, 413, 405, 398, 393, 388, 385, 381,

380, 378, 378, 379, 382, 386, 392, 398, 406, 417, 429, 446, 464, 486, 509, 535,

562, 589, 617, 645, 673, 702, 732, 763, 796, 830, 866, 903, 941, 979, 1017,
1055,

1094, 1132, 1171, 1210, 1248, 1285, 1322, 1360, 1397, 1434, 1470, 1507, 1542, 1576,
1608, 1638,

1667, 1694, 1721, 1747, 1771, 1795, 1816, 1836, 1854, 1870, 1886, 1900, 1913, 1926,
1938, 1951,

1963, 1974, 1983, 1991, 1997, 2000, 1999, 1992, 1979, 1958, 1934, 1907, 1880, 1847,
1802, 1741,

1665, 1591, 1518, 1449, 1403, 1409, 1430, 1473, 1534, 1600, 1678, 1783, 1898, 2019,
2136, 2255,

2372, 2467, 2550, 2619, 2674, 2701, 2697, 2667, 2619, 2561, 2480, 2375, 2259, 2143,
2011, 1874,

1751, 1650, 1559, 1470, 1406, 1372, 1352, 1355, 1395, 1471, 1555, 1668, 1806, 1954,
2104, 2244,

2353, 2432, 2495, 2536, 2546, 2521, 2456, 2323, 2131, 1932, 1801, 1706, 1660, 1645,
1644, 1682,

1743, 1810, 1890, 1995, 2112, 2231, 2348, 2468, 2581, 2676, 2756, 2815, 2856, 2880,
2883, 2866,

2831, 2778, 2713, 2635, 2542, 2441, 2333, 2222, 2108, 1989, 1871, 1764, 1674, 1600,
1535, 1482,

1440, 1408, 1387, 1375, 1372, 1375, 1380, 1389, 1402, 1419, 1440, 1466, 1498, 1533,
1570, 1607,

1646, 1684, 1723, 1763, 1804, 1848, 1893, 1942, 1993, 2045, 2099, 2154, 2209, 2264,
2318, 2371,

2425, 2478, 2531, 2585, 2639, 2694, 2748, 2802, 2855, 2906, 2956, 3004, 3050, 3093,
3135, 3174,

3212, 3248, 3282, 3315, 3344, 3372, 3398, 3422, 3444, 3464, 3483, 3499, 3514, 3527,
3536, 3544,

3550, 3552, 3553, 3551, 3546, 3538, 3529, 3518, 3505, 3490, 3474, 3456, 3437, 3416,
3393, 3369,

3344, 3317, 3289, 3259, 3227, 3194, 3160, 3126, 3090, 3055, 3018, 2981, 2944, 2906,
2867, 2828,

2789, 2749, 2708, 2668, 2626, 2585, 2543, 2500, 2457, 2414, 2371, 2328, 2285, 2242, 2199, 2157,

2114, 2071, 2029, 1986, 1943, 1900, 1856, 1812, 1768, 1724, 1679, 1635, 1590, 1546, 1503, 1461,

1419, 1379, 1339, 1301, 1264, 1227, 1192, 1157, 1123, 1090, 1057, 1024, 993, 963, 934, 907,

883, 860, 841, 824, 808, 795, 784, 775, 768, 764, 765, 770, 781, 797, 818, 844, 874, 909, 947, 988, 1033, 1083, 1138, 1197, 1259, 1323, 1386, 1450, 1512, 1574, 1637, 1699,

1759, 1817, 1871, 1921, 1966, 2007, 2045, 2079, 2110, 2137, 2161, 2180, 2193, 2200, 2199, 2191,

2169, 2135, 2085, 2019, 1924, 1796, 1653, 1521, 1476, 1484, 1537, 1623, 1732, 1869, 2013, 2159,

2307, 2430, 2522, 2590, 2623, 2628, 2608, 2549, 2460, 2349, 2224, 2080, 1922, 1782, 1666, 1568,

1510, 1487, 1480, 1506, 1572, 1660, 1767, 1898, 2039, 2190, 2340, 2464, 2560, 2628, 2666, 2678,

2668, 2626, 2544, 2429, 2285, 2121, 1959, 1817, 1708, 1619, 1560, 1536, 1538, 1550, 1577, 1624,

1690, 1767, 1852, 1948, 2055, 2168, 2278, 2381, 2482, 2578, 2666, 2740, 2801, 2852, 2885, 2900,

2897, 2883, 2857, 2814, 2756, 2682, 2595, 2494, 2385, 2278, 2174, 2073, 1974, 1882, 1799, 1726,

1665, 1613, 1572, 1541, 1521, 1508, 1501, 1498, 1500, 1506, 1515, 1528, 1546, 1568, 1593, 1621,

1650, 1681, 1714, 1747, 1782, 1819, 1859, 1900, 1944, 1991, 2038, 2087, 2138, 2188, 2237, 2287,

2336, 2385, 2433, 2482, 2530, 2579, 2628, 2678, 2728, 2778, 2828, 2878, 2926, 2974, 3021, 3067,

3111, 3155, 3198, 3238, 3278, 3316, 3352, 3387, 3420, 3450, 3478, 3503, 3525, 3544, 3562, 3577,

3590, 3601, 3609, 3614, 3617, 3618, 3617, 3615, 3609, 3602, 3593, 3582, 3570, 3555, 3539, 3521,

3501, 3481, 3459, 3436, 3411, 3385, 3357, 3329, 3299, 3268, 3236, 3202, 3169, 3134, 3099, 3063,

3027, 2991, 2954, 2917, 2879, 2841, 2803, 2764, 2725, 2685, 2645, 2605, 2565, 2524, 2483, 2442,

2401, 2360, 2319, 2277, 2236, 2195, 2155, 2114, 2074, 2034, 1995, 1955, 1916, 1876, 1836, 1796,

1756, 1716, 1676, 1636, 1597, 1558, 1519, 1481, 1443, 1407, 1370, 1335, 1300, 1267, 1233, 1201,

1170, 1139, 1109, 1080, 1052, 1025, 1000, 975, 953, 932, 912, 894, 876, 861, 847, 834,

822, 811, 801, 793, 787, 784, 785, 788, 795, 806, 820, 836, 854, 874, 896, 920, 948, 978, 1012, 1049, 1087, 1126, 1166, 1204, 1242, 1280, 1317, 1355, 1394, 1432, 1470, 1508,

1544, 1578, 1610, 1640, 1667, 1692, 1716, 1737, 1756, 1774, 1790, 1804, 1817, 1829, 1839, 1846,

1846, 1842, 1832, 1816, 1794, 1760, 1715, 1664, 1611, 1577, 1586, 1635, 1724, 1851, 2007, 2173,

2327, 2446, 2517, 2524, 2481, 2388, 2257, 2097, 1925, 1757, 1624, 1529, 1486, 1491, 1536, 1635,

1766, 1925, 2087, 2242, 2376, 2469, 2525, 2533, 2504, 2424, 2296, 2137, 1955, 1788, 1672, 1593,

1559, 1557, 1571, 1602, 1655, 1726, 1808, 1899, 2001, 2110, 2221, 2330, 2436, 2538, 2629, 2708,

2777, 2833, 2873, 2895, 2901, 2894, 2873, 2838, 2789, 2731, 2664, 2589, 2505, 2413, 2314, 2213,

2111, 2010, 1911, 1815, 1723, 1638, 1560, 1492, 1433, 1385, 1348, 1321, 1304, 1293, 1289, 1293,

1304, 1323, 1349, 1381, 1419, 1461, 1505, 1552, 1602, 1654, 1709, 1766, 1826, 1888, 1951, 2014,

2077, 2140, 2202, 2264, 2326, 2388, 2450, 2511, 2572, 2632, 2691, 2749, 2806, 2860, 2913, 2964,

3012, 3057, 3098, 3136, 3171, 3202, 3229, 3253, 3273, 3290, 3304, 3313, 3318, 3318, 3314, 3305,

3292, 3275, 3254, 3231, 3206, 3178, 3148, 3114, 3078, 3040, 2999, 2957, 2914, 2869, 2824, 2777,

2730, 2682, 2633, 2583, 2531, 2479, 2427, 2374, 2322, 2269, 2216, 2164, 2112, 2060, 2007, 1955,

1903, 1851, 1800, 1748, 1698, 1648, 1599, 1552, 1504, 1458, 1413, 1369, 1326, 1285, 1247, 1212,

1180, 1151, 1126, 1105, 1087, 1072, 1062, 1057, 1059, 1069, 1088, 1115, 1151, 1194, 1244, 1303,

1368, 1437, 1505, 1571, 1635, 1696, 1754, 1808, 1855, 1896, 1929, 1953, 1966, 1970, 1967, 1956,

1931, 1885, 1815, 1717, 1640, 1631, 1694, 1813, 1957, 2121, 2284, 2429, 2529, 2577, 2574, 2519,

2422, 2294, 2144, 1985, 1832, 1703, 1613, 1563, 1557, 1590, 1660, 1767, 1899, 2050, 2205, 2353,

2481, 2570, 2623, 2640, 2616, 2560, 2470, 2349, 2206, 2058, 1907, 1765, 1647, 1564, 1511, 1488,

1500, 1541, 1603, 1691, 1797, 1913, 2038, 2165, 2287, 2403, 2507, 2592, 2657, 2704, 2729, 2729,

2701, 2648, 2571, 2475, 2363, 2240, 2110, 1983, 1863, 1760, 1676, 1611, 1563, 1535, 1523, 1524,

1533, 1551, 1576, 1609, 1649, 1694, 1742, 1793, 1846, 1902, 1961, 2022, 2085, 2149, 2215, 2281,

2346, 2411, 2476, 2541, 2605, 2669, 2731, 2791, 2849, 2905, 2958, 3008, 3055, 3098, 3137, 3171,

3199, 3222, 3240, 3252, 3259, 3260, 3257, 3249, 3235, 3217, 3194, 3168, 3138, 3104, 3067, 3027,

2985, 2940, 2893, 2844, 2794, 2742, 2689, 2634, 2579, 2522, 2465, 2407, 2349, 2291, 2233, 2175,

2118, 2062, 2006, 1951, 1896, 1843, 1790, 1738, 1687, 1638, 1591, 1546, 1502, 1460, 1420, 1382,

1348, 1316, 1290, 1270, 1256, 1248, 1247, 1253, 1267, 1292, 1327, 1369, 1415, 1465, 1517, 1568,

1613, 1655, 1694, 1728, 1755, 1775, 1787, 1790, 1786, 1775, 1760, 1740, 1712, 1671,
1623, 1572,

1544, 1541, 1569, 1619, 1695, 1800, 1920, 2054, 2196, 2327, 2445, 2530, 2580, 2588,
2552, 2477,

2369, 2239, 2089, 1932, 1782, 1652, 1554, 1490, 1467, 1488, 1557, 1676, 1830, 1999,
2160, 2299,

2405, 2463, 2467, 2409, 2293, 2124, 1926, 1754, 1639, 1587, 1576, 1605, 1665, 1752,
1860, 1980,

2109, 2244, 2373, 2492, 2595, 2670, 2718, 2741, 2739, 2713, 2663, 2592, 2502, 2399,
2283, 2159,

2031, 1906, 1788, 1682, 1596, 1529, 1479, 1445, 1429, 1434, 1456, 1492, 1538, 1594,
1659, 1728,

1800, 1874, 1950, 2028, 2108, 2188, 2267, 2344, 2419, 2493, 2564, 2632, 2695, 2754,
2808, 2856,

2900, 2938, 2969, 2990, 3003, 3008, 3004, 2991, 2973, 2945, 2910, 2869, 2822, 2769,
2711, 2650,

2586, 2521, 2453, 2384, 2315, 2244, 2174, 2103, 2034, 1965, 1895, 1826, 1758, 1694,
1635, 1583,

1536, 1495, 1465, 1449, 1450, 1471, 1526, 1599, 1689, 1771, 1838, 1880, 1886, 1851,
1764, 1613

};

// Guitar: 8192 samples, 32,768 bytes (32.00KB)

uint32_t Guitar_LUT[NS_WAVE] = {

2048, 2165, 2043, 2021, 1835, 1753, 1825, 1803, 1763, 1786, 1893, 1994, 2125, 2174,
2191, 2150,

2193, 2249, 2240, 2324, 2372, 2382, 2397, 2428, 2408, 2526, 2571, 2485, 2509, 2553,
2550, 2488,

2423, 2456, 2477, 2420, 2425, 2367, 2280, 2293, 2227, 2215, 2188, 2080, 2047, 1939,
1895, 1902,

1794, 1792, 1748, 1655, 1648, 1587, 1489, 1490, 1473, 1406, 1388, 1413, 1430, 1411,
1376, 1369,

1363, 1324, 1301, 1335, 1378, 1363, 1335, 1373, 1408, 1372, 1336, 1361, 1378, 1321, 1256, 1234,

1232, 1217, 1208, 1223, 1254, 1284, 1309, 1316, 1309, 1315, 1324, 1308, 1301, 1338, 1364, 1355,

1385, 1441, 1463, 1467, 1519, 1549, 1553, 1603, 1620, 1632, 1650, 1681, 1717, 1734, 1784, 1739,

1800, 1756, 1849, 1932, 1957, 1949, 2071, 2116, 2118, 2170, 2222, 2258, 2262, 2270, 2252, 2248,

2296, 2374, 2399, 2435, 2518, 2535, 2536, 2603, 2655, 2630, 2634, 2712, 2770, 2760, 2758, 2798,

2832, 2825, 2807, 2813, 2830, 2831, 2823, 2825, 2828, 2818, 2816, 2840, 2861, 2853, 2840, 2861,

2901, 2907, 2874, 2854, 2875, 2901, 2887, 2842, 2813, 2825, 2848, 2844, 2823, 2832, 2877, 2909,

2887, 2841, 2827, 2847, 2860, 2840, 2807, 2794, 2799, 2788, 2750, 2717, 2713, 2724, 2724, 2696,

2651, 2630, 2622, 2588, 2533, 2506, 2501, 2462, 2386, 2354, 2355, 2318, 2257, 2229, 2199, 2131,

2069, 2031, 1980, 1920, 1864, 1837, 1840, 1798, 1730, 1713, 1683, 1609, 1597, 1622, 1550, 1494,

1508, 1474, 1436, 1442, 1424, 1392, 1375, 1365, 1366, 1337, 1305, 1315, 1294, 1273, 1300, 1303,

1284, 1313, 1336, 1324, 1352, 1382, 1385, 1410, 1441, 1453, 1461, 1467, 1472, 1474, 1465, 1494,

1537, 1539, 1601, 1653, 1657, 1724, 1716, 1750, 1787, 1770, 1782, 1783, 1765, 1825, 1864, 1963,

2003, 2035, 2055, 2073, 2118, 2102, 2164, 2169, 2259, 2268, 2338, 2432, 2459, 2513, 2565, 2601,

2637, 2671, 2705, 2714, 2705, 2728, 2748, 2732, 2729, 2758, 2767, 2751, 2753, 2789, 2810, 2804,

2807, 2842, 2869, 2859, 2850, 2864, 2871, 2843, 2803, 2797, 2819, 2835, 2824, 2807, 2807, 2812,

2804, 2786, 2778, 2792, 2787, 2753, 2719, 2717, 2723, 2697, 2668, 2669, 2663, 2621, 2582, 2568,

2537, 2489, 2447, 2404, 2358, 2313, 2268, 2236, 2214, 2168, 2119, 2106, 2083, 2019, 1972, 1951,

1901, 1848, 1836, 1817, 1748, 1679, 1648, 1622, 1578, 1536, 1505, 1476, 1436, 1399, 1391, 1390,

1361, 1333, 1339, 1342, 1322, 1322, 1336, 1322, 1293, 1285, 1285, 1271, 1262, 1271, 1279, 1278,

1283, 1289, 1290, 1291, 1298, 1305, 1312, 1329, 1349, 1353, 1358, 1384, 1408, 1399, 1389, 1404,

1416, 1405, 1399, 1426, 1454, 1458, 1474, 1515, 1551, 1570, 1577, 1593, 1624, 1623, 1625, 1645,

1615, 1617, 1633, 1620, 1630, 1651, 1686, 1699, 1739, 1787, 1824, 1820, 1838, 1880, 1916, 1935,

1971, 1969, 2015, 2053, 2076, 2101, 2154, 2198, 2252, 2326, 2366, 2406, 2486, 2553, 2585, 2634,

2695, 2720, 2722, 2740, 2769, 2784, 2779, 2777, 2793, 2809, 2807, 2799, 2814, 2846, 2869, 2873,

2870, 2876, 2895, 2921, 2939, 2940, 2925, 2912, 2911, 2918, 2918, 2910, 2905, 2914, 2932, 2941,

2929, 2906, 2897, 2913, 2934, 2930, 2906, 2893, 2899, 2901, 2882, 2857, 2845, 2841, 2835, 2829,

2821, 2802, 2776, 2758, 2746, 2724, 2691, 2668, 2659, 2636, 2584, 2535, 2516, 2493, 2436, 2393,

2374, 2345, 2300, 2265, 2240, 2192, 2132, 2107, 2090, 2046, 1988, 1937, 1868, 1812, 1771, 1705,

1692, 1650, 1628, 1592, 1583, 1563, 1540, 1512, 1511, 1533, 1513, 1550, 1562, 1596, 1631, 1628,

1645, 1651, 1718, 1760, 1788, 1845, 1833, 1818, 1808, 1822, 1887, 1875, 1917, 1948, 1988, 2033,

2060, 2138, 2217, 2292, 2378, 2467, 2513, 2569, 2587, 2607, 2634, 2608, 2633, 2648, 2639, 2658,

2702, 2712, 2739, 2751, 2719, 2732, 2740, 2732, 2719, 2716, 2696, 2687, 2692, 2661, 2636, 2625,

2604, 2600, 2589, 2545, 2534, 2518, 2463, 2431, 2375, 2315, 2265, 2206, 2143, 2114, 2057, 1992,

1965, 1920, 1891, 1811, 1759, 1695, 1629, 1574, 1543, 1522, 1508, 1509, 1518, 1525, 1523, 1505,

1517, 1517, 1528, 1533, 1550, 1563, 1592, 1618, 1630, 1639, 1644, 1674, 1728, 1767, 1809, 1846,

1845, 1814, 1819, 1843, 1880, 1842, 1847, 1895, 1935, 1965, 2034, 2125, 2216, 2327, 2391, 2469,

2486, 2530, 2528, 2561, 2571, 2635, 2649, 2657, 2665, 2664, 2672, 2637, 2643, 2618, 2617, 2588,

2580, 2566, 2536, 2518, 2489, 2474, 2411, 2369, 2294, 2227, 2158, 2089, 2014, 1984, 1927, 1835,

1720, 1674, 1688, 1729, 1761, 1826, 1938, 2028, 2084, 2016, 2064, 2157, 2310, 2396, 2426, 2420,

2472, 2487, 2455, 2424, 2378, 2339, 2291, 2264, 2207, 2157, 2077, 2000, 1929, 1849, 1777, 1740,

1699, 1658, 1593, 1540, 1485, 1469, 1445, 1461, 1479, 1522, 1552, 1562, 1577, 1577, 1603, 1629,

1670, 1726, 1777, 1814, 1830, 1887, 1975, 2085, 2015, 2004, 2064, 2211, 2333, 2396, 2398, 2454,

2476, 2452, 2426, 2379, 2332, 2293, 2246, 2190, 2090, 2008, 1903, 1829, 1778, 1712, 1625, 1589,

1596, 1676, 1721, 1752, 1811, 1900, 2000, 2037, 2142, 2312, 2250, 2191, 2100, 2089, 2164, 2282,

2365, 2378, 2415, 2387, 2359, 2305, 2240, 2186, 2143, 2070, 1998, 1916, 1833, 1772, 1723, 1688,

1631, 1581, 1525, 1501, 1487, 1477, 1497, 1533, 1563, 1583, 1593, 1618, 1644, 1676, 1744, 1814,

1861, 1879, 1942, 2056, 2134, 1965, 2026, 2200, 2318, 2361, 2413, 2420, 2395, 2354, 2292, 2245,

2181, 2094, 1995, 1892, 1828, 1765, 1677, 1620, 1595, 1645, 1702, 1732, 1779, 1865, 1980, 2031,

2112, 2263, 2286, 2241, 2126, 2074, 2097, 2215, 2318, 2344, 2393, 2435, 2419, 2394, 2353, 2298,

2245, 2195, 2116, 2057, 1979, 1897, 1829, 1780, 1741, 1683, 1617, 1560, 1524, 1494, 1478, 1481,

1505, 1531, 1548, 1559, 1577, 1596, 1623, 1665, 1722, 1792, 1829, 1872, 1915, 1998, 2098, 1971,

1987, 2058, 2193, 2317, 2372, 2404, 2439, 2489, 2508, 2513, 2498, 2484, 2459, 2419, 2373, 2343,

2279, 2213, 2145, 2059, 1980, 1919, 1864, 1789, 1699, 1632, 1582, 1583, 1627, 1642, 1658, 1681,

1729, 1828, 1887, 1937, 2040, 2090, 2018, 2048, 2098, 2206, 2302, 2393, 2421, 2447, 2473, 2507,

2543, 2560, 2570, 2567, 2563, 2551, 2534, 2515, 2479, 2450, 2424, 2380, 2327, 2273, 2209, 2135,

2067, 2007, 1953, 1910, 1847, 1777, 1707, 1640, 1584, 1547, 1552, 1565, 1566, 1571, 1570, 1584,

1598, 1650, 1705, 1741, 1771, 1802, 1844, 1897, 1944, 1926, 1923, 1872, 1895, 1929, 1997, 2094,

2197, 2302, 2364, 2415, 2444, 2476, 2513, 2550, 2578, 2597, 2607, 2611, 2613, 2606, 2598, 2590,

2563, 2539, 2524, 2501, 2465, 2422, 2371, 2313, 2242, 2176, 2112, 2067, 2015, 1934, 1850, 1770,

1696, 1686, 1684, 1678, 1688, 1783, 1840, 1885, 1970, 1971, 1927, 1895, 1927, 1977, 2070, 2170,

2277, 2349, 2399, 2430, 2463, 2506, 2541, 2572, 2593, 2604, 2613, 2615, 2609, 2612, 2607, 2585,

2559, 2542, 2526, 2499, 2461, 2418, 2367, 2306, 2237, 2169, 2116, 2074, 2015, 1939, 1863, 1787,

1711, 1690, 1675, 1655, 1646, 1670, 1739, 1780, 1814, 1853, 1914, 1929, 1870, 1853, 1856, 1779,

1802, 1834, 1903, 1987, 2108, 2203, 2281, 2331, 2374, 2420, 2465, 2511, 2541, 2564, 2580, 2584,

2590, 2585, 2566, 2549, 2533, 2513, 2479, 2440, 2390, 2323, 2249, 2177, 2127, 2073, 1993, 1905,

1805, 1756, 1745, 1751, 1881, 1962, 2038, 1936, 1853, 1895, 1987, 2125, 2240, 2305, 2355, 2402,

2451, 2488, 2516, 2525, 2534, 2538, 2512, 2487, 2468, 2437, 2398, 2341, 2263, 2174, 2105, 2044,

1974, 1889, 1804, 1728, 1700, 1698, 1698, 1747, 1841, 1906, 2010, 1985, 1912, 1850, 1900, 2028,

2176, 2257, 2313, 2375, 2419, 2460, 2480, 2496, 2479, 2452, 2428, 2384, 2330, 2254, 2152, 2070,

2005, 1926, 1836, 1739, 1687, 1705, 1719, 1739, 1847, 1935, 2059, 2048, 1951, 2045, 2202, 2299,

2341, 2392, 2431, 2466, 2478, 2488, 2475, 2443, 2414, 2375, 2323, 2261, 2169, 2078, 2008, 1947,

1872, 1799, 1717, 1651, 1629, 1635, 1647, 1657, 1660, 1725, 1792, 1852, 1919, 2039, 2033, 1918,

1987, 2154, 2271, 2322, 2376, 2420, 2449, 2465, 2468, 2444, 2413, 2371, 2321, 2260, 2163, 2069,

1997, 1933, 1854, 1770, 1686, 1638, 1648, 1662, 1682, 1685, 1761, 1839, 1898, 2003, 2120, 1981,

2013, 2136, 2277, 2336, 2383, 2417, 2456, 2480, 2483, 2494, 2467, 2444, 2413, 2370, 2328, 2267,

2173, 2085, 2018, 1950, 1880, 1802, 1720, 1653, 1631, 1636, 1655, 1656, 1661, 1726, 1796, 1848,

1921, 2043, 2032, 1917, 1993, 2165, 2273, 2328, 2385, 2436, 2445, 2459, 2443, 2424, 2391, 2346,

2292, 2211, 2113, 2037, 1963, 1883, 1797, 1714, 1648, 1646, 1658, 1686, 1684, 1748, 1824, 1887,

1981, 2128, 1980, 1995, 2128, 2266, 2323, 2380, 2421, 2455, 2470, 2478, 2478, 2453, 2428, 2394,

2355, 2301, 2217, 2122, 2050, 1979, 1901, 1818, 1732, 1671, 1652, 1667, 1682, 1683, 1744, 1822,

1881, 1973, 2091, 1935, 1965, 2142, 2250, 2323, 2381, 2416, 2440, 2437, 2413, 2379, 2336, 2286,

2192, 2097, 2011, 1938, 1858, 1772, 1682, 1654, 1661, 1691, 1692, 1740, 1816, 1890, 1977, 2120,

1971, 2004, 2158, 2271, 2335, 2378, 2431, 2447, 2461, 2469, 2440, 2422, 2375, 2340, 2283, 2180,

2097, 2017, 1939, 1860, 1761, 1693, 1677, 1705, 1703, 1751, 1829, 1894, 2008, 2059, 1921, 2045,

2203, 2280, 2356, 2393, 2423, 2419, 2395, 2357, 2313, 2240, 2135, 2045, 1966, 1880, 1780, 1691,

1681, 1717, 1712, 1771, 1855, 1926, 2065, 1972, 1954, 2109, 2239, 2308, 2367, 2414, 2434, 2447,

2429, 2408, 2363, 2337, 2253, 2150, 2063, 1976, 1892, 1778, 1720, 1742, 1747, 1829, 1921, 2053,

2011, 1909, 2097, 2225, 2304, 2362, 2393, 2384, 2348, 2310, 2231, 2117, 2024, 1930, 1835, 1728,

1700, 1740, 1752, 1832, 1925, 2048, 2035, 1943, 2120, 2242, 2320, 2371, 2406, 2408, 2393, 2351,

2321, 2233, 2126, 2026, 1943, 1833, 1740, 1727, 1760, 1791, 1897, 1996, 2116, 1914, 2066, 2210,

2296, 2353, 2382, 2363, 2318, 2272, 2166, 2064, 1962, 1881, 1770, 1697, 1695, 1735, 1753, 1843,

1930, 2044, 2081, 1948, 2125, 2245, 2317, 2370, 2405, 2400, 2374, 2334, 2284, 2177, 2079, 1983,

1894, 1784, 1714, 1730, 1749, 1819, 1917, 2026, 2093, 1932, 2131, 2238, 2318, 2366, 2367, 2327,

2285, 2194, 2080, 1982, 1897, 1782, 1709, 1704, 1738, 1768, 1863, 1943, 2087, 1968, 2000, 2183,

2270, 2343, 2383, 2384, 2366, 2322, 2270, 2159, 2053, 1967, 1868, 1760, 1710, 1741, 1752, 1840,

1937, 2062, 2063, 1980, 2183, 2281, 2338, 2359, 2330, 2291, 2215, 2097, 1994, 1913, 1809, 1724,

1680, 1718, 1735, 1781, 1879, 1964, 2118, 2002, 2027, 2199, 2292, 2349, 2372, 2352, 2295, 2247,

2137, 2028, 1940, 1841, 1743, 1691, 1719, 1739, 1787, 1887, 1975, 2139, 2009, 2042, 2211, 2303,

2340, 2336, 2286, 2226, 2112, 2002, 1914, 1826, 1733, 1674, 1683, 1714, 1728, 1797, 1887, 1967,

2121, 2056, 2024, 2204, 2295, 2348, 2362, 2338, 2291, 2223, 2116, 2003, 1919, 1819, 1731, 1686,

1702, 1725, 1760, 1842, 1936, 2057, 2146, 1969, 2166, 2280, 2333, 2348, 2307, 2263, 2179, 2064,

1960, 1879, 1778, 1703, 1672, 1704, 1723, 1764, 1853, 1929, 2050, 2149, 1968, 2153, 2274, 2330,

2357, 2336, 2289, 2213, 2106, 1996, 1909, 1807, 1722, 1691, 1721, 1735, 1805, 1899, 1991, 2152,

1992, 2077, 2231, 2308, 2338, 2315, 2271, 2181, 2065, 1964, 1872, 1771, 1696, 1695, 1727, 1746,

1825, 1918, 2012, 2160, 1951, 2135, 2259, 2319, 2344, 2304, 2256, 2151, 2026, 1932, 1821, 1730,

1723, 1749, 1798, 1897, 1995, 2167, 2024, 2058, 2230, 2301, 2308, 2265, 2176, 2049, 1944, 1840,

1741, 1698, 1732, 1757, 1830, 1926, 2034, 2189, 1977, 2111, 2248, 2308, 2292, 2241, 2136, 2013,

1912, 1802, 1719, 1715, 1747, 1793, 1896, 1991, 2149, 2133, 2000, 2199, 2290, 2290, 2245, 2145,

2019, 1917, 1812, 1718, 1701, 1738, 1772, 1855, 1947, 2067, 2199, 1981, 2121, 2261, 2306, 2269,

2197, 2073, 1964, 1854, 1751, 1699, 1729, 1761, 1833, 1931, 2031, 2202, 2109, 2022, 2211, 2287,

2267, 2201, 2080, 1966, 1865, 1764, 1695, 1717, 1740, 1788, 1887, 1976, 2111, 2186, 1967, 2154,

2269, 2298, 2248, 2154, 2027, 1929, 1822, 1727, 1698, 1728, 1758, 1837, 1934, 2041, 2203, 2109,

2043, 2217, 2293, 2267, 2199, 2076, 1966, 1863, 1759, 1695, 1716, 1742, 1793, 1889, 1973, 2119,

2185, 1963, 2166, 2269, 2274, 2225, 2109, 1991, 1883, 1780, 1705, 1718, 1744, 1789, 1887, 1976,

2116, 2211, 2023, 2119, 2254, 2281, 2235, 2130, 2006, 1901, 1797, 1713, 1706, 1732, 1769, 1852,

1938, 2060, 2208, 2063, 2081, 2240, 2291, 2246, 2152, 2025, 1917, 1818, 1722, 1697, 1727, 1753,

1833, 1925, 2026, 2185, 2186, 1971, 2178, 2274, 2253, 2189, 2059, 1949, 1845, 1747, 1697, 1714,

1738, 1795, 1888, 1975, 2111, 2220, 2054, 2105, 2246, 2262, 2206, 2081, 1966, 1863, 1760, 1700,

1716, 1737, 1790, 1885, 1975, 2105, 2229, 2095, 2064, 2236, 2262, 2216, 2094, 1972, 1869, 1770,

1702, 1713, 1737, 1781, 1871, 1961, 2083, 2227, 2140, 2023, 2222, 2255, 2196, 2070, 1952, 1848,

1749, 1700, 1718, 1746, 1804, 1896, 1986, 2127, 2239, 2123, 2036, 2227, 2250, 2193, 2064, 1945,

1840, 1744, 1700, 1720, 1744, 1811, 1899, 1993, 2134, 2238, 2118, 2046, 2235, 2229, 2150, 2022,

1914, 1807, 1719, 1707, 1731, 1770, 1856, 1937, 2047, 2203, 2220, 2042, 2138, 2257, 2210, 2095,

1964, 1859, 1759, 1708, 1727, 1749, 1810, 1901, 1999, 2142, 2239, 2115, 2060, 2237, 2217, 2121,

1991, 1879, 1774, 1710, 1720, 1741, 1795, 1884, 1970, 2103, 2241, 2197, 1992, 2187, 2239, 2172,

2047, 1931, 1825, 1729, 1704, 1723, 1754, 1827, 1912, 2009, 2153, 2244, 2130, 2049, 2231, 2210,

2104, 1977, 1871, 1767, 1707, 1712, 1738, 1792, 1881, 1968, 2093, 2233, 2207, 2005, 2175, 2242,

2174, 2049, 1932, 1828, 1729, 1704, 1724, 1752, 1830, 1917, 2011, 2156, 2243, 2134, 2062, 2228,

2190, 2069, 1953, 1849, 1743, 1700, 1716, 1741, 1800, 1886, 1969, 2103, 2231, 2215, 2031, 2163,

2242, 2190, 2056, 1945, 1837, 1737, 1706, 1716, 1744, 1814, 1882, 1961, 2091, 2243, 2226, 2051,

2137, 2225, 2141, 2008, 1892, 1805, 1719, 1699, 1706, 1728, 1777, 1843, 1932, 2030, 2160, 2264,

2274, 2153, 2076, 2173, 2252, 2294, 2182, 2032, 1929, 1852, 1774, 1700, 1683, 1695, 1687, 1676,

1711, 1766, 1826, 1966, 2091, 2184, 1949, 2076, 2197, 2301, 2334, 2401, 2425, 2398, 2320, 2219,

2098, 2021, 1937, 1851, 1774, 1717, 1692, 1658, 1662, 1656, 1680, 1730, 1744, 1811, 1953, 2100,

2095, 2108, 2142, 2185, 2235, 2281, 2316, 2352, 2407, 2446, 2482, 2521, 2546, 2547, 2553, 2594,

2639, 2643, 2617, 2593, 2560, 2507, 2449, 2418, 2397, 2375, 2340, 2287, 2222, 2153, 2080, 2023,

1987, 1958, 1927, 1876, 1800, 1716, 1647, 1593, 1551, 1530, 1530, 1543, 1539, 1525, 1520, 1529,

1526, 1512, 1498, 1491, 1499, 1502, 1504, 1507, 1506, 1503, 1504, 1502, 1510, 1547, 1590, 1658,

1705, 1753, 1822, 1905, 1973, 2004, 2066, 2121, 2154, 2227, 2306, 2165, 2167, 2267, 2160, 2119,

2021, 1925, 1841, 1721, 1646, 1630, 1647, 1673, 1661, 1716, 1823, 1912, 2000, 2073, 2117, 2185,

2272, 2344, 2394, 2453, 2506, 2547, 2570, 2531, 2455, 2392, 2344, 2299, 2239, 2117, 2077, 2052,

2007, 1948, 1986, 1965, 1867, 1771, 1684, 1697, 1785, 1861, 1949, 2043, 2158, 2252, 2302, 2361,

2429, 2487, 2475, 2450, 2409, 2305, 2202, 2069, 1999, 1888, 1877, 2018, 2099, 2035, 2122, 2236,

2323, 2397, 2413, 2361, 2343, 2274, 2182, 2144, 2085, 1998, 1918, 1814, 1743, 1714, 1697, 1680,

1658, 1628, 1604, 1586, 1570, 1572, 1577, 1571, 1570, 1590, 1627, 1679, 1744, 1803, 1869, 1977,

2083, 2149, 2205, 2254, 2123, 2024, 2210, 2214, 2124, 2011, 1917, 1805, 1705, 1630, 1611, 1607,

1661, 1739, 1811, 1874, 1938, 2015, 2104, 2172, 2225, 2291, 2359, 2420, 2485, 2526, 2509, 2465,

2446, 2450, 2420, 2356, 2288, 2255, 2220, 2188, 2005, 1961, 1925, 1869, 1834, 1917, 2116, 2011,

1773, 1822, 1930, 2073, 2160, 2227, 2321, 2394, 2445, 2477, 2477, 2475, 2415, 2296, 2181, 2111,

2044, 1920, 1816, 1751, 1764, 1780, 1891, 2029, 2102, 2171, 2198, 2254, 2301, 2388, 2384, 2322,

2304, 2221, 2150, 2088, 2053, 1994, 1913, 1829, 1762, 1719, 1690, 1664, 1633, 1601, 1569, 1541,

1523, 1515, 1511, 1506, 1503, 1500, 1490, 1480, 1480, 1495, 1520, 1550, 1581, 1611, 1639, 1669,

1708, 1767, 1844, 1925, 2000, 2068, 2133, 2198, 2263, 2320, 2377, 2432, 2435, 2448, 2425, 2313,

2215, 2128, 2066, 2094, 2151, 2069, 1952, 1819, 1732, 1635, 1641, 1729, 1793, 1869, 1941, 2036,

2135, 2200, 2259, 2328, 2412, 2455, 2450, 2450, 2445, 2413, 2374, 2302, 2233, 2177, 2129, 1953,

1887, 1899, 2019, 1921, 2150, 2064, 1872, 1742, 1851, 1975, 2081, 2151, 2224, 2311, 2386, 2428,

2474, 2523, 2515, 2449, 2380, 2292, 2214, 2150, 2059, 1943, 1843, 1837, 1802, 1916, 2036, 2150,

2239, 2277, 2263, 2199, 2150, 2089, 1980, 1902, 1865, 1825, 1776, 1720, 1664, 1619, 1588, 1569,

1551, 1532, 1513, 1498, 1493, 1498, 1506, 1509, 1516, 1536, 1569, 1607, 1649, 1702, 1763, 1827,

1891, 1956, 2018, 2076, 2140, 2208, 2279, 2339, 2385, 2428, 2483, 2514, 2488, 2432, 2363, 2298,

2235, 2174, 2096, 2093, 2203, 2123, 2008, 1901, 1770, 1685, 1665, 1704, 1757, 1837, 1927, 2017,

2143, 2232, 2291, 2331, 2356, 2381, 2338, 2266, 2215, 2087, 1959, 1922, 2074, 1959, 1806, 1862,

2036, 2172, 1879, 1984, 2079, 2176, 2261, 2314, 2383, 2459, 2514, 2528, 2507, 2471, 2422, 2355,

2288, 2221, 2153, 2083, 1990, 1896, 1822, 1763, 1712, 1684, 1710, 1744, 1799, 1832, 1903, 1994,

2090, 2209, 2250, 2278, 2281, 2240, 2244, 2155, 2080, 2006, 1955, 1900, 1848, 1805, 1766, 1723,

1673, 1619, 1571, 1536, 1515, 1504, 1498, 1488, 1478, 1469, 1458, 1454, 1459, 1476, 1505, 1544,

1589, 1641, 1695, 1744, 1789, 1834, 1886, 1944, 2003, 2061, 2116, 2170, 2226, 2287, 2352, 2415,

2476, 2524, 2550, 2562, 2551, 2505, 2444, 2392, 2355, 2324, 2273, 2191, 2122, 2100, 2128, 2065,

1965, 1809, 1749, 1716, 1752, 1812, 1920, 2063, 2176, 2235, 2257, 2338, 2347, 2300, 2206, 2111,

2004, 1847, 1859, 1984, 2149, 1919, 1857, 1993, 2076, 1871, 1984, 2069, 2147, 2240, 2329, 2409,

2473, 2515, 2519, 2491, 2441, 2397, 2359, 2306, 2239, 2164, 2089, 2020, 1925, 1815, 1751, 1720,

1700, 1724, 1745, 1758, 1797, 1867, 1915, 2034, 2179, 2221, 2287, 2279, 2216, 2237, 2230, 2128,

2061, 2007, 1957, 1903, 1843, 1784, 1728, 1673, 1623, 1587, 1562, 1541, 1518, 1490, 1467, 1458,

1462, 1475, 1498, 1531, 1573, 1621, 1674, 1729, 1783, 1831, 1874, 1915, 1962, 2015, 2079, 2149,

2226, 2304, 2376, 2441, 2495, 2526, 2523, 2489, 2441, 2395, 2359, 2314, 2262, 2202, 2122, 2060,

2090, 1940, 1825, 1793, 1785, 1922, 2096, 2140, 2238, 2299, 2193, 2094, 1911, 1839, 1986, 2156,

2231, 1995, 1845, 1847, 1922, 2104, 1951, 2034, 2180, 2292, 2328, 2407, 2456, 2383, 2453, 2419,

2426, 2369, 2364, 2263, 2125, 2001, 1964, 1868, 1760, 1710, 1742, 1752, 1724, 1701, 1681, 1666,

1692, 1768, 1776, 1820, 1849, 1995, 2013, 2264, 2250, 2291, 2238, 2227, 2323, 2191, 2201, 2122,

2173, 2193, 2044, 1935, 1774, 1724, 1602, 1546, 1505, 1622, 1707, 1727, 1806, 1839, 1813, 1837,

1822, 1943, 2091, 2143, 2292, 2366, 2381, 2464, 2496, 2444, 2451, 2089, 2039, 1979, 1854, 2033,

2067, 1779, 1805, 1852, 1942, 2144, 2076, 2217, 1869, 2354, 1827, 2055, 2359, 1922, 1657, 1583,

1954, 1876, 2139, 2540, 2180, 2248, 2452, 2640, 2894, 2754, 2772, 2645, 2369, 2215, 2149, 2364,

2474, 2446, 2760, 2863, 2734, 2635, 2495, 2587, 2557, 2628, 2436, 2437, 2034, 1846, 1967, 1932,

1756, 1658, 1712, 1624, 1840, 1989, 1702, 1526, 1449, 1230, 1370, 1486, 1600, 1694, 1770, 1812,

1399, 1406, 1159, 1138, 1509, 1639, 1676, 1635, 1977, 1776, 1601, 1439, 1531, 1377, 1733, 1931,

1823, 1820, 1715, 1535, 1443, 1932, 2182, 2619, 2824, 2573, 2196, 2219, 2299, 2303, 2274, 1999,

2158, 2490, 2002, 1755, 1874, 2238, 1739, 2198, 2381, 2699, 2626, 2429, 2047, 2051, 2270, 2246,

2091, 2146, 1927, 1976, 1896, 2307, 2505, 2755, 2962, 2902, 2894, 2736, 2601, 2722, 2644, 2595,

2891, 2800, 2462, 2136, 2108, 2157, 2025, 1885, 2121, 2349, 2196, 2051, 2156, 2424, 2228, 1949,

1960, 1712, 1487, 1409, 1199, 1212, 1327, 1222, 1298, 1493, 1359, 1090, 1348, 1015, 1411, 1532,

1787, 1718, 1718, 1940, 2033, 1870, 1903, 1669, 1591, 1458, 1294, 1693, 1979, 1717, 1837, 1957,

1817, 1781, 1713, 1643, 1582, 1761, 1984, 2331, 2227, 2444, 2872, 2697, 2827, 2747, 2578, 2423,

2590, 2644, 2715, 3097, 2684, 2612, 2522, 2442, 2374, 2177, 2339, 2276, 2339, 2577, 2664, 3010,

2979, 3105, 2682, 2346, 2340, 2050, 1552, 1705, 1653, 1529, 1461, 1620, 1818, 1468, 1476, 1414,

1219, 1313, 1163, 1167, 1551, 1619, 1576, 1912, 1785, 2076, 2220, 1967, 1945, 1866, 1774, 1386,

1322, 1399, 1642, 1614, 1555, 1425, 1473, 1435, 1150, 1029, 1071, 1272, 1418, 1414, 1471, 1841,

1835, 1794, 2079, 1925, 1773, 2022, 1809, 1861, 2277, 2165, 2306, 2252, 2198, 2228, 2158, 1973,

2276, 2204, 2387, 2608, 2482, 2997, 3111, 3117, 3066, 2982, 2440, 2631, 2018, 1849, 2020, 2329,

2274, 2771, 2437, 2461, 2656, 2183, 2296, 2530, 2616, 2622, 2332, 2865, 2990, 3018, 2770, 3033,

2473, 2180, 2332, 2463, 2455, 1951, 2131, 2045, 1828, 1898, 1827, 1819, 2126, 2230, 2334, 2159,

1738, 1943, 1737, 1483, 1396, 2148, 2010, 2154, 1930, 1908, 2138, 1883, 1597, 1774, 1841, 1646,

1706, 1962, 2072, 2088, 2021, 1924, 2048, 2403, 2748, 2897, 2860, 2751, 2668, 2674, 2591, 2201,

1760, 1622, 1633, 1412, 1140, 1260, 1347, 1193, 1015, 941, 981, 843, 717, 629, 775, 861,

886, 718, 575, 568, 656, 828, 805, 726, 558, 455, 600, 739, 865, 868, 751, 732,

770, 972, 1228, 1379, 1513, 1428, 1338, 1326, 1351, 1637, 1896, 2033, 2106, 1977, 1919, 2042,

2173, 2323, 2386, 2291, 2257, 2325, 2372, 2387, 2283, 1987, 1766, 1665, 1645, 1761, 1805, 1729,

1607, 1415, 1291, 1262, 1298, 1409, 1448, 1418, 1331, 1202, 1185, 1229, 1301, 1385, 1387, 1369,

1296, 1215, 1018, 819, 690, 600, 724, 841, 938, 938, 797, 596, 349, 232, 157, 313,

471, 570, 580, 466, 400, 355, 525, 712, 926, 1024, 1039, 930, 868, 943, 1180, 1444,

1576, 1521, 1399, 1302, 1266, 1247, 1244, 1340, 1390, 1383, 1301, 1378, 1488, 1573, 1507, 1362,

1307, 1396, 1760, 1946, 1915, 1651, 1710, 1969, 2264, 2394, 2385, 2295, 2138, 1940, 1855, 1783,

1733, 1755, 1852, 1888, 1895, 1972, 2088, 2165, 2186, 2283, 2417, 2580, 2764, 2891, 2928, 2795,

2816, 3018, 3231, 3276, 3122, 2821, 2552, 2579, 2498, 2083, 1741, 1751, 1955, 1864, 1703, 1940,

2089, 1843, 1721, 1857, 1843, 1798, 1605, 1626, 2200, 2118, 2020, 2509, 2643, 2329, 2806, 3386,

3200, 3125, 3587, 3841, 3455, 3120, 3210, 2965, 2537, 2503, 2544, 2329, 2223, 2406, 2460, 2571,

2486, 2348, 2602, 2613, 2220, 1978, 2076, 2190, 2173, 2282, 2340, 2260, 2108, 2061, 2088, 2070,

1992, 1956, 1993, 2096, 2296, 2454, 2569, 2506, 2485, 2503, 2635, 2734, 2803, 2824, 2773, 2665,

2526, 2529, 2584, 2597, 2373, 2226, 2289, 2610, 2847, 2618, 2426, 2732, 3077, 2959, 2623, 2545,

2746, 2814, 2630, 2511, 2634, 2854, 2984, 3075, 3098, 3063, 2994, 3089, 3257, 3454, 3466, 3418,

3320, 3412, 3547, 3741, 3757, 3691, 3487, 3379, 3339, 3434, 3451, 3340, 3070, 2775, 2578, 2464,

2416, 2330, 2232, 2114, 2068, 2058, 2112, 2135, 2135, 2108, 2122, 2191, 2237, 2241, 2167, 2134,

2114, 2092, 1966, 1789, 1644, 1627, 1712, 1919, 2089, 2220, 2270, 2343, 2454, 2544, 2583, 2558,

2663, 2813, 2993, 2978, 2976, 3018, 3245, 3456, 3627, 3712, 3771, 3813, 3823, 3714, 3510, 3319,

3198, 3131, 2900, 2633, 2440, 2366, 2326, 2351, 2431, 2333, 2102, 2127, 2190, 1930, 1991, 2069,

1768, 1582, 1531, 1472, 1705, 1905, 1784, 1725, 1775, 1698, 1773, 2200, 2527, 2329, 2129, 2368,

2507, 2217, 2013, 2112, 2109, 1748, 1343, 1415, 1743, 1692, 1456, 1429, 1591, 1767, 1758, 1562,

1438, 1530, 1666, 1647, 1565, 1636, 1830, 1850, 1609, 1468, 1633, 1920, 2176, 2327, 2416, 2552,

2789, 3001, 3052, 2959, 2751, 2531, 2359, 2182, 1991, 1843, 1712, 1413, 1049, 949, 996, 989,

907, 809, 717, 608, 529, 601, 670, 647, 539, 484, 610, 701, 759, 825, 993, 1163,

1246, 1326, 1448, 1611, 1739, 1840, 1946, 2103, 2325, 2557, 2623, 2731, 2886, 2961, 2772, 2529,

2659, 2766, 2586, 2101, 1777, 2046, 2262, 2250, 2144, 2172, 2296, 2320, 2287, 2114, 1879, 1760,

1742, 1695, 1551, 1336, 1093, 1077, 1236, 1204, 1148, 1308, 1405, 1265, 1837, 2174, 1944, 2320,

2583, 2412, 2568, 2537, 2408, 2157, 2135, 2365, 2431, 2330, 2133, 2019, 2230, 2168, 1767, 1863,

2071, 2010, 2028, 2036, 1970, 2163, 2381, 2537, 2775, 2701, 2496, 2411, 1806, 1519, 1170, 941,

1057, 1067, 970, 1284, 1091, 880, 972, 1119, 1007, 1193, 1548, 1419, 1779, 1949, 2106, 2430,

2445, 2454, 2696, 2475, 2181, 1786, 1749, 1908, 1858, 2050, 2252, 2476, 2334, 2052, 1968, 1958,

1588, 1539, 1597, 2050, 2236, 2092, 1835, 1657, 1604, 1591, 1686, 2010, 2158, 1993, 1905, 2078,

2081, 1882, 1860, 1972, 2058, 2147, 2201, 2236, 2341, 2477, 2514, 2430, 2372, 2455, 2557, 2518,

2478, 2558, 2568, 2431, 2218, 1997, 1754, 1467, 1216, 1042, 865, 705, 716, 850, 861, 710,

647, 733, 701, 514, 435, 493, 600, 732, 894, 1061, 1203, 1341, 1514, 1655, 1671, 1778,

2100, 2389, 2502, 2549, 2679, 2899, 3063, 3067, 3026, 3017, 2917, 2690, 2452, 2328, 2303, 2257,

2171, 2097, 2034, 2007, 2025, 2026, 1964, 1885, 1862, 1861, 1745, 1503, 1376, 1494, 1549, 1451,

1503, 1639, 1654, 1618, 1616, 1682, 1854, 2036, 2061, 2054, 2290, 2611, 2686, 2612, 2595, 2561,

2435, 2313, 2211, 2030, 1832, 1780, 1823, 1856, 1964, 2007, 1885, 1912, 2098, 2143, 2054, 1992,

1946, 1944, 2041, 2153, 2232, 2375, 2564, 2758, 2949, 3152, 3363, 3568, 3698, 3751, 3732, 3677,

3635, 3687, 3743, 3805, 3769, 3677, 3534, 3379, 3251, 3122, 2978, 2831, 2664, 2504, 2371, 2249,

2168, 2088, 2012, 1934, 1853, 1784, 1764, 1751, 1772, 1789, 1788, 1787, 1787, 1785, 1779, 1757,

1684, 1605, 1503, 1423, 1408, 1410, 1462, 1493, 1467, 1432, 1375, 1358, 1437, 1534, 1669, 1769,

1794, 1798, 1759, 1747, 1790, 1871, 2026, 2178, 2323, 2446, 2513, 2582, 2657, 2740, 2870, 3002,

3160, 3318, 3465, 3611, 3713, 3796, 3860, 3905, 3952, 4008, 4064, 4079, 4095, 4046, 3971, 3884,

3780, 3682, 3638, 3595, 3597, 3612, 3615, 3607, 3584, 3520, 3455, 3392, 3329, 3311, 3298, 3290,

3284, 3249, 3204, 3173, 3146, 3141, 3142, 3107, 3066, 3014, 2962, 2909, 2853, 2739, 2626, 2521,

2429, 2426, 2428, 2460, 2481, 2448, 2404, 2299, 2195, 2102, 2014, 1992, 1970, 1994, 2018, 2068,

2122, 2170, 2217, 2221, 2203, 2179, 2150, 2151, 2197, 2251, 2326, 2397, 2447, 2499, 2567, 2635,

2719, 2804, 2847, 2879, 2887, 2884, 2917, 2982, 3065, 3179, 3270, 3293, 3307, 3221, 3135, 3073,

3012, 3012, 3024, 3038, 3052, 3064, 3076, 3107, 3142, 3163, 3179, 3177, 3176, 3174, 3158, 3134,

3105, 3079, 3062, 3022, 2970, 2868, 2769, 2677, 2650, 2653, 2724, 2767, 2776, 2710, 2644, 2624,

2668, 2742, 2775, 2762, 2715, 2733, 2809, 2945, 3042, 3076, 3030, 2963, 2926, 2946, 3009, 3067,

3092, 3028, 2918, 2727, 2559, 2415, 2322, 2232, 2119, 1990, 1849, 1729, 1620, 1539, 1469, 1406,

1342, 1290, 1265, 1276, 1305, 1318, 1314, 1282, 1247, 1211, 1170, 1114, 1044, 986, 938, 928,

912, 885, 844, 797, 807, 825, 854, 853, 793, 724, 649, 643, 658, 748, 842, 950, 1043, 1124, 1175, 1217, 1261, 1328, 1455, 1635, 1862, 2063, 2256, 2325, 2372, 2345, 2295, 2207,

2092, 1956, 1828, 1702, 1637, 1578, 1537, 1458, 1339, 1199, 1060, 1061, 1108, 1232, 1320, 1387,

1393, 1409, 1434, 1470, 1495, 1481, 1447, 1399, 1353, 1304, 1252, 1220, 1199, 1204, 1186, 1149,

1064, 981, 904, 851, 803, 741, 645, 525, 472, 484, 588, 683, 746, 735, 756, 819,

967, 1129, 1252, 1316, 1301, 1308, 1355, 1441, 1503, 1437, 1287, 1128, 1041, 1093, 1158, 1191,

1191, 1201, 1292, 1418, 1564, 1620, 1658, 1693, 1795, 1901, 1980, 1992, 1979, 1989, 2028, 2061,

2078, 2113, 2183, 2268, 2313, 2346, 2383, 2438, 2468, 2488, 2519, 2609, 2712, 2811, 2832, 2826,

2800, 2809, 2825, 2845, 2863, 2893, 2895, 2822, 2605, 2328, 2128, 1947, 1718, 1521, 1433, 1359,

1227, 1156, 1203, 1211, 1091, 969, 931, 961, 1092, 1269, 1368, 1420, 1492, 1569, 1660, 1839,

2054, 2174, 2285, 2538, 2864, 3109, 3254, 3323, 3296, 3167, 3003, 2899, 2811, 2605, 2342, 2224,

2235, 2180, 2061, 2052, 2094, 1933, 1663, 1531, 1521, 1479, 1445, 1411, 1350, 1355, 1405, 1484,

1591, 1696, 1954, 2180, 2217, 2245, 2295, 2308, 2231, 2008, 1736, 1606, 1598, 1553, 1498, 1496,

1480, 1460, 1576, 1753, 1786, 1694, 1630, 1625, 1614, 1565, 1506, 1458, 1450, 1540, 1741, 1954,

2131, 2308, 2477, 2617, 2714, 2794, 2935, 3104, 3250, 3293, 3197, 3111, 3046, 2990, 2851, 2583,

2274, 1975, 1726, 1538, 1372, 1231, 1131, 1098, 1147, 1172, 1146, 1059, 959, 917, 914, 939,

925, 872, 808, 773, 804, 840, 866, 845, 848, 878, 966, 1064, 1162, 1253, 1360, 1476,

1600, 1739, 1929, 2164, 2454, 2701, 2905, 3060, 3170, 3229, 3239, 3195, 3170, 3163, 3169, 3145,

3083, 2992, 2875, 2749, 2627, 2525, 2470, 2499, 2522, 2487, 2380, 2236, 2114, 1987, 1858, 1754,

1686, 1621, 1469, 1302, 1185, 1136, 1145, 1182, 1208, 1210, 1195, 1179, 1204, 1247, 1297, 1340,

1360, 1382, 1416, 1504, 1614, 1735, 1860, 2023, 2198, 2345, 2422, 2373, 2317, 2294, 2310, 2328,

2305, 2279, 2275, 2317, 2400, 2453, 2462, 2460, 2470, 2473, 2439, 2377, 2371, 2463, 2550, 2524,

2446, 2439, 2511, 2612, 2724, 2838, 2935, 2999, 3065, 3166, 3270, 3361, 3432, 3504, 3560, 3577,

3568, 3549, 3530, 3459, 3316, 3127, 2931, 2761, 2623, 2465, 2314, 2237, 2237, 2282, 2316, 2331,

2311, 2238, 2135, 2058, 2085, 2131, 2126, 2069, 1982, 1918, 1863, 1822, 1821, 1879, 1982, 2023,

1991, 1929, 1901, 1996, 2251, 2552, 2838, 3069, 3232, 3332, 3406, 3454, 3448, 3367, 3218, 3054,

2898, 2738, 2623, 2605, 2643, 2658, 2685, 2778, 2766, 2612, 2584, 2461, 2208, 1989, 1716, 1487,

1488, 1537, 1533, 1585, 1708, 1854, 2007, 2118, 2152, 2114, 2014, 1900, 1802, 1693, 1613, 1631,

1709, 1765, 1812, 1912, 2036, 2122, 2167, 2197, 2223, 2286, 2415, 2504, 2459, 2403, 2466, 2579,

2658, 2719, 2773, 2808, 2849, 2943, 3060, 3114, 3061, 2930, 2786, 2680, 2580, 2409, 2154, 1897,

1704, 1552, 1397, 1257, 1123, 944, 752, 652, 669, 732, 784, 824, 860, 896, 930, 959,

991, 1071, 1198, 1319, 1394, 1430, 1487, 1652, 1886, 2063, 2179, 2304, 2489, 2709, 2872, 2919,

2901, 2843, 2764, 2681, 2611, 2563, 2496, 2404, 2313, 2248, 2205, 2160, 2141, 2141, 2080, 1963,

1845, 1761, 1709, 1662, 1610, 1549, 1473, 1388, 1318, 1274, 1223, 1116, 1018, 1052, 1179, 1281,

1374, 1512, 1668, 1832, 2018, 2115, 2041, 1883, 1744, 1637, 1504, 1314, 1146, 1084, 1116, 1176,

1222, 1263, 1321, 1379, 1426, 1451, 1493, 1543, 1581, 1564, 1472, 1360, 1242, 1200, 1204, 1259,

1329, 1394, 1452, 1531, 1622, 1723, 1826, 1928, 2028, 2131, 2242, 2375, 2515, 2661, 2744, 2807,

2829, 2868, 2917, 2987, 3032, 3024, 2938, 2814, 2630, 2440, 2242, 2056, 1880, 1737, 1606, 1487,

1371, 1259, 1161, 1082, 1016, 968, 922, 876, 834, 793, 778, 769, 771, 761, 746, 725,

706, 695, 676, 653, 618, 596, 589, 614, 648, 688, 725, 759, 798, 837, 875, 926,

992, 1136, 1330, 1570, 1808, 2025, 2205, 2373, 2543, 2725, 2893, 3045, 3150, 3226, 3286, 3335,

3354, 3348, 3282, 3198, 3108, 3034, 2974, 2923, 2879, 2827, 2769, 2704, 2666, 2652, 2663, 2632,

2547, 2419, 2270, 2152, 2046, 1946, 1835, 1741, 1670, 1625, 1576, 1502, 1428, 1383, 1391, 1426,

1399, 1313, 1229, 1197, 1236, 1343, 1515, 1759, 2040, 2185, 2213, 2269, 2255, 2162, 2126, 2189,

2281, 2262, 2197, 2195, 2188, 2181, 2071, 1980, 2045, 2126, 2205, 2284, 2380, 2467, 2571, 2747,

2848, 2845, 2859, 2808, 2551, 2178, 1856, 1689, 1645, 1580, 1441, 1347, 1413, 1486, 1347, 1188,

1172, 1186, 1148, 1074, 975, 978, 1148, 1401, 1715, 2127, 2565, 2866, 2934, 2921, 2895, 2780,

2621, 2559, 2506, 2445, 2480, 2602, 2681, 2699, 2713, 2755, 2798, 2769, 2665, 2538, 2394, 2236,

2113, 2055, 2025, 1970, 1893, 1856, 1858, 1889, 1950, 2050, 2191, 2334, 2475, 2611, 2722, 2770,

2734, 2649, 2585, 2539, 2450, 2324, 2221, 2182, 2210, 2279, 2351, 2389, 2395, 2399, 2431, 2505,

2615, 2742, 2849, 2901, 2889, 2845, 2822, 2865, 2950, 2999, 3001, 2982, 2975, 3006, 3080, 3173,

3260, 3333, 3381, 3396, 3364, 3325, 3307, 3309, 3318, 3271, 3178, 3050, 2916, 2819, 2732, 2643,

2501, 2337, 2176, 2038, 1912, 1750, 1557, 1347, 1219, 1164, 1159, 1170, 1144, 1101, 1056, 1022,

1039, 1082, 1133, 1177, 1200, 1234, 1282, 1347, 1405, 1456, 1513, 1590, 1691, 1782, 1875, 1986,

2158, 2386, 2596, 2765, 2882, 2956, 2994, 3033, 3094, 3137, 3108, 3011, 2915, 2865, 2819, 2740,

2659, 2589, 2505, 2414, 2305, 2206, 2141, 2038, 1873, 1697, 1589, 1613, 1633, 1586, 1658, 1891,

2123, 2294, 2414, 2462, 2461, 2384, 2144, 1843, 1719, 1668, 1569, 1509, 1552, 1624, 1694, 1778,

1842, 1820, 1705, 1575, 1510, 1504, 1521, 1526, 1506, 1488, 1527, 1651, 1797, 1921, 2046, 2184,

2336, 2491, 2646, 2812, 2976, 3128, 3203, 3229, 3217, 3174, 3085, 2966, 2820, 2660, 2496, 2330,

2157, 1989, 1833, 1690, 1557, 1413, 1267, 1131, 1056, 1020, 1017, 1016, 993, 952, 892, 831,

777, 734, 719, 710, 706, 693, 675, 655, 642, 635, 635, 650, 685, 739, 828, 935,

1056, 1199, 1370, 1566, 1805, 2051, 2288, 2484, 2632, 2760, 2875, 2989, 3089, 3168, 3191, 3188,

3160, 3105, 3044, 2980, 2915, 2865, 2837, 2836, 2851, 2819, 2737, 2616, 2499, 2426, 2347, 2237,

2095, 1949, 1850, 1781, 1717, 1653, 1595, 1559, 1553, 1520, 1429, 1307, 1202, 1157, 1172, 1218,

1288, 1438, 1676, 1901, 2057, 2162, 2171, 2130, 2142, 2193, 2243, 2216, 2220, 2226, 2187, 2158,

2121, 2044, 1983, 1980, 2016, 2062, 2099, 2116, 2127, 2159, 2218, 2284, 2339, 2397, 2479, 2567,

2654, 2745, 2849, 2953, 3015, 2980, 2847, 2672, 2516, 2394, 2267, 2111, 1945, 1787, 1655, 1576,

1550, 1530, 1458, 1316, 1174, 1097, 1059, 1002, 919, 834, 772, 732, 669, 579, 473, 378,

324, 299, 307, 360, 489, 672, 899, 1152, 1371, 1547, 1682, 1783, 1877, 1961, 2028, 2041,

2002, 1938, 1861, 1801, 1750, 1710, 1683, 1677, 1679, 1686, 1688, 1689, 1703, 1732, 1774, 1816,

1853, 1878, 1882, 1849, 1790, 1741, 1700, 1660, 1586, 1498, 1406, 1320, 1246, 1174, 1101, 1021,

937, 853, 795, 774, 785, 853, 935, 1031, 1134, 1239, 1348, 1461, 1570, 1671, 1750, 1802,

1818, 1775, 1675, 1567, 1514, 1520, 1546, 1548, 1513, 1496, 1564, 1691, 1816, 1923, 2016, 2095,

2158, 2223, 2323, 2452, 2551, 2576, 2569, 2609, 2715, 2824, 2876, 2908, 2960, 3027, 3094, 3141,

3174, 3214, 3264, 3331, 3384, 3418, 3406, 3365, 3319, 3283, 3266, 3238, 3194, 3103, 2989, 2862,

2726, 2582, 2433, 2280, 2142, 2034, 1945, 1875, 1810, 1757, 1717, 1701, 1684, 1665, 1641, 1622,

1606, 1598, 1610, 1670, 1753, 1845, 1901, 1906, 1895, 1886, 1922, 1994, 2091, 2207, 2338, 2469,

2600, 2727, 2856, 2985, 3110, 3208, 3290, 3360, 3423, 3466, 3483, 3469, 3432, 3395, 3351, 3299,

3240, 3197, 3182, 3145, 3051, 2931, 2849, 2828, 2811, 2728, 2581, 2428, 2303, 2202, 2138, 2074,

1979, 1920, 1948, 2025, 2095, 2144, 2191, 2275, 2393, 2507, 2608, 2691, 2766, 2814, 2820, 2777,

2686, 2567, 2447, 2334, 2230, 2115, 1976, 1846, 1759, 1744, 1798, 1880, 1959, 2021, 2047, 2037,

2013, 1991, 1962, 1906, 1834, 1781, 1772, 1794, 1847, 1924, 1994, 2047, 2125, 2269, 2463, 2665,

2853, 3034, 3219, 3398, 3530, 3594, 3592, 3540, 3463, 3369, 3251, 3103, 2925, 2722, 2506, 2306,

2127, 1972, 1840, 1736, 1651, 1565, 1482, 1411, 1357, 1315, 1266, 1202, 1129, 1054, 1014, 1002,

998, 988, 940, 879, 807, 766, 755, 773, 836, 922, 1030, 1186, 1348, 1517, 1665, 1812,

1960, 2146, 2343, 2549, 2710, 2828, 2908, 2942, 2972, 2996, 3008, 3005, 2987, 2949, 2912, 2877,

2848, 2796, 2720, 2625, 2532, 2462, 2382, 2253, 2072, 1886, 1743, 1651, 1588, 1541, 1500, 1450,

1383, 1295, 1186, 1064, 955, 876, 826, 818, 876, 995, 1147, 1294, 1417, 1525, 1625, 1704,

1750, 1780, 1808, 1839, 1862, 1857, 1850, 1883, 1932, 1937, 1908, 1878, 1844, 1809, 1798, 1805,

1818, 1855, 1938, 2042, 2120, 2159, 2214, 2331, 2509, 2735, 2960, 3107, 3156, 3171, 3152, 3018,

2797, 2630, 2491, 2364, 2268, 2136, 1991, 1865, 1682, 1590, 1561, 1379, 1105, 873, 848, 990,

1220, 1514, 1822, 2110, 2345, 2459, 2470, 2467, 2481, 2477, 2428, 2378, 2372, 2370, 2397, 2459,

2465, 2421, 2320, 2175, 1984, 1686, 1411, 1251, 1163, 1220, 1491, 1756, 1856, 1759, 1704, 1481,

1341, 1412, 1432, 1448, 1530, 1626, 1728, 1845, 1914, 1951, 2014, 2095, 2158, 2201, 2243, 2294,

2360, 2439, 2513, 2576, 2640, 2693, 2733, 2763, 2746, 2647, 2488, 2308, 2098, 1860, 1639, 1479,

1378, 1283, 1147, 997, 884, 848, 882, 945, 989, 993, 983, 1010, 1110, 1268, 1475, 1707,

1934, 2187, 2435, 2585, 2643, 2680, 2729, 2760, 2774, 2735, 2621, 2563, 2584, 2568, 2468, 2355,

2270, 2155, 2010, 1894, 1794, 1704, 1699, 1713, 1687, 1737, 1879, 2059, 2226, 2355, 2438, 2405,

2253, 1983, 1698, 1577, 1595, 1643, 1692, 1756, 1742, 1633, 1561, 1565, 1574, 1595, 1651, 1712,

1792, 1904, 2083, 2328, 2595, 2831, 3022, 3177, 3304, 3402, 3463, 3463, 3410, 3315, 3188, 3060,

2934, 2809, 2669, 2519, 2371, 2234, 2128, 2044, 1981, 1941, 1904, 1868, 1809, 1730, 1637, 1538,

1467, 1423, 1421, 1437, 1445, 1432, 1370, 1297, 1219, 1167, 1140, 1132, 1140, 1160, 1190, 1235,

1308, 1393, 1492, 1617, 1784, 1982, 2217, 2452, 2666, 2855, 3002, 3129, 3243, 3344, 3431, 3502,

3554, 3567, 3567, 3559, 3554, 3551, 3540, 3518, 3480, 3419, 3328, 3207, 3069, 2934, 2816, 2719,

2637, 2552, 2471, 2419, 2387, 2336, 2244, 2121, 1995, 1887, 1806, 1751, 1725, 1733, 1777, 1852,

1944, 2041, 2134, 2226, 2316, 2392, 2442, 2467, 2457, 2410, 2369, 2377, 2414, 2430, 2412, 2387,

2369, 2324, 2233, 2125, 2023, 1953, 1947, 1999, 2014, 1977, 2001, 2080, 2203, 2414, 2634, 2780,

2908, 2916, 2794, 2672, 2563, 2420, 2266, 2183, 2138, 2043, 1880, 1717, 1621, 1575, 1464, 1224,

931, 722, 610, 579, 663, 890, 1165, 1423, 1670, 1901, 2086, 2219, 2296, 2302, 2288, 2299,

2314, 2357, 2447, 2480, 2502, 2556, 2551, 2466, 2326, 2141, 1912, 1657, 1553, 1669, 1809, 1961,

2112, 2066, 1827, 1670, 1651, 1735, 1782, 1846, 1911, 1955, 2008, 2073, 2136, 2184, 2215, 2233,

2228, 2227, 2287, 2385, 2476, 2559, 2637, 2681, 2651, 2525, 2326, 2100, 1896, 1741, 1616, 1457,

1242, 1034, 886, 810, 805, 827, 800, 751, 728, 734, 771, 844, 940, 1068, 1253, 1495,

1715, 1860, 1979, 2137, 2278, 2337, 2341, 2352, 2363, 2348, 2343, 2357, 2366, 2347, 2304, 2260,

2209, 2120, 2012, 1925, 1867, 1823, 1779, 1731, 1679, 1629, 1609, 1641, 1731, 1860, 2013, 2197,

2350, 2342, 2200, 1991, 1761, 1581, 1558, 1571, 1531, 1475, 1433, 1414, 1385, 1357, 1379, 1425,

1448, 1471, 1542, 1646, 1774, 1947, 2177, 2434, 2693, 2933, 3129, 3255, 3324, 3355, 3359, 3334,

3272, 3134, 2951, 2766, 2605, 2511, 2452, 2398, 2338, 2254, 2155, 2047, 1934, 1826, 1734, 1660,

1609, 1571, 1536, 1499, 1452, 1396, 1339, 1284, 1247, 1214, 1177, 1130, 1071, 1029, 1013, 1031,

1097, 1212, 1356, 1522, 1700, 1894, 2095, 2297, 2496, 2691, 2864, 3004, 3110, 3184, 3239, 3299,

3362, 3422, 3462, 3467, 3447, 3409, 3360, 3302, 3225, 3122, 3000, 2880, 2784, 2720, 2674, 2630,

2581, 2518, 2436, 2348, 2263, 2171, 2067, 1968, 1897, 1863, 1853, 1859, 1885, 1944, 2037, 2153,

2272, 2371, 2448, 2508, 2550, 2574, 2588, 2602, 2617, 2624, 2615, 2601, 2611, 2642, 2662, 2641,

2577, 2493, 2419, 2358, 2303, 2257, 2229, 2221, 2220, 2216, 2215, 2224, 2244, 2267, 2289, 2325,

2404, 2531, 2680, 2816, 2926, 3013, 3090, 3156, 3200, 3213, 3179, 3088, 2964, 2843, 2758, 2711,

2672, 2602, 2489, 2362, 2258, 2187, 2125, 2040, 1923, 1786, 1637, 1464, 1269, 1078, 905, 757,

643, 576, 555, 571, 610, 675, 782, 929, 1088, 1243, 1392, 1533, 1650, 1731, 1775, 1795,

1806, 1815, 1822, 1830, 1849, 1877, 1899, 1925, 1978, 2049, 2079, 2084, 2108, 2112, 2044, 1912,

1744, 1575, 1442, 1358, 1309, 1281, 1282, 1336, 1442, 1566, 1676, 1761, 1836, 1899, 1905, 1821,

1714, 1650, 1652, 1671, 1729, 1841, 1922, 1991, 2146, 2329, 2443, 2495, 2511, 2528, 2601, 2715,

2890, 3102, 3257, 3302, 3199, 3005, 2850, 2724, 2511, 2266, 2058, 1932, 1848, 1731, 1641, 1616,

1545, 1481, 1528, 1631, 1762, 1942, 2146, 2377, 2525, 2582, 2627, 2652, 2640, 2623, 2620, 2580,

2489, 2349, 2228, 2118, 1981, 1876, 1767, 1669, 1701, 1844, 2013, 2160, 2232, 2216, 2078, 1873,

1707, 1589, 1506, 1439, 1385, 1333, 1288, 1258, 1227, 1201, 1188, 1176, 1160, 1161, 1186, 1219,

1238, 1235, 1235, 1276, 1368, 1505, 1666, 1832, 1998, 2175, 2372, 2571, 2757, 2911, 3018, 3060,

3021, 2934, 2826, 2724, 2642, 2579, 2504, 2397, 2262, 2113, 1968, 1842, 1738, 1643, 1553, 1467,

1387, 1316, 1255, 1201, 1150, 1099, 1038, 954, 853, 752, 662, 593, 551, 529, 530, 557,

619, 730, 905, 1107, 1304, 1487, 1661, 1834, 2001, 2154, 2286, 2399, 2488, 2554, 2597, 2615,

2594, 2521, 2406, 2286, 2178, 2083, 2003, 1944, 1904, 1870, 1815, 1733, 1648, 1552, 1444, 1349,

1264, 1182, 1139, 1170, 1260, 1369, 1466, 1557, 1661, 1776, 1872, 1924, 1959, 2007, 2060, 2112,

2154, 2162, 2126, 2087, 2059, 2009, 1941, 1897, 1900, 1926, 1955, 1969, 1968, 1981, 2045, 2164,

2302, 2436, 2585, 2784, 3011, 3201, 3301, 3327, 3312, 3277, 3228, 3169, 3095, 3012, 2927, 2842,

2772, 2710, 2630, 2514, 2356, 2169, 1977, 1785, 1607, 1463, 1358, 1281, 1247, 1280, 1377, 1524,

1709, 1912, 2086, 2209, 2300, 2381, 2440, 2458, 2457, 2474, 2509, 2541, 2588, 2663, 2734, 2806,

2805, 2649, 2394, 2048, 1908, 1949, 2076, 2242, 2294, 2208, 2080, 1896, 1863, 1871, 1842, 1903,

2047, 2224, 2314, 2266, 2240, 2442, 2867, 3034, 2932, 2830, 2590, 2319, 2079, 1863, 1735, 1639,

1551, 1478, 1468, 1449, 1498, 1695, 1949, 2218, 2441, 2572, 2616, 2650, 2686, 2682, 2672, 2647,

2559, 2473, 2432, 2375, 2246, 2105, 2024, 2005, 2098, 2258, 2408, 2485, 2443, 2333, 2194, 2035,

1872, 1742, 1620, 1535, 1487, 1434, 1348, 1273, 1245, 1237, 1221, 1190, 1153, 1126, 1129, 1171,

1244, 1331, 1440, 1606, 1844, 2124, 2394, 2610, 2760, 2847, 2880, 2880, 2861, 2821, 2754, 2656,

2530, 2395, 2261, 2120, 1970, 1828, 1717, 1640, 1583, 1524, 1455, 1381, 1311, 1240, 1157, 1056,

938, 821, 733, 706, 746, 833, 952, 1103, 1294, 1521, 1759, 1989, 2218, 2442, 2641, 2783,

2831, 2798, 2710, 2610, 2505, 2405, 2340, 2289, 2217, 2147, 2106, 2055, 1943, 1790, 1657, 1548,

1472, 1440, 1457, 1542, 1670, 1787, 1863, 1948, 2045, 2039, 1984, 1879, 1707, 1571, 1530, 1442,

1363, 1338, 1335, 1348, 1402, 1533, 1732, 1967, 2190, 2394, 2523, 2578, 2598, 2567, 2483, 2401,

2326, 2269, 2236, 2191, 2101, 1971, 1810, 1625, 1441, 1272, 1094, 905, 755, 713, 755, 850,

1009, 1220, 1458, 1666, 1824, 1944, 2046, 2137, 2218, 2268, 2329, 2436, 2561, 2662, 2735, 2787,

2811, 2772, 2645, 2493, 2358, 2223, 2125, 2138, 2225, 2337, 2426, 2501, 2571, 2515, 2348, 2259,

2202, 2106, 2033, 2118, 2270, 2363, 2348, 2233, 2101, 2282, 2598, 2848, 2933, 2862, 2685, 2405,

2109, 1924, 1723, 1624, 1580, 1441, 1334, 1353, 1519, 1860, 2131, 2289, 2393, 2427, 2468, 2509,

2449, 2444, 2473, 2385, 2209, 2073, 2143, 2332, 2500, 2564, 2538, 2486, 2396, 2220, 2046, 1918,

1842, 1766, 1655, 1568, 1526, 1482, 1437, 1422, 1423, 1414, 1403, 1454, 1639, 1936, 2269, 2569,

2828, 3015, 3119, 3168, 3182, 3168, 3116, 3007, 2851, 2678, 2522, 2400, 2289, 2175, 2081, 2015,

1961, 1886, 1779, 1649, 1505, 1371, 1268, 1202, 1176, 1198, 1286, 1434, 1626, 1852, 2119, 2392,

2626, 2820, 2948, 2975, 2929, 2862, 2805, 2741, 2646, 2567, 2539, 2528, 2427, 2264, 2070, 1889,

1875, 1967, 2068, 2185, 2351, 2497, 2525, 2487, 2403, 2270, 2149, 2054, 1960, 1884, 1818, 1761,

1725, 1700, 1675, 1654, 1651, 1678, 1739, 1830, 1943, 2074, 2214, 2351, 2474, 2576, 2657, 2720,

2766, 2791, 2792, 2771, 2734, 2693, 2648, 2599, 2547, 2497, 2456, 2423, 2395, 2361, 2302, 2220,

2124, 2021, 1921, 1828, 1733, 1630, 1512, 1376, 1230, 1079, 929, 786, 662, 554, 462, 388,

335, 300, 285, 300, 347, 412, 491, 573, 653, 730, 805, 880, 954, 1025, 1091, 1144,

1185, 1212, 1226, 1226, 1228, 1237, 1257, 1294, 1340, 1391, 1443, 1488, 1530, 1575, 1626, 1683,

1737, 1782, 1814, 1830, 1831, 1827, 1815, 1793, 1755, 1694, 1609, 1519, 1438, 1374, 1333, 1311,

1302, 1310, 1336, 1387, 1448, 1514, 1582, 1650, 1720, 1781, 1828, 1854, 1857, 1841, 1818, 1797,

1789, 1788, 1737, 1641, 1639, 1730, 1888, 2068, 2117, 2044, 1960, 2026, 2251, 2570, 2840, 3002,

3082, 2984, 2780, 2585, 2343, 2188, 2157, 2090, 1931, 1695, 1558, 1682, 1894, 2144, 2331, 2432,

2473, 2499, 2473, 2431, 2447, 2519, 2405, 2151, 2090, 2262, 2412, 2511, 2556, 2492, 2342, 2190,

2076, 1974, 1847, 1707, 1591, 1513, 1473, 1460, 1440, 1387, 1331, 1297, 1331, 1459, 1698, 2012,

2316, 2564, 2771, 2939, 3063, 3113, 3085, 3009, 2887, 2741, 2578, 2407, 2268, 2175, 2082, 1979,

1867, 1725, 1551, 1377, 1218, 1093, 1040, 1084, 1212, 1422, 1735, 2072, 2332, 2471, 2513, 2512,

2457, 2348, 2266, 2245, 2240, 2217, 1990, 1729, 1658, 1702, 1850, 2074, 2271, 2412, 2477, 2464,

2450, 2435, 2371, 2287, 2221, 2162, 2101, 2038, 1995, 1970, 1915, 1836, 1786, 1819, 1923, 2097,

2339, 2577, 2786, 2965, 3096, 3156, 3158, 3144, 3143, 3150, 3136, 3079, 2997, 2912, 2816, 2686,

2510, 2315, 2104, 1877, 1637, 1416, 1261, 1174, 1163, 1203, 1295, 1455, 1634, 1772, 1848, 1882,

1913, 1947, 1987, 2040, 2120, 2216, 2289, 2359, 2439, 2456, 2320, 2040, 1777, 1744, 1888, 2069,

2037, 1898, 1568, 1582, 1719, 1634, 1406, 1478, 1826, 2222, 2460, 2418, 2222, 1922, 1801, 1808,

1699, 1391, 1252, 1447, 1850, 2143, 2214, 2189, 2239, 2332, 2405, 2356, 2235, 2116, 2066, 2071,

2118, 2210, 2317, 2389, 2419, 2373, 2279, 2159, 2038, 1906, 1732, 1539, 1386, 1308, 1248, 1185,

1130, 1057, 956, 912, 975, 1144, 1388, 1659, 1931, 2199, 2433, 2612, 2727, 2794, 2817, 2783,

2702, 2589, 2461, 2350, 2265, 2198, 2142, 2093, 2038, 1966, 1859, 1728, 1586, 1441, 1322, 1242,

1212, 1273, 1476, 1800, 2151, 2431, 2625, 2693, 2650, 2590, 2524, 2492, 2544, 2516, 2354, 2133,

1932, 1870, 1954, 2184, 2424, 2550, 2570, 2553, 2460, 2319, 2187, 2041, 1925, 1844, 1733, 1557,

1402, 1370, 1508, 1760, 2085, 2390, 2567, 2645, 2739, 2811, 2818, 2787, 2712, 2611, 2486, 2359,

2203, 1973, 1709, 1464, 1272, 1137, 1095, 1209, 1398, 1585, 1744, 1861, 1943, 2035, 2158, 2290,

2391, 2494, 2601, 2699, 2736, 2631, 2395, 2210, 2157, 2298, 2486, 2568, 2566, 2481, 2162, 2062,

2114, 2085, 1810, 1600, 1616, 1873, 2290, 2485, 2482, 2266, 1963, 1876, 1845, 1511, 1199, 1307,

1658, 1899, 1924, 1927, 2040, 2120, 2091, 2032, 1984, 1953, 1986, 2060, 2189, 2316, 2363, 2333,

2275, 2177, 1996, 1779, 1607, 1473, 1397, 1316, 1183, 1070, 1064, 1173, 1390, 1675, 1992, 2313,

2566, 2746, 2824, 2815, 2738, 2619, 2497, 2398, 2310, 2253, 2187, 2084, 1955, 1803, 1605, 1393,

1262, 1293, 1483, 1768, 2081, 2330, 2461, 2499, 2458, 2368, 2377, 2445, 2462, 2321, 2104, 1902,

1883, 2150, 2432, 2636, 2740, 2714, 2612, 2441, 2256, 2131, 2068, 1931, 1758, 1602, 1495, 1449,

1477, 1569, 1706, 1866, 2023, 2162, 2274, 2366, 2456, 2548, 2627, 2688, 2733, 2764, 2775, 2758,

2714, 2648, 2568, 2483, 2401, 2327, 2253, 2158, 2038, 1901, 1754, 1592, 1413, 1223, 1033, 863,

736, 665, 649, 678, 745, 834, 915, 974, 1030, 1094, 1165, 1244, 1333, 1418, 1488, 1555,

1619, 1674, 1730, 1802, 1891, 1988, 2057, 2063, 2029, 1967, 1879, 1791, 1717, 1679, 1696, 1770,

1882, 1997, 2093, 2172, 2245, 2321, 2395, 2458, 2481, 2451, 2386, 2324, 2268, 2241, 2279, 2381,

2443, 2355, 2178, 2002, 1944, 2114, 2389, 2676, 2909, 3053, 3107, 3048, 2890, 2714, 2634, 2640,

2658, 2586, 2394, 2154, 1925, 1822, 1856, 2005, 2147, 2245, 2287, 2268, 2246, 2257, 2267, 2268,

2263, 2220, 2188, 2109, 2103, 2253, 2344, 2395, 2389, 2260, 2071, 1839, 1618, 1460, 1346, 1229,

1123, 1047, 1049, 1138, 1302, 1555, 1837, 2115, 2339, 2476, 2531, 2533, 2478, 2368, 2263, 2170,

2089, 2021, 1955, 1885, 1744, 1528, 1307, 1132, 1056, 1121, 1327, 1605, 1836, 2007, 2016, 1946,

1906, 1935, 1970, 1912, 1789, 1545, 1397, 1550, 1829, 2119, 2336, 2476, 2532, 2529, 2446, 2292,

2152, 2080, 2038, 1947, 1797, 1623, 1491, 1482, 1574, 1727, 1903, 2105, 2320, 2537, 2738, 2916,

3039, 3062, 3019, 2946, 2855, 2757, 2656, 2539, 2358, 2104, 1811, 1537, 1358, 1311, 1346, 1408,

1495, 1615, 1754, 1895, 1990, 2055, 2124, 2246, 2354, 2317, 2152, 1870, 1875, 2061, 2311, 2329,

2065, 1994, 1782, 1353, 1798, 2389, 2454, 2232, 2339, 2305, 1940, 1716, 1882, 2187, 2346, 2356,

2385, 2409, 2458, 2427, 2315, 2277, 2306, 2376, 2463, 2520, 2484, 2383, 2227, 2016, 1764, 1570,

1434, 1319, 1210, 1107, 1084, 1191, 1438, 1759, 2070, 2327, 2488, 2554, 2535, 2459, 2347, 2275,

2260, 2221, 2062, 1812, 1543, 1443, 1613, 1932, 2241, 2357, 2331, 2325, 2305, 2221, 1905, 1657,

1717, 1991, 2299, 2514, 2603, 2549, 2307, 2047, 1919, 1775, 1546, 1329, 1229, 1246, 1371, 1552,

1771, 2041, 2315, 2522, 2637, 2639, 2555, 2453, 2367, 2301, 2198, 1981, 1681, 1425, 1275, 1244,

1294, 1412, 1591, 1832, 2007, 2132, 2283, 2375, 2400, 2263, 2079, 2005, 2054, 2243, 2480, 2593,

2494, 2247, 2180, 1726, 1413, 1499, 1832, 2175, 2190, 2002, 2000, 2086, 1762, 1406, 1533, 1817,

1944, 1946, 2096, 2136, 2092, 2127, 2291, 2481, 2543, 2516, 2295, 1987, 1768, 1627, 1507, 1414,

1480, 1734, 2070, 2402, 2627, 2761, 2791, 2675, 2564, 2524, 2514, 2358, 2020, 1743, 1686, 1852,

2072, 2229, 2355, 2416, 2431, 2427, 2410, 2389, 2415, 2440, 2319, 2063, 1861, 1862, 2075, 2378,

2654, 2859, 2974, 2908, 2648, 2423, 2293, 2166, 2015, 1846, 1684, 1571, 1510, 1479, 1468, 1482,

1523, 1588, 1677, 1785, 1902, 2025, 2148, 2269, 2383, 2482, 2556, 2600, 2608, 2584, 2537, 2469,

2382, 2289, 2206, 2141, 2088, 2041, 1990, 1922, 1826, 1701, 1551, 1389, 1233, 1087, 957, 850,

769, 713, 681, 677, 692, 727, 791, 887, 998, 1109, 1210, 1294, 1370, 1446, 1516, 1576,

1628, 1672, 1721, 1772, 1811, 1831, 1831, 1812, 1770, 1716, 1660, 1600, 1549, 1520, 1525, 1566,

1638, 1739, 1864, 2002, 2132, 2239, 2325, 2401, 2466, 2508, 2504, 2462, 2456, 2497, 2499, 2324,

2070, 1899, 1883, 2062, 2386, 2648, 2678, 2565, 2522, 2614, 2614, 2252, 1832, 1891, 2120, 2206,

2104, 2063, 2137, 2130, 2048, 1931, 2032, 2249, 2376, 2303, 1947, 1613, 1432, 1277, 1201, 1267,

1488, 1785, 2071, 2323, 2358, 2254, 2168, 2187, 2106, 1871, 1593, 1439, 1545, 1743, 1929, 2050,

2089, 2036, 2040, 2089, 1894, 1608, 1537, 1759, 2109, 2426, 2710, 2890, 2881, 2741, 2586, 2449,

2354, 2277, 2172, 2034, 1902, 1797, 1720, 1685, 1717, 1804, 1952, 2158, 2409, 2680, 2905, 3027,

3027, 2938, 2828, 2747, 2696, 2629, 2470, 2241, 1969, 1723, 1541, 1446, 1492, 1654, 1849, 2036,

2168, 2234, 2262, 2267, 2175, 1914, 1702, 1923, 2198, 2288, 2146, 1832, 1420, 1690, 2157, 2115,

2330, 2124, 1787, 2175, 2202, 2276, 2032, 1980, 2157, 2311, 1961, 1608, 1422, 1421, 1706, 2138,

2249, 2164, 2178, 1957, 1758, 1951, 2240, 2229, 1983, 1674, 2058, 2519, 2430, 2193, 1897, 1637,

1566, 1848, 2353, 2459, 2327, 2223, 1960, 1619, 1711, 2107, 2292, 2315, 1986, 2064, 2364, 2384,

1930, 1658, 2106, 2068, 1816, 2044, 2018, 2112, 1954, 2176, 2035, 2174, 2053, 2261, 1814, 1770,

2165, 2106, 1892, 1549, 1721, 2025, 2120, 1934, 2044, 2327, 2002, 2233, 1989, 2032, 1966, 2077,

2222, 2173, 2051, 1891, 2089, 2231, 2042, 1852, 1944, 2090, 1964, 1911, 1987, 2001, 1986, 2088,

2025, 2010, 1993, 2168, 2169, 1949, 1997, 1978, 2101, 2126, 2029, 1990, 2065, 2139, 2029, 2003,

1980, 1974, 2107, 2019, 2090, 2110, 2066, 2099, 2097, 1997, 2061, 2056, 2062, 2035, 2048, 2088

};

// Drum: 8192 samples, 32,768 bytes (32.00KB)

uint32_t Drum_LUT[NS_WAVE] = {

1185, 1217, 1063, 941, 904, 945, 1090, 1366, 1608, 1704, 1838, 2060, 2221, 2337, 2342, 2593,

2656, 2818, 2949, 2962, 3115, 3013, 3131, 3201, 3192, 3250, 3221, 3203, 3338, 3432, 3487, 3455,

3472, 3508, 3568, 3617, 3418, 3401, 3773, 3707, 3517, 3633, 3666, 3579, 3567, 3576, 3567, 3623,

3606, 3566, 3596, 3611, 3581, 3591, 3653, 3703, 3660, 3571, 3555, 3474, 3377, 3371, 3229, 3031,

3012, 2999, 2823, 2692, 2707, 2627, 2442, 2460, 2257, 2007, 2018, 1907, 1815, 1737, 1686, 1600,

1433, 1360, 1268, 1152, 1072, 1044, 968, 845, 793, 802, 831, 713, 580, 611, 637, 532,

413, 360, 358, 350, 319, 300, 298, 298, 292, 281, 278, 293, 311, 312, 294, 270,
261, 276, 293, 287, 272, 268, 275, 283, 285, 284, 286, 295, 303, 302, 292, 280,
280, 294, 310, 309, 291, 277, 272, 273, 279, 281, 272, 253, 236, 222, 208, 202,
205, 206, 202, 200, 205, 214, 210, 198, 199, 216, 229, 232, 236, 252, 266, 281,
323, 383, 414, 438, 520, 618, 649, 694, 775, 795, 840, 954, 1052, 1127, 1194,
1247,

1306, 1433, 1536, 1608, 1719, 1786, 1809, 2010, 2118, 2155, 2282, 2389, 2439, 2563,
2663, 2651,

2703, 2823, 2935, 3001, 3063, 3127, 3213, 3247, 3277, 3360, 3437, 3471, 3482, 3510,
3554, 3598,

3623, 3626, 3641, 3693, 3758, 3798, 3813, 3837, 3871, 3898, 3921, 3946, 3965, 3976,
3980, 3980,

3979, 3986, 4004, 4016, 4017, 4010, 4001, 3995, 3995, 3999, 4009, 4021, 4024, 4020,
4011, 4003,

4006, 4012, 4018, 4020, 4022, 4023, 4022, 4020, 4021, 4026, 4033, 4035, 4034, 4032,
4032, 4033,

4034, 4032, 4028, 4026, 4029, 4035, 4040, 4042, 4040, 4036, 4033, 4032, 4034, 4036,
4041, 4045,

4049, 4051, 4051, 4048, 4045, 4043, 4044, 4047, 4050, 4053, 4058, 4063, 4067, 4067,
4063, 4056,

4052, 4052, 4057, 4063, 4068, 4071, 4071, 4069, 4067, 4064, 4061, 4062, 4065, 4070,
4074, 4077,

4078, 4081, 4083, 4086, 4086, 4086, 4085, 4087, 4089, 4090, 4089, 4086, 4083, 4082,
4084, 4086,

4087, 4087, 4087, 4086, 4082, 4076, 4075, 4081, 4087, 4082, 4066, 4046, 4031, 4019,
4011, 4000,

3967, 3896, 3799, 3731, 3715, 3690, 3598, 3487, 3412, 3361, 3294, 3215, 3145, 3079,
3022, 2962,

2919, 2865, 2777, 2703, 2661, 2579, 2476, 2401, 2287, 2159, 2092, 2008, 1890, 1807,
1744, 1674,

1568, 1447, 1387, 1228, 1183, 1134, 1009, 955, 888, 828, 772, 736, 734, 672, 615,
601,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,
4090, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,
4090, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,
4090, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,
4090, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,
4090, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4087,
4084, 4081,

4079, 4075, 4069, 4065, 4063, 4063, 4062, 4059, 4055, 4049, 4042, 4036, 4032, 4028,
4023, 4016,

4012, 4010, 4012, 4015, 4015, 4009, 3995, 3981, 3968, 3962, 3956, 3949, 3939, 3931,
3923, 3916,

3908, 3898, 3891, 3887, 3883, 3874, 3865, 3858, 3854, 3846, 3837, 3830, 3825, 3822,
3817, 3808,

3793, 3780, 3769, 3757, 3743, 3733, 3725, 3714, 3701, 3689, 3677, 3664, 3643, 3628,
3630, 3618,

3587, 3564, 3552, 3543, 3530, 3498, 3462, 3434, 3406, 3379, 3361, 3347, 3328, 3302,
3276, 3254,

3223, 3195, 3178, 3169, 3146, 3120, 3089, 3097, 3082, 3049, 3010, 2989, 2987, 2963,
2914, 2893,

2884, 2821, 2790, 2762, 2725, 2730, 2718, 2702, 2697, 2716, 2731, 2719, 2717, 2710,
2726, 2727,

2742, 2761, 2782, 2808, 2834, 2875, 2907, 2908, 2927, 2936, 2941, 2940, 2956, 2947,
2937, 2927,

2883, 2857, 2846, 2819, 2814, 2779, 2749, 2653, 2500, 2336, 2165, 1991, 1847, 1706,
1603, 1471,

1389, 1315, 1258, 1175, 1121, 1076, 1038, 994, 946, 908, 881, 848, 816, 774, 735,
700,

654, 616, 592, 565, 539, 514, 488, 459, 428, 405, 392, 375, 361, 350, 334, 315,

292, 264, 235, 211, 196, 185, 177, 166, 152, 135, 123, 111, 101, 93, 89, 84,
77, 69, 61, 57, 53, 49, 44, 40, 36, 33, 28, 21, 13, 7, 4, 3,
2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,
1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0,
0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 1, 3, 9,
13, 16, 15, 18, 24, 33, 40, 46, 53, 61, 68, 69, 68, 69, 75, 80,
83, 84, 88, 95, 105, 114, 121, 127, 131, 135, 143, 154, 164, 169, 171, 174,
181, 190, 201, 210, 218, 225, 231, 235, 238, 242, 249, 257, 262, 271, 285, 301,
309, 311, 312, 313, 318, 328, 342, 352, 364, 378, 391, 400, 405, 410, 419, 433,
445, 452, 454, 460, 469, 477, 482, 487, 496, 505, 510, 520, 536, 549, 555, 554,
547, 551, 567, 582, 599, 608, 612, 623, 632, 641, 650, 656, 663, 675, 691, 704,
718, 727, 739, 737, 721, 727, 740, 741, 759, 761, 763, 764, 774, 784, 779, 786,
798, 797, 806, 828, 836, 830, 825, 837, 852, 846, 856, 864, 864, 873, 858, 869,
881, 898, 927, 937, 942, 955, 998, 1021, 1032, 1050, 1077, 1085, 1116, 1146,
1169, 1185,
1250, 1274, 1312, 1343, 1401, 1450, 1513, 1595, 1710, 1918, 2236, 2420, 2494, 2584,
2648, 2708,
2758, 2830, 2865, 2913, 2935, 2970, 2995, 3013, 3032, 3052, 3062, 3077, 3091, 3109,
3115, 3130,

3150, 3176, 3177, 3193, 3211, 3232, 3257, 3279, 3293, 3303, 3318, 3341, 3356, 3374, 3395, 3410,

3429, 3444, 3453, 3468, 3487, 3505, 3522, 3536, 3543, 3544, 3553, 3573, 3592, 3605, 3619, 3631,

3636, 3638, 3643, 3645, 3652, 3669, 3685, 3693, 3704, 3719, 3728, 3727, 3729, 3741, 3753, 3761,

3769, 3781, 3791, 3797, 3802, 3805, 3807, 3813, 3824, 3830, 3836, 3844, 3852, 3853, 3851, 3850,

3851, 3853, 3859, 3869, 3876, 3881, 3886, 3891, 3895, 3899, 3906, 3915, 3921, 3927, 3933, 3939,

3942, 3945, 3948, 3952, 3958, 3966, 3971, 3972, 3971, 3971, 3973, 3976, 3979, 3985, 3990, 3997,

4000, 4001, 4001, 3999, 3999, 4000, 4001, 4002, 4004, 4009, 4015, 4022, 4022, 4022, 4021, 4023,

4024, 4025, 4029, 4032, 4034, 4033, 4032, 4033, 4034, 4032, 4029, 4028, 4029, 4029, 4025, 4023,

4024, 4030, 4035, 4039, 4044, 4050, 4058, 4061, 4062, 4060, 4060, 4061, 4063, 4062, 4058, 4053,

4049, 4049, 4050, 4053, 4059, 4064, 4068, 4067, 4065, 4064, 4065, 4062, 4056, 4050, 4050, 4056,

4064, 4066, 4061, 4053, 4047, 4048, 4049, 4050, 4046, 4042, 4039, 4041, 4042, 4043, 4036, 4029,

4021, 4015, 4010, 4006, 4004, 4002, 4002, 4002, 4001, 4000, 3996, 3993, 3990, 3993, 3994, 3994,

3988, 3985, 3983, 3985, 3984, 3981, 3977, 3977, 3977, 3976, 3972, 3966, 3960, 3959, 3958, 3956,

3953, 3953, 3958, 3963, 3959, 3952, 3946, 3947, 3947, 3942, 3927, 3918, 3912, 3912, 3910, 3908,

3905, 3901, 3896, 3891, 3887, 3879, 3870, 3865, 3864, 3868, 3865, 3860, 3853, 3849, 3841, 3832,

3824, 3823, 3826, 3825, 3822, 3820, 3819, 3815, 3813, 3811, 3807, 3797, 3788, 3784, 3779, 3774,

3772, 3772, 3767, 3764, 3765, 3763, 3757, 3750, 3748, 3747, 3742, 3735, 3729, 3720, 3706, 3694,

3686, 3680, 3673, 3666, 3659, 3652, 3647, 3641, 3625, 3610, 3605, 3598, 3586, 3577, 3570, 3563,

3559, 3558, 3556, 3551, 3544, 3534, 3519, 3504, 3499, 3496, 3486, 3477, 3469, 3463, 3457, 3453,

3444, 3426, 3424, 3421, 3413, 3400, 3376, 3372, 3374, 3363, 3347, 3329, 3317, 3317, 3318, 3299,

3275, 3262, 3258, 3248, 3230, 3204, 3196, 3176, 3158, 3125, 3099, 3082, 3071, 3021, 2977, 2953,

2904, 2865, 2808, 2741, 2708, 2621, 2518, 2412, 2241, 1998, 1752, 1622, 1512, 1416, 1367, 1318,

1272, 1229, 1198, 1170, 1147, 1129, 1112, 1082, 1072, 1069, 1054, 1035, 1031, 1009, 1002, 995,

980, 968, 969, 974, 966, 972, 968, 967, 961, 964, 964, 970, 977, 969, 958, 977,

991, 982, 989, 999, 1012, 1017, 1028, 1039, 1049, 1048, 1061, 1078, 1081, 1094, 1115, 1128,

1154, 1159, 1176, 1190, 1204, 1229, 1255, 1275, 1299, 1327, 1347, 1376, 1401, 1469, 1528, 1575,

1674, 1778, 1920, 1819, 1671, 1567, 1498, 1447, 1403, 1381, 1350, 1340, 1330, 1302, 1298, 1290,

1288, 1302, 1314, 1323, 1364, 1370, 1422, 1454, 1514, 1593, 1687, 1858, 2167, 2382, 2493, 2555,

2619, 2672, 2701, 2717, 2723, 2765, 2776, 2769, 2768, 2765, 2767, 2758, 2752, 2754, 2741, 2738,

2712, 2705, 2687, 2669, 2657, 2635, 2622, 2616, 2591, 2575, 2562, 2558, 2555, 2569, 2564, 2584,

2570, 2566, 2578, 2583, 2566, 2579, 2567, 2559, 2550, 2524, 2513, 2461, 2406, 2308, 1920, 1735,

1607, 1502, 1476, 1450, 1416, 1385, 1368, 1342, 1342, 1323, 1300, 1324, 1276, 1284, 1256, 1267,

1257, 1264, 1258, 1284, 1273, 1260, 1266, 1255, 1258, 1243, 1245, 1249, 1254, 1268, 1263, 1251,

1257, 1253, 1244, 1243, 1245, 1239, 1235, 1238, 1237, 1234, 1218, 1213, 1215, 1198, 1213, 1186,

1178, 1178, 1176, 1170, 1160, 1157, 1147, 1148, 1136, 1152, 1138, 1132, 1129, 1136, 1139, 1114,

1102, 1115, 1105, 1099, 1109, 1113, 1117, 1132, 1129, 1132, 1132, 1139, 1147, 1149, 1174, 1185,

1189, 1221, 1221, 1233, 1247, 1278, 1301, 1314, 1348, 1397, 1433, 1495, 1548, 1624, 1736, 1991,

2315, 2391, 2454, 2504, 2558, 2602, 2648, 2690, 2710, 2752, 2773, 2804, 2833, 2873, 2886, 2914,

2947, 2958, 2979, 3005, 3017, 3032, 3051, 3069, 3079, 3096, 3124, 3138, 3152, 3158, 3170, 3188,

3207, 3214, 3229, 3238, 3252, 3271, 3274, 3276, 3281, 3297, 3310, 3312, 3312, 3316, 3326, 3337,

3348, 3351, 3344, 3344, 3349, 3356, 3353, 3349, 3357, 3362, 3368, 3365, 3366, 3359, 3358, 3354,

3357, 3360, 3356, 3353, 3350, 3348, 3348, 3353, 3346, 3342, 3346, 3345, 3344, 3331, 3320, 3320,

3320, 3316, 3309, 3305, 3299, 3292, 3279, 3275, 3266, 3253, 3245, 3239, 3225, 3209, 3195, 3193,

3190, 3175, 3154, 3141, 3127, 3120, 3105, 3092, 3071, 3057, 3037, 3021, 3016, 2998, 2976, 2961,

2933, 2915, 2891, 2871, 2843, 2822, 2790, 2776, 2743, 2725, 2684, 2650, 2620, 2571, 2505, 2423,

2299, 1929, 1716, 1611, 1534, 1477, 1440, 1407, 1378, 1348, 1325, 1294, 1276, 1266, 1233, 1229,

1212, 1202, 1199, 1191, 1192, 1179, 1169, 1172, 1155, 1150, 1143, 1140, 1142, 1133, 1134, 1141,

1132, 1129, 1133, 1133, 1132, 1142, 1138, 1139, 1146, 1146, 1145, 1153, 1157, 1158, 1168, 1165,

1175, 1176, 1178, 1188, 1198, 1202, 1217, 1226, 1241, 1249, 1266, 1289, 1306, 1327, 1340, 1367,

1388, 1423, 1460, 1503, 1552, 1612, 1712, 1898, 2296, 2385, 2427, 2444, 2455, 2461, 2458, 2452,

2436, 2410, 2371, 2288, 1966, 1896, 1817, 1801, 1804, 1844, 2199, 2320, 2377, 2416, 2444, 2459,

2461, 2450, 2433, 2397, 2329, 2180, 1837, 1761, 1715, 1686, 1662, 1655, 1648, 1662, 1685, 1735,

1811, 2186, 2212, 2409, 2294, 2439, 2400, 2176, 1612, 1818, 1581, 2233, 1791, 2385, 1416, 1583,

2635, 2008, 2486, 2229, 2033, 2603, 2169, 2499, 1301, 1631, 1829, 1610, 1689, 1260, 632, 1179,

1339, 1332, 1606, 1884, 2568, 1778, 2386, 2967, 2642, 2571, 2554, 3022, 2903, 2488, 1921, 1861,

2441, 2840, 3271, 3580, 3526, 3482, 3464, 3195, 3162, 3487, 3410, 3247, 3144, 3179, 3537, 3921,

4072, 4088, 4090, 4084, 4081, 4035, 3784, 3693, 3928, 4074, 4082, 4088, 4090, 4071, 4009, 3982,

4020, 4065, 4087, 4090, 4088, 4074, 3837, 3487, 3393, 3532, 3693, 3872, 4057, 4088, 4039, 3763,

3426, 3458, 3484, 3389, 3429, 3555, 3550, 3233, 2979, 2848, 2494, 2429, 2705, 2890, 2661, 2428,

2417, 2380, 2390, 2384, 2149, 1785, 1633, 1862, 1964, 1603, 1426, 1466, 1124, 791, 891, 1067,

968, 652, 650, 1024, 1166, 1142, 1178, 1118, 882, 597, 593, 733, 464, 125, 45, 53,

53, 35, 20, 17, 6, 33, 179, 408, 429, 280, 141, 70, 35, 1, 19, 76,

130, 148, 109, 33, 6, 32, 102, 154, 124, 71, 20, 9, 5, 11, 37, 56,

56, 31, 10, 2, 10, 53, 87, 107, 81, 45, 11, 3, 3, 8, 16, 22,

26, 22, 13, 1, 5, 6, 3, 8, 24, 65, 63, 43, 12, 9, 17, 22,

26, 26, 14, 10, 9, 11, 4, 0, 1, 7, 12, 12, 3, 1, 10, 25,

38, 38, 28, 13, 1, 4, 40, 60, 53, 37, 43, 88, 124, 149, 164, 156,

133, 110, 94, 104, 121, 107, 97, 134, 217, 318, 357, 343, 395, 591, 829, 924,

865, 828, 918, 1100, 1255, 1304, 1341, 1519, 1856, 2184, 2296, 2192, 2113, 2246, 2510, 2749,

2857, 2863, 2876, 2877, 2950, 3113, 3147, 3187, 3313, 3365, 3385, 3398, 3422, 3600, 3741, 3588,

3361, 3381, 3579, 3662, 3661, 3757, 3840, 3786, 3662, 3670, 3783, 3863, 3909, 3939, 3892, 3815,

3825, 3885, 3902, 3871, 3834, 3825, 3832, 3830, 3842, 3880, 3886, 3862, 3842, 3854, 3904, 3942,

3909, 3819, 3712, 3637, 3641, 3651, 3635, 3664, 3745, 3837, 3839, 3818, 3822, 3856, 3897, 3902,

3837, 3721, 3631, 3634, 3752, 3839, 3870, 3907, 3953, 3987, 3997, 4007, 4022, 4001, 3976, 3998,

4023, 3990, 3917, 3880, 3952, 4042, 4058, 4046, 4021, 3985, 4019, 4079, 4005, 3791, 3657, 3649,

3726, 3780, 3692, 3612, 3620, 3588, 3536, 3548, 3515, 3336, 3114, 2929, 2774, 2688, 2689, 2703,

2668, 2603, 2508, 2351, 2212, 2127, 2006, 1834, 1676, 1587, 1560, 1499, 1398, 1321, 1142, 839,

661, 615, 566, 580, 548, 400, 369, 444, 423, 312, 212, 170, 186, 193, 214, 272,

279, 236, 204, 179, 177, 198, 216, 228, 202, 169, 195, 237, 244, 213, 173, 152,

154, 177, 213, 249, 267, 244, 195, 159, 143, 149, 156, 168, 179, 187, 190, 196,

201, 203, 201, 206, 223, 237, 237, 222, 188, 154, 131, 127, 128, 122, 108, 102,

103, 101, 85, 79, 85, 100, 110, 106, 97, 91, 98, 120, 132, 123, 85, 72,

83, 102, 112, 108, 80, 61, 65, 80, 90, 84, 59, 42, 45, 46, 44, 57,

72, 76, 76, 78, 81, 88, 99, 126, 173, 195, 216, 313, 470, 585, 593, 602,

745, 912, 976, 1022, 1122, 1222, 1235, 1204, 1288, 1461, 1555, 1596, 1678, 1801, 1945, 2101,

2258, 2367, 2375, 2324, 2327, 2448, 2647, 2825, 2911, 2903, 2854, 2830, 2874, 2970, 3061, 3129,

3258, 3490, 3678, 3698, 3645, 3651, 3713, 3749, 3772, 3870, 4010, 4081, 4088, 4074, 4059, 4063,

4082, 4089, 4078, 4072, 4077, 4083, 4083, 4079, 4080, 4084, 4088, 4089, 4086, 4080, 4080, 4085,

4090, 4090, 4090, 4090, 4090, 4089, 4079, 4064, 4047, 4047, 4055, 4069, 4078, 4083, 4085, 4086,

4085, 4084, 4081, 4082, 4085, 4088, 4089, 4086, 4082, 4080, 4087, 4090, 4090, 4090, 4090, 4088,

4087, 4088, 4081, 4075, 4070, 4061, 4050, 4038, 4035, 4049, 4069, 4089, 4090, 4090, 4089, 4090,

4090, 4090, 4090, 4090, 4090, 4090, 4089, 4083, 4074, 4083, 4089, 4090, 4090, 4089, 4086, 4083,

4082, 4087, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4086, 4085, 4088,

4090, 4090, 4089, 4086, 4088, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090, 4090,

4090, 4090, 4090, 4086, 4081, 4081, 4066, 3978, 3900, 3925, 3958, 3910, 3830, 3778, 3736, 3664,

3553, 3489, 3521, 3511, 3450, 3396, 3306, 3200, 3194, 3202, 3069, 2971, 2988, 2933, 2869, 2790,

2734, 2679, 2476, 2366, 2446, 2321, 2144, 2174, 2033, 1943, 1797, 1723, 1514, 1469, 1310, 1240,

1120, 1108, 1149, 1034, 894, 781, 820, 801, 644, 544, 457, 451, 555, 623, 539, 396,

378, 378, 445, 459, 431, 356, 366, 350, 354, 417, 386, 374, 413, 465, 413, 456, 539, 507, 480, 489, 507, 472, 454, 600, 765, 708, 523, 540, 730, 840, 820, 834, 819, 807, 709, 630, 741, 894, 764, 863, 905, 1051, 838, 1016, 1122, 1181, 1328, 1740,

1855, 1935, 2030, 2060, 2293, 2565, 2820, 2993, 2503, 2278, 1995, 2006, 1959, 1961, 1550, 1629,

1564, 1411, 1174, 1168, 1041, 975, 1016, 857, 834, 694, 851, 817, 652, 786, 826, 620,

509, 576, 631, 531, 428, 244, 172, 322, 371, 267, 123, 98, 246, 350, 290, 182, 133, 207, 345, 386, 372, 427, 439, 305, 158, 178, 355, 480, 430, 337, 344, 417,

468, 493, 502, 461, 384, 350, 403, 468, 466, 411, 350, 292, 243, 234, 281, 358,
446, 483, 419, 350, 321, 304, 317, 370, 419, 437, 421, 378, 362, 407, 441, 433,
427, 428, 429, 434, 447, 471, 424, 300, 339, 515, 607, 600, 569, 522, 463, 442,
469, 466, 535, 655, 703, 695, 687, 679, 622, 784, 980, 874, 728, 808, 813, 852,
1030, 1175, 1250, 1238, 1251, 1405, 1426, 1312, 1581, 1551, 1673, 1654, 1603, 1702,
1652, 1589,

1567, 1411, 1498, 1560, 1901, 1787, 1844, 2047, 2215, 2374, 2606, 2709, 2666, 2839,
2733, 2865,

2997, 3006, 3007, 3020, 3065, 3078, 3130, 3232, 3246, 3249, 3344, 3352, 3273, 3259,
3416, 3491,

3407, 3326, 3322, 3361, 3405, 3401, 3427, 3494, 3591, 3640, 3578, 3558, 3594, 3602,
3615, 3647,

3698, 3753, 3741, 3701, 3685, 3693, 3752, 3824, 3836, 3809, 3824, 3883, 3921, 3930,
3910, 3874,

3862, 3875, 3894, 3904, 3906, 3911, 3935, 3960, 3967, 3944, 3879, 3809, 3771, 3758,
3749, 3699,

3682, 3721, 3816, 3904, 3920, 3899, 3856, 3819, 3801, 3791, 3782, 3788, 3833, 3894,
3934, 3907,

3868, 3830, 3836, 3860, 3897, 3936, 3934, 3904, 3840, 3836, 3867, 3933, 3970, 3996,
4013, 4025,

4034, 4041, 4045, 4038, 4017, 3993, 3989, 4009, 4046, 4072, 4085, 4088, 4090, 4088,
4066, 4053,

4059, 4073, 4084, 4085, 4086, 4088, 4090, 4090, 4089, 4088, 4088, 4087, 4082, 4078,
4076, 4070,

4063, 4055, 4065, 4076, 4085, 4088, 4087, 4088, 4088, 4086, 4087, 4088, 4087, 4088,
4090, 4090,

4090, 4089, 4086, 4083, 4074, 4073, 4077, 4085, 4090, 4090, 4085, 4078, 4077, 4077,
4080, 4077,

4074, 4081, 4086, 4090, 4090, 4090, 4090, 4089, 4087, 4088, 4089, 4088, 4086, 4084,
4082, 4083,

4088, 4089, 4087, 4078, 4072, 4081, 4085, 4087, 4085, 4085, 4087, 4086, 4084, 4084,
4085, 4087,

4089, 4090, 4090, 4090, 4090, 4090, 4090, 4088, 4088, 4090, 4090, 4090, 4090, 4090,
4090, 4089,

4088, 4089, 4090, 4090, 4090, 4090, 4090, 4088, 4084, 4084, 4081, 4072, 4064, 4059,
4065, 4073,

4084, 4087, 4088, 4086, 4088, 4090, 4090, 4090, 4090, 4089, 4088, 4086, 4089, 4090,
4089, 4089,

4090, 4090, 4088, 4083, 4077, 4074, 4073, 4080, 4086, 4090, 4090, 4090, 4090, 4090,
4086, 4078,

4043, 4034, 4046, 4073, 4057, 4013, 3949, 3923, 3914, 3914, 3898, 3898, 3920, 3944,
3933, 3903,

3869, 3871, 3895, 3925, 3921, 3855, 3795, 3767, 3804, 3857, 3891, 3913, 3924, 3902,
3871, 3870,

3887, 3902, 3903, 3891, 3877, 3873, 3885, 3901, 3902, 3875, 3832, 3778, 3728, 3694,
3685, 3693,

3679, 3657, 3649, 3622, 3582, 3535, 3464, 3448, 3515, 3568, 3526, 3377, 3285, 3288,
3298, 3306,

3346, 3404, 3340, 3330, 3316, 3208, 3129, 3154, 3270, 3079, 3017, 3015, 2946, 2915,
2904, 2712,

2839, 2711, 2817, 2754, 2623, 2725, 2531, 2553, 2587, 2508, 2339, 2251, 2331, 2158,
2064, 1990,

1886, 1762, 1538, 1524, 1416, 1329, 1325, 1246, 1250, 1128, 1049, 1036, 976, 891,
853, 833,

800, 788, 743, 630, 519, 477, 485, 495, 458, 361, 279, 246, 203, 124, 62, 50,

72, 86, 64, 20, 6, 6, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 1, 4, 6, 5, 2, 1, 0, 0, 0, 0, 1, 2, 3,

3, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

0, 0, 0, 1, 2, 4, 3, 0, 0, 0, 0, 0, 1, 0, 0, 0,

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,
4, 6, 10, 8, 6, 3, 3, 3, 2, 1, 0, 1, 2, 6, 10, 9,
6, 1, 0, 0, 0, 1, 2, 1, 1, 0, 0, 0, 0, 0, 0, 0,
0, 1, 2, 5, 6, 4, 2, 0, 0, 0, 0, 0, 0, 1, 2, 3,
3, 2, 2, 2, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1,
2, 1, 0, 0, 0, 0, 0, 0, 1, 2, 3, 5, 5, 3, 2, 0,
0, 0, 0, 0, 0, 0, 0, 1, 2, 2, 2, 1, 1, 1, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0,
0, 0, 0, 1, 1, 2, 2, 2, 1, 0, 0, 0, 1, 3, 2, 1,
0, 0, 0, 0, 0, 2, 7, 15, 23, 27, 31, 39, 65, 108, 157, 201,
226, 232, 222, 205, 196, 216, 250, 280, 295, 292, 301, 333, 353, 355, 348, 333,
313, 318, 363, 402, 427, 469, 509, 538, 558, 557, 547, 550, 558, 562, 562, 611,
710, 763, 780, 792, 808, 852, 867, 922, 1006, 999, 981, 971, 908, 999, 1052,
1056,
1240, 1153, 1245, 1304, 1360, 1552, 1615, 1794, 1948, 2196, 2294, 2529, 2546, 2650,
2629, 2612,
2672, 2680, 2617, 2708, 2626, 2516, 2432, 2467, 2527, 2553, 2548, 2581, 2656, 2645,
2627, 2674,
2710, 2703, 2797, 2838, 2848, 2908, 2789, 2832, 2887, 2889, 2906, 2987, 3030, 2986,
3016, 3110,
3138, 3138, 3205, 3185, 3176, 3249, 3235, 3266, 3327, 3273, 3218, 3270, 3298, 3275,
3301, 3299,
3271, 3304, 3369, 3358, 3365, 3411, 3422, 3407, 3404, 3413, 3393, 3422, 3407, 3403,
3434, 3430,
3399, 3421, 3427, 3375, 3326, 3344, 3355, 3380, 3419, 3396, 3355, 3359, 3369, 3362,
3346, 3351,
3408, 3394, 3339, 3332, 3363, 3355, 3333, 3341, 3360, 3335, 3270, 3259, 3286, 3319,
3286, 3243,
3230, 3224, 3255, 3237, 3198, 3187, 3189, 3174, 3220, 3246, 3189, 3123, 3116, 3197,
3257, 3215,

3170, 3152, 3168, 3173, 3233, 3205, 3138, 3167, 3172, 3178, 3187, 3156, 3142, 3196, 3194, 3178,

3201, 3193, 3167, 3186, 3255, 3231, 3189, 3166, 3176, 3220, 3250, 3242, 3229, 3224, 3237, 3246,

3291, 3306, 3243, 3200, 3239, 3316, 3321, 3291, 3305, 3315, 3297, 3301, 3320, 3344, 3382, 3386,

3350, 3332, 3357, 3352, 3343, 3423, 3496, 3494, 3496, 3481, 3455, 3463, 3474, 3474, 3473, 3469,

3462, 3443, 3417, 3438, 3460, 3478, 3524, 3570, 3539, 3480, 3473, 3483, 3470, 3436, 3417, 3423,

3403, 3351, 3341, 3378, 3385, 3346, 3342, 3357, 3345, 3307, 3306, 3288, 3286, 3305, 3278, 3240,

3182, 3149, 3160, 3124, 3125, 3056, 3024, 2967, 2958, 2885, 2834, 2870, 2749, 2695, 2541, 2447,

2231, 2075, 1945, 1742, 1628, 1496, 1364, 1231, 1264, 1143, 1136, 1129, 1061, 1066, 981, 975,

937, 894, 871, 875, 829, 749, 705, 706, 723, 723, 689, 644, 652, 667, 656, 667, 673, 647, 639, 650, 646, 619, 590, 599, 629, 642, 626, 587, 558, 556, 571, 578, 560, 547, 545, 544, 548, 553, 549, 555, 569, 568, 555, 519, 488, 512, 568, 612, 617, 596, 589, 620, 659, 658, 632, 624, 643, 653, 649, 646, 644, 656, 648, 634, 640, 641, 618, 639, 682, 683, 677, 675, 666, 682, 704, 708, 722, 744, 747, 725, 736, 774, 800, 817, 815, 814, 845, 856, 862, 893, 914, 915, 931, 973, 998, 1004,

1038, 1068, 1098, 1110, 1135, 1187, 1216, 1221, 1252, 1274, 1323, 1359, 1413, 1418, 1498, 1520,

1559, 1575, 1794, 2026, 2341, 2327, 2092, 1846, 2272, 2123, 2275, 2408, 2414, 2430, 2393, 2300,

2185, 1946, 1743, 1628, 1580, 1523, 1473, 1407, 1356, 1314, 1274, 1233, 1212, 1189, 1173, 1143,

1137, 1138, 1113, 1080, 1088, 1072, 1056, 1051, 1055, 1035, 1038, 1030, 1029, 1025, 1040, 1043,

1046, 1051, 1064, 1077, 1078, 1096, 1098, 1108, 1120, 1130, 1150, 1169, 1186, 1217, 1238, 1266,

1288, 1306, 1351, 1387, 1428, 1490, 1539, 1590, 1677, 1810, 2161, 2348, 2439, 2510, 2550, 2593,

2638, 2667, 2688, 2707, 2733, 2738, 2756, 2759, 2774, 2772, 2798, 2795, 2798, 2788, 2784, 2771,

2755, 2741, 2714, 2696, 2685, 2646, 2611, 2587, 2555, 2514, 2466, 2420, 2370, 2368, 2394, 2436,

2473, 2497, 2518, 2524, 2533, 2542, 2530, 2512, 2482, 2402, 2238, 1823, 1714, 1658, 1623, 1583,

1552, 1517, 1493, 1480, 1462, 1433, 1419, 1400, 1383, 1364, 1348, 1335, 1327, 1316, 1309, 1306,

1300, 1299, 1296, 1288, 1295, 1290, 1301, 1309, 1320, 1337, 1345, 1358, 1372, 1395, 1408, 1435,

1461, 1494, 1530, 1566, 1611, 1686, 1744, 1835, 2186, 2369, 2454, 2516, 2559, 2588, 2620, 2642,

2663, 2683, 2690, 2693, 2694, 2687, 2685, 2671, 2660, 2646, 2628, 2604, 2583, 2550, 2514, 2471,

2415, 2341, 2237, 1860, 1843, 1913, 1882, 1903, 1901, 2250, 2389, 2370, 2480, 2407, 2535, 2475,

2457, 2445, 2458, 2406, 2430, 2369, 2364, 2331, 2159, 1718, 1757, 1721, 1598, 1545, 1581, 1562,

1489, 1469, 1453, 1505, 1479, 1477, 1453, 1489, 1516, 1487, 1479, 1494, 1566, 1558, 1610, 1608,

1654, 1724, 1854, 2192, 2327, 2391, 2408, 2400, 2433, 2453, 2416, 2438, 2409, 2409, 2405, 2370,

2375, 2354, 2324, 2230, 1848, 1750, 1700, 1671, 1676, 1688, 1680, 1681, 1698, 1730, 1828, 2217,

2250, 2224, 1906, 1876, 2178, 2277, 2285, 2242, 1971, 1832, 1810, 1806, 1827, 1934, 2235, 2275,

2257, 2202, 1902, 1857, 1890, 2180, 1944, 1965, 1993, 1969, 2090, 1837, 2261, 1869, 1993, 1470,

2139, 2472, 1518, 1610, 1781, 1718, 1649, 954, 1135, 1164, 1132, 1584, 1861, 2520, 1836, 2304,

2981, 2610, 2541, 2559, 3046, 2878, 2425, 1878, 1909, 2482, 2825, 3264, 3569, 3522, 3496, 3470,

3180, 3172, 3480, 3377, 3229, 3153, 3221, 3588, 3944, 4075, 4065, 4076, 4072, 4074, 4022, 3783,

3715, 3951, 4077, 4073, 4082, 4080, 4085, 4039, 4040, 4075, 4076, 4074, 4089, 4075, 4037, 3786,

3456, 3387, 3525, 3717, 3904, 4058, 4082, 4049, 3746, 3412, 3458, 3469, 3386, 3435, 3534, 3478,

3169, 2966, 2815, 2451, 2435, 2743, 2878, 2634, 2416, 2427, 2397, 2407, 2351, 2088, 1732, 1639,

1892, 1931, 1582, 1426, 1415, 1041, 771, 911, 1066, 906, 611, 717, 1106, 1160, 1162, 1201,

1102, 811, 552, 618, 652, 358, 92, 43, 67, 72, 42, 17, 18, 22, 75, 245,

444, 434, 249, 147, 81, 34, 15, 43, 92, 133, 134, 97, 37, 25, 53, 100,

147, 110, 66, 28, 16, 7, 13, 35, 46, 32, 10, 6, 8, 22, 59, 80,

85, 68, 40, 18, 7, 15, 20, 12, 9, 8, 12, 9, 5, 9, 8, 2,

13, 33, 63, 45, 24, 7, 5, 4, 1, 0, 4, 13, 8, 4, 0, 0,

0, 0, 2, 5, 11, 18, 19, 15, 9, 5, 10, 28, 25, 7, 1, 17,

30, 28, 24, 41, 83, 103, 137, 162, 155, 120, 92, 88, 103, 119, 99, 91,

129, 194, 277, 333, 337, 372, 537, 779, 904, 847, 798, 889, 1062, 1224, 1290, 1328,

1480, 1777, 2122, 2299, 2211, 2090, 2181, 2443, 2694, 2821, 2851, 2871, 2894, 2948, 3069, 3126,

3139, 3263, 3365, 3383, 3402, 3417, 3537, 3699, 3633, 3409, 3360, 3541, 3653, 3633, 3712, 3805,

3782, 3672, 3643, 3776, 3865, 3903, 3925, 3906, 3835, 3815, 3876, 3902, 3868, 3831, 3821, 3838,

3855, 3871, 3885, 3887, 3871, 3849, 3850, 3905, 3933, 3913, 3845, 3740, 3624, 3638, 3659, 3658,

3660, 3716, 3787, 3821, 3829, 3833, 3853, 3897, 3907, 3872, 3780, 3663, 3618, 3736, 3811, 3841,

3867, 3914, 3959, 3979, 4001, 4029, 4011, 3973, 3980, 4014, 4002, 3925, 3879, 3936, 4013, 4032,

4043, 4052, 4016, 3999, 4048, 4005, 3822, 3674, 3638, 3698, 3768, 3709, 3617, 3627, 3598, 3539,

3573, 3553, 3362, 3126, 2949, 2794, 2682, 2685, 2722, 2685, 2609, 2528, 2383, 2228, 2129, 2003,

1838, 1681, 1579, 1571, 1523, 1399, 1321, 1181, 895, 687, 606, 541, 558, 553, 413, 364,

432, 416, 307, 214, 183, 191, 177, 217, 292, 274, 217, 192, 172, 188, 229, 256, 242, 196, 173, 201, 247, 258, 229, 194, 183, 181, 196, 215, 236, 253, 259, 211, 172, 150, 145, 141, 154, 173, 192, 206, 214, 216, 210, 206, 209, 220, 237, 239, 228, 202, 172, 154, 147, 143, 118, 93, 89, 106, 109, 98, 88, 85, 93, 116, 132, 114, 94, 92, 118, 117, 100, 78, 70, 85, 105, 114, 105, 81, 67, 71, 93, 103, 90, 63, 47, 50, 50, 54, 72, 74, 63, 80, 97, 89, 81, 102, 147, 192, 205, 233, 359, 525, 617, 623, 675, 823, 930, 981, 1053, 1153, 1221, 1202,

1208, 1343, 1495, 1556, 1608, 1713, 1845, 2021, 2206, 2342, 2389, 2339, 2300, 2343, 2500, 2727,

2886, 2922, 2869, 2821, 2833, 2913, 3031, 3096, 3168, 3358, 3585, 3691, 3661, 3647, 3686, 3722,

3757, 3812, 3934, 4060, 4084, 4089, 4079, 4060, 4067, 4081, 4085, 4082, 4081, 4083, 4084, 4083,

4082, 4082, 4083, 4078, 4070, 4074, 4083, 4090, 4088, 4086, 4084, 4083, 4085, 4087, 4087, 4082,

4073, 4066, 4062, 4067, 4070, 4072, 4068, 4070, 4076, 4083, 4087, 4088, 4084, 4083, 4087, 4087,

4086, 4083, 4078, 4073, 4070, 4069, 4071, 4077, 4082, 4083, 4083, 4083, 4080, 4075, 4064, 4057,

4051, 4055, 4063, 4078, 4082, 4081, 4073, 4068, 4070, 4076, 4084, 4087, 4088, 4087, 4088, 4088,

4087, 4086, 4087, 4084, 4081, 4080, 4085, 4089, 4087, 4088, 4088, 4089, 4089, 4088,
4087, 4088,

4089, 4090, 4090, 4089, 4083, 4080, 4080, 4086, 4087, 4086, 4082, 4085, 4087, 4086,
4079, 4077,

4077, 4079, 4088, 4088, 4084, 4083, 4087, 4081, 4084, 4086, 4085, 4084, 4087, 4090,
4083, 3972,

3893, 3932, 3964, 3903, 3812, 3761, 3736, 3663, 3539, 3459, 3494, 3506, 3432, 3376,
3299, 3184,

3170, 3198, 3071, 2964, 2979, 2937, 2873, 2779, 2720, 2657, 2457, 2361, 2432, 2296,
2135, 2165,

2011, 1945, 1776, 1702, 1480, 1478, 1281, 1242, 1105, 1118, 1116, 996, 879, 786,
845, 778,

624, 540, 441, 472, 580, 601, 488, 377, 376, 408, 458, 460, 409, 342, 380, 355,
390, 421, 370, 384, 435, 458, 407, 487, 543, 499, 471, 488, 504, 457, 497, 654,
782, 653, 510, 569, 773, 843, 829, 835, 816, 801, 676, 635, 802, 857, 774, 873,
956, 1024, 809, 1081, 1109, 1227, 1426, 1814, 1826, 1927, 2090, 2071, 2312, 2509,
2731, 2707,

2315, 2159, 2227, 1888, 2023, 1932, 1552, 1573, 1561, 1353, 1136, 1193, 965, 1021,
951, 860,

792, 743, 891, 744, 670, 835, 780, 574, 519, 623, 594, 497, 386, 203, 235, 358,
339, 199, 63, 135, 328, 357, 252, 163, 167, 278, 379, 390, 404, 445, 393, 246,
168, 264, 435, 483, 380, 326, 395, 452, 482, 513, 486, 411, 359, 383, 451, 483,
444, 376, 322, 262, 229, 262, 330, 416, 473, 455, 387, 333, 308, 306, 344, 400,
431, 437, 402, 358, 385, 429, 428, 430, 438, 432, 438, 453, 463, 436, 342, 323,
467, 586, 599, 579, 538, 481, 426, 441, 452, 481, 611, 700, 714, 674, 689, 644,
703, 924, 945, 743, 756, 823, 806, 988, 1092, 1275, 1203, 1252, 1362, 1448, 1365,
1342,

1577, 1577, 1730, 1564, 1686, 1573, 1669, 1580, 1584, 1454, 1494, 1852, 1805, 1823,
1858, 2117,

2329, 2473, 2670, 2688, 2834, 2731, 2809, 2875, 3094, 2938, 3058, 3036, 3079, 3096,
3182, 3250,

3239, 3298, 3360, 3330, 3259, 3318, 3477, 3476, 3369, 3309, 3328, 3379, 3407, 3396, 3446, 3530,

3613, 3618, 3561, 3570, 3597, 3597, 3622, 3664, 3721, 3764, 3733, 3689, 3675, 3695, 3776, 3841,

3828, 3814, 3849, 3897, 3922, 3922, 3902, 3869, 3866, 3885, 3893, 3889, 3888, 3905, 3944, 3969,

3962, 3921, 3853, 3783, 3765, 3756, 3732, 3681, 3676, 3754, 3848, 3921, 3920, 3882, 3841, 3809,

3802, 3790, 3783, 3787, 3852, 3902, 3925, 3884, 3848, 3829, 3835, 3871, 3911, 3949, 3929, 3882,

3832, 3841, 3890, 3946, 3977, 3994, 4008, 4025, 4036, 4043, 4045, 4028, 4008, 3988, 4001, 4026,

4062, 4076, 4084, 4089, 4086, 4076, 4055, 4058, 4070, 4083, 4084, 4081, 4086, 4089, 4090, 4089,

4089, 4090, 4088, 4083, 4076, 4074, 4071, 4067, 4058, 4053, 4067, 4077, 4085, 4088, 4090, 4090,

4088, 4087, 4089, 4089, 4088, 4088, 4088, 4087, 4089, 4090, 4090, 4083, 4074, 4077, 4081, 4086,

4088, 4089, 4086, 4083, 4080, 4080, 4080, 4077, 4080, 4087, 4089, 4090, 4090, 4090, 4090, 4086,

4083, 4085, 4086, 4085, 4086, 4088, 4089, 4090, 4090, 4089, 4081, 4064, 4068, 4082, 4087, 4090,

4089, 4089, 4089, 4086, 4086, 4089, 4089, 4087, 4084, 4084, 4086, 4087, 4088, 4089, 4089, 4089,

4089, 4088, 4088, 4089, 4090, 4090, 4089, 4086, 4086, 4087, 4089, 4090, 4090, 4090, 4090, 4089,

4089, 4090, 4080, 4068, 4060, 4060, 4064, 4074, 4081, 4084, 4085, 4085, 4086, 4085, 4084, 4087,

4088, 4089, 4089, 4090, 4090, 4088, 4086, 4087, 4088, 4089, 4090, 4089, 4086, 4078, 4079, 4086,

4086, 4087, 4089, 4087, 4086, 4086, 4081, 4061, 4030, 4038, 4057, 4075, 4031, 3976, 3927, 3921,

3921, 3913, 3890, 3904, 3924, 3941, 3916, 3885, 3869, 3880, 3918, 3918, 3879, 3804,
3763, 3789,

3837, 3880, 3898, 3916, 3911, 3888, 3864, 3873, 3894, 3901, 3895, 3881, 3866, 3866,
3893, 3902,

3884, 3843, 3790, 3742, 3708, 3683, 3678, 3680, 3663, 3651, 3634, 3595, 3555, 3492,
3442, 3482,

3553, 3561, 3436, 3305, 3282, 3292, 3300, 3308, 3394, 3363, 3313, 3334, 3247, 3133,
3140, 3239,

3169, 2994, 3036, 2959, 2928, 2898, 2815, 2774, 2749, 2787, 2798, 2622, 2693, 2610,
2784, 2621,

2489, 2329, 2247, 2242, 2295, 2186, 2006, 1973, 1777, 1619, 1536, 1495, 1334, 1340,
1279, 1262,

1189, 1047, 1042, 1012, 923, 860, 851, 815, 792, 771, 678, 549, 477, 475, 498,
479,

393, 296, 249, 222, 155, 73, 44, 67, 101, 81, 36, 8, 0, 1, 0, 1,

1, 1, 0, 0, 0, 0, 1, 3, 2, 2, 2, 5, 4, 4, 4, 1,

0, 0, 0, 0, 0, 0, 1, 1, 2, 2, 1, 0, 0, 1, 1, 1,

1, 1, 1, 2, 5, 7, 7, 4, 0, 0, 0, 0, 1, 2, 2, 2,

1, 0, 0, 0, 0, 0, 1, 3, 2, 2, 2, 2, 2, 3, 4, 3,

2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,

1, 2, 3, 2, 2, 1, 1, 2, 3, 3, 4, 5, 6, 4, 3, 2,

1, 1, 0, 0, 0, 2, 5, 4, 3, 2, 1, 0, 0, 0, 0, 1,

1, 2, 2, 1, 1, 1, 1, 1, 0, 1, 3, 4, 4, 3, 3, 2,

3, 3, 3, 3, 3, 4, 5, 4, 3, 2, 1, 1, 1, 3, 7, 7,

6, 3, 2, 1, 2, 2, 3, 4, 5, 6, 7, 5, 2, 1, 2, 5,

7, 9, 9, 9, 7, 5, 3, 2, 1, 1, 2, 3, 5, 5, 5, 3,

2, 3, 4, 5, 5, 5, 5, 4, 2, 2, 3, 7, 8, 8, 5, 3,

2, 2, 4, 4, 3, 3, 2, 3, 2, 2, 2, 1, 1, 2, 4, 4,

4, 2, 2, 1, 2, 3, 3, 3, 3, 4, 5, 5, 3, 1, 0, 1,

1, 2, 2, 1, 0, 0, 0, 1, 2, 2, 2, 1, 1, 0, 0, 1,

2, 1, 0, 1, 1, 2, 2, 3, 3, 1, 1, 0, 1, 3, 6, 6,
5, 3, 1, 0, 0, 0, 1, 2, 3, 2, 1, 0, 0, 4, 12, 25,
30, 33, 39, 66, 108, 155, 196, 227, 235, 225, 204, 193, 212, 249, 282, 298,
292, 299, 337, 359, 360, 349, 331, 312, 320, 364, 404, 432, 478, 522, 552, 570,
562, 547, 546, 554, 559, 564, 616, 711, 760, 780, 792, 807, 854, 868, 927,
1000,
996, 982, 967, 915, 998, 1055, 1070, 1241, 1149, 1246, 1301, 1359, 1565, 1607,
1778, 1940,
2200, 2327, 2551, 2533, 2641, 2621, 2625, 2649, 2684, 2646, 2726, 2639, 2570, 2451,
2480, 2495,
2548, 2519, 2573, 2655, 2641, 2612, 2686, 2736, 2699, 2803, 2814, 2844, 2909, 2785,
2852, 2892,
2884, 2916, 2997, 3017, 2987, 3027, 3134, 3129, 3147, 3207, 3171, 3189, 3252, 3228,
3283, 3325,
3253, 3227, 3282, 3293, 3274, 3302, 3291, 3271, 3319, 3369, 3355, 3366, 3418, 3421,
3403, 3409,
3411, 3394, 3424, 3405, 3408, 3435, 3421, 3398, 3430, 3423, 3357, 3322, 3352, 3364,
3394, 3422,
3385, 3351, 3361, 3367, 3359, 3347, 3367, 3416, 3383, 3339, 3348, 3365, 3353, 3337,
3347, 3357,
3326, 3263, 3266, 3298, 3317, 3278, 3242, 3228, 3227, 3256, 3234, 3193, 3187, 3185,
3174, 3232,
3233, 3177, 3114, 3125, 3214, 3246, 3210, 3164, 3148, 3172, 3175, 3246, 3183, 3141,
3166, 3165,
3178, 3191, 3151, 3150, 3199, 3186, 3182, 3199, 3186, 3158, 3197, 3257, 3217, 3188,
3164, 3183,
3231, 3252, 3239, 3222, 3226, 3236, 3253, 3306, 3299, 3228, 3194, 3261, 3322, 3312,
3293, 3311,
3307, 3293, 3304, 3319, 3350, 3388, 3382, 3342, 3333, 3359, 3342, 3356, 3451, 3504,
3494, 3499,
3474, 3457, 3463, 3470, 3468, 3471, 3470, 3462, 3433, 3418, 3448, 3463, 3480, 3536,
3570, 3523,

3476, 3476, 3480, 3462, 3431, 3423, 3427, 3388, 3339, 3350, 3387, 3377, 3340, 3343, 3357, 3335,

3306, 3305, 3286, 3294, 3297, 3270, 3229, 3163, 3160, 3151, 3125, 3110, 3052, 3010, 2968, 2954,

2870, 2839, 2833, 2746, 2675, 2521, 2408, 2200, 2047, 1857, 1739, 1598, 1475, 1347, 1210, 1252,

1128, 1149, 1099, 1067, 1043, 980, 966, 932, 887, 869, 866, 810, 736, 701, 706, 718,

722, 680, 640, 658, 665, 651, 669, 672, 641, 636, 652, 640, 609, 589, 602, 630, 639, 616, 575, 553, 557, 575, 576, 553, 544, 544, 543, 548, 552, 547, 552, 567, 564, 548, 512, 485, 510, 567, 612, 614, 592, 591, 624, 660, 656, 629, 623, 642, 648, 645, 644, 643, 654, 646, 635, 639, 635, 616, 642, 680, 681, 676, 673, 665, 679, 701, 708, 723, 744, 749, 725, 731, 770, 798, 814, 814, 813, 841, 855, 863, 894, 912, 913, 930, 971, 998, 1002, 1033, 1068, 1095, 1107, 1136, 1178, 1218, 1215, 1249,

1271, 1324, 1350, 1415, 1414, 1503, 1504, 1548, 1591, 1784, 1996, 2282, 2306, 2108, 1818, 2271,

2115, 2298, 2401, 2419, 2427, 2399, 2291, 2188, 1951, 1747, 1636, 1584, 1525, 1476, 1408, 1360,

1314, 1275, 1234, 1211, 1191, 1174, 1143, 1137, 1137, 1114, 1081, 1090, 1073, 1057, 1050, 1054,

1035, 1037, 1031, 1030, 1025, 1039, 1041, 1045, 1051, 1064, 1077, 1078, 1093, 1097, 1108, 1119,

1129, 1148, 1169, 1185, 1215, 1236, 1265, 1286, 1302, 1350, 1383, 1426, 1487, 1535, 1581, 1669,

1798, 2105, 2342, 2433, 2505, 2546, 2594, 2638, 2665, 2687, 2706, 2733, 2737, 2756, 2758, 2776,

2771, 2798, 2795, 2800, 2789, 2786, 2772, 2758, 2743, 2717, 2702, 2688, 2651, 2619, 2593, 2559,

2521, 2468, 2420, 2375, 2355, 2389, 2433, 2465, 2491, 2518, 2524, 2531, 2540, 2534, 2515, 2489,

2415, 2265, 1843, 1726, 1664, 1627, 1589, 1556, 1521, 1495, 1482, 1466, 1436, 1421, 1402, 1385,

1365, 1349, 1337, 1328, 1318, 1310, 1305, 1301, 1298, 1297, 1287, 1295, 1291, 1299, 1308, 1317,

1337, 1344, 1355, 1371, 1393, 1408, 1432, 1457, 1491, 1526, 1562, 1606, 1676, 1739, 1825, 2119,

2354, 2446, 2510, 2556, 2586, 2615, 2641, 2660, 2682, 2689, 2692, 2695, 2688, 2685, 2673, 2662,

2647, 2629, 2609, 2588, 2555, 2519, 2479, 2425, 2353, 2254, 1885, 1839, 1901, 1893, 1906, 1889,

2205, 2321, 2430, 2412, 2513, 2485, 2486, 2455, 2445, 2470, 2432, 2405, 2371, 2383, 2371, 2192,

1709, 1724, 1660, 1623, 1555, 1595, 1568, 1499, 1458, 1442, 1505, 1490, 1478, 1453, 1488, 1509,

1485, 1477, 1495, 1566, 1552, 1611, 1604, 1651, 1717, 1820, 2192, 2313, 2398, 2396, 2405, 2430,

2454, 2414, 2439, 2410, 2408, 2403, 2369, 2376, 2355, 2323, 2231, 1846, 1749, 1699, 1669, 1678,

1689, 1680, 1680, 1697, 1731, 1828, 2218, 2250, 2220, 1906, 1876, 2182, 2275, 2288, 2249, 1962,

1834, 1811, 1805, 1828, 1931, 2236, 2264, 2262, 2200, 1901, 1854, 1900, 2183, 1949, 1960, 1988,

1970, 2054, 2040, 2036, 1903, 1859, 1825, 1922, 1883, 1857, 1453, 773, 339, 247, 154, 291,

382, 573, 741, 1062, 1427, 1453, 1501, 1500, 1347, 1218, 972, 751, 859, 1107, 1424, 1687,

1826, 1908, 2024, 2350, 3001, 3690, 3941, 3870, 3410, 2881, 2412, 2106, 1903, 1698, 1517, 1376,

1391, 1621, 1965, 2227, 2288, 2228, 2240, 2439, 2810, 3269, 3707, 3994, 4081, 4088, 4088, 4059,

4026, 4026, 4062, 4088, 4087, 4086, 4090, 4090, 4089, 4086, 4086, 4087, 4087, 4085, 4087, 4088,

4088, 4089, 4088, 4088, 4088, 4088, 4090, 4090, 4090, 4090, 4088, 4086, 4087, 4087,
4089, 4089,

4090, 4088, 4054, 4010, 3955, 3846, 3703, 3538, 3355, 3176, 3005, 2826, 2644, 2450,
2216, 1976,

1773, 1578, 1379, 1203, 1086, 974, 792, 615, 488, 360, 250, 257, 342, 440, 572,
823,

1083, 1308, 1529, 1824, 2172, 2341, 2446, 2585, 2591, 2516, 2592, 2797, 2882, 2828,
2741, 2673,

2653, 2667, 2699, 2655, 2472, 2180, 1815, 1421, 1057, 751, 515, 399, 376, 343,
257, 166,

116, 85, 65, 40, 14, 3, 1, 4, 4, 3, 3, 4, 4, 6, 2, 1,

0, 1, 3, 5, 4, 2, 2, 1, 0, 0, 1, 1, 0, 0, 0, 2,

2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 2, 3, 3, 2,

1, 0, 0, 1, 2, 2, 3, 4, 3, 12, 35, 65, 104, 168, 223, 272,

323, 372, 413, 450, 468, 457, 449, 454, 461, 451, 420, 385, 366, 352, 326, 321,
346, 390, 434, 481, 551, 645, 733, 809, 871, 934, 1009, 1096, 1198, 1322, 1463,
1595,

1690, 1781, 1895, 1993, 2073, 2150, 2238, 2356, 2507, 2658, 2771, 2874, 2990, 3099,
3207, 3315,

3428, 3560, 3704, 3814, 3893, 3992, 4062, 4084, 4084, 4083, 4085, 4082, 4084, 4084,
4082, 4081,

4083, 4083, 4083, 4082, 4082, 4083, 4087, 4083, 4082, 4086, 4090, 4090, 4090, 4085,
4081, 4081,

4087, 4088, 4088, 4087, 4086, 4088, 4090, 4090, 4090, 4089, 4085, 4084, 4085, 4089,
4090, 4089,

4086, 4082, 4079, 4080, 4083, 4087, 4089, 4090, 4089, 4088, 4087, 4088, 4088, 4086,
4081, 4078,

4080, 4081, 4082, 4082, 4083, 4086, 4086, 4087, 4089, 4090, 4090, 4090, 4089, 4086,
4087, 4087,

4084, 4083, 4084, 4089, 4090, 4090, 4089, 4089, 4090, 4087, 4087, 4087, 4088, 4086,
4087, 4089,

4090, 4090, 4089, 4087, 4080, 4080, 4080, 4081, 4082, 4079, 4074, 4073, 4075, 4079, 4084, 4086,

4086, 4079, 4071, 4065, 4060, 4045, 4014, 4005, 3990, 3890, 3781, 3710, 3639, 3587, 3559, 3539,

3510, 3438, 3297, 3122, 2979, 2878, 2751, 2548, 2318, 2117, 1937, 1751, 1608, 1533, 1456, 1390,

1325, 1219, 1137, 1127, 1143, 1149, 1143, 1111, 1075, 1062, 1033, 966, 907, 862, 807, 762,

760, 831, 950, 1007, 914, 751, 624, 472, 299, 191, 138, 95, 65, 42, 21, 28,

74, 97, 37, 3, 0, 5, 12, 15, 9, 3, 3, 4, 5, 5, 4, 5,

7, 13, 22, 12, 6, 13, 66, 101, 115, 90, 64, 40, 18, 9, 5, 6,

7, 7, 9, 14, 15, 11, 5, 1, 4, 10, 16, 16, 12, 6, 2, 0,

1, 2, 3, 1, 0, 0, 6, 12, 17, 15, 8, 3, 0, 0, 7, 18,

22, 23, 20, 16, 13, 9, 9, 11, 10, 9, 9, 10, 9, 3, 2, 4,

5, 6, 8, 16, 17, 16, 18, 19, 18, 12, 11, 12, 16, 20, 22, 21,

14, 15, 16, 18, 19, 16, 12, 8, 11, 9, 5, 2, 0, 1, 8, 17,

12, 14, 13, 11, 48, 161, 267, 349, 433, 492, 532, 483, 384, 432, 579, 694,

782, 842, 892, 968, 1038, 1097, 1088, 1079, 1175, 1217, 1132, 1130, 1185, 1203, 1253, 1198,

1081, 1097, 1164, 1173, 1115, 1081, 1168, 1303, 1334, 1372, 1487, 1622, 1785, 1831, 1933, 2165,

2308, 2424, 2647, 2861, 3059, 3395, 3706, 3749, 3647, 3633, 3733, 3869, 3966, 4006, 4026, 4039,

4041, 4042, 4043, 4027, 3989, 3962, 3965, 3981, 4002, 4025, 4028, 4008, 4021, 4045, 4035, 4010,

3998, 4010, 4024, 4027, 4017, 4003, 3994, 3988, 3966, 3951, 3945, 3945, 3938, 3945, 3966, 3998,

3992, 3964, 3918, 3891, 3866, 3839, 3782, 3728, 3680, 3687, 3723, 3790, 3838, 3877, 3906, 3971,

4030, 4072, 4074, 4071, 4073, 4080, 4086, 4083, 4075, 4065, 4058, 4029, 3977, 3857, 3778, 3743,

3862, 3979, 4080, 4085, 4085, 4081, 4083, 4085, 4081, 4049, 4005, 3973, 3980, 4015, 4051, 4081,

4085, 4086, 4086, 4083, 4081, 4080, 4081, 4083, 4087, 4086, 4078, 4059, 4063, 4077, 4082, 4080,

4080, 4088, 4084, 4077, 4078, 4077, 4067, 4045, 4016, 3992, 4022, 4073, 4078, 4062, 3955, 3792,

3712, 3701, 3670, 3523, 3367, 3487, 3777, 3919, 3797, 3510, 3372, 3392, 3243, 2908, 2577, 2367,

2254, 2263, 2292, 2206, 2117, 2031, 1899, 1783, 1665, 1626, 1773, 1875, 1939, 2198, 2448, 2402,

2195, 2089, 2035, 1792, 1398, 1085, 988, 1077, 1014, 736, 506, 426, 429, 434, 390, 225,

48, 2, 11, 78, 194, 236, 226, 268, 322, 319, 274, 207, 146, 118, 75, 42,

49, 82, 106, 96, 47, 19, 32, 46, 44, 23, 14, 15, 9, 0, 0, 0,

0, 6, 25, 53, 67, 81, 110, 179, 230, 256, 250, 229, 203, 172, 143, 94,

43, 1, 1, 4, 5, 4, 6, 4, 2, 0, 0, 0, 1, 2, 2, 20,

124, 241, 268, 297, 331, 298, 160, 43, 57, 149, 257, 325, 362, 428, 532, 536,

378, 220, 165, 224, 353, 498, 698, 939, 1020, 858, 781, 959, 1219, 1268, 1358, 1083,

1122, 1055, 1157, 1383, 1302, 1351, 1638, 1649, 1931, 1915, 1743, 1795, 1944, 2180, 2270, 2705,

2492, 2870, 3044, 3081, 3262, 3260, 3542, 3579, 3577, 3137, 2995, 2981, 2846, 3048, 3070, 3036,

2982, 2873, 2836, 2872, 2980, 3324, 3729, 3877, 3727, 3599, 3612, 3584, 3513, 3457, 3442, 3500,

3575, 3645, 3705, 3675, 3551, 3448, 3393, 3421, 3563, 3693, 3724, 3697, 3684, 3761, 3930, 4069,

4085, 4070, 4078, 4089, 4089, 4044, 3991, 4034, 4080, 4067, 4017, 3976, 3979, 4021, 4050, 4055,

4064, 4076, 4031, 3945, 3920, 3935, 3889, 3821, 3806, 3785, 3789, 3914, 4002, 3994, 3986, 3973,

3934, 3922, 3953, 3935, 3695, 3335, 3153, 3198, 3364, 3246, 2740, 2292, 2086, 1904, 1927, 2120,

2204, 1975, 1629, 1646, 1942, 1830, 1780, 1363, 1118, 977, 1573, 2522, 2130, 2029, 1987, 1079,

800, 950, 966, 1434, 1019, 376, 811, 1217, 656, 1428, 846, 1169, 658, 655, 1208, 2015,

846, 1582, 1875, 1384, 1850, 1542, 2213, 1615, 1209, 2496, 2001, 1461, 2165, 2931, 2111, 2064,

2591, 3188, 3080, 2244, 2100, 1567, 2447, 2865, 2918, 3311, 3897, 3204, 3171, 3125, 3029, 3074,

3157, 3141, 3329, 2929, 2589, 2725, 3337, 2451, 1986, 2085, 1828, 2452, 1791, 1340, 1572, 2292,

1981, 2138, 1536, 2728, 2787, 2398, 1916, 1934, 1307, 931, 671, 794, 991, 880, 817, 304,

268, 590, 473, 444, 509, 743, 936, 944, 750, 1038, 1508, 2009, 2138, 2484, 2576, 2848,

2993, 2817, 3035, 2503, 2782, 2747, 2665, 2076, 2573, 2008, 2460, 2639, 2310, 2663, 2133, 2013,

1926, 1545, 2018, 1251, 1634, 1320, 1285, 1599, 1590, 1636, 1104, 1426, 1266, 1515, 1846, 1698,

1176, 2241, 2748, 2633, 2604, 2284, 2557, 1586, 1672, 1401, 2772, 2021, 2542, 1997, 1671, 1497,

1224, 2720, 2743, 2225, 1773, 923, 537, 1250, 2273, 1733, 1105, 1465, 1521, 1952, 2285, 1779,

1964, 2166, 2972, 1868, 3088, 2540, 1965, 2020, 2556, 2153, 1995, 1840, 1397, 1370, 1577, 1321,

1986, 1619, 1433, 1755, 1654, 1314, 1038, 1188, 1904, 2011, 2519, 2150, 2789, 2079, 2870, 2555,

2504, 2323, 2491, 2515, 2825, 2598, 2562, 2590, 2284, 2369, 1446, 2073, 1741, 1416, 1545, 1453,

2141, 1699, 2348, 2756, 2476, 2341, 2378, 2202, 2518, 2263, 1988, 1578, 1861, 1727, 2079, 2255,

2767, 2610, 2510, 2398, 2190, 2587, 2099, 2047, 1660, 1505, 1539, 1550, 2190, 1767,
1601, 2781,

2563, 1345, 1543, 2428, 2881, 2936, 2502, 2205, 2256, 2474, 1922, 1843, 1908, 1490,
1481, 1676,

1208, 1424, 1336, 2084, 1896, 2170, 2341, 2396, 2615, 2681, 2334, 2041, 1990, 1873,
1343, 1428,

1570, 1534, 1787, 1567, 1810, 1992, 2263, 2513, 2095, 2091, 1618, 1564, 1543, 1703,
1646, 1656,

2030, 2253, 2420, 2239, 1944, 1798, 1630, 1738, 1884, 2273, 2559, 2321, 2610, 1945,
1866, 1744,

1686, 1964, 2169, 2519, 2440, 2399, 1956, 1724, 2222, 2465, 2295, 1882, 2181, 2274,
2027, 2201

};

// **TODO:** Equation to calculate TIM2_Ticks

uint32_t TIM2_Ticks = TIM2CLK / (NS * F_SIGNAL); // How often to write new LUT value

uint32_t DestAddress = (**uint32_t**) &(TIM3->CCR3); // Write LUT TO TIM3->CCR3 to
modify PWM duty cycle

/* USER CODE END PV */

/* Private function prototypes -----*/

void SystemClock_Config(void);

static void MX_GPIO_Init(void);

static void MX_DMA_Init(void);

static void MX_TIM2_Init(void);

static void MX_TIM3_Init(void);

```

/* USER CODE BEGIN PFP */

void EXTI0_IRQHandler(void);

/* USER CODE END PFP */


/* Private user code -----*/

/* USER CODE BEGIN 0 */


/* USER CODE END 0 */


/**
 * @brief The application entry point.
 * @retval int
 */
int main(void)
{

/* USER CODE BEGIN 1 */


/* USER CODE END 1 */


/* MCU Configuration-----*/

/* Reset of all peripherals, Initializes the Flash interface and the Systick. */
HAL_Init();


/* USER CODE BEGIN Init */

// init_LCD();

// HAL_Delay(50); // Give LCD time to initialize

```

```
/* USER CODE END Init */
```

```
/* Configure the system clock */
```

```
SystemClock_Config();
```

```
/* USER CODE BEGIN SysInit */
```

```
HAL_Delay(100);
```

```
init_LCD();
```

```
HAL_Delay(100);
```

```
/* USER CODE END SysInit */
```

```
/* Initialize all configured peripherals */
```

```
MX_GPIO_Init();
```

```
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_4, GPIO_PIN_RESET);
```

```
MX_DMA_Init();
```

```
MX_TIM2_Init();
```

```
MX_TIM3_Init();
```

```
/* USER CODE BEGIN 2 */
```

```
// TODO: Start TIM3 in PWM mode on channel 3
```

```
HAL_TIM_PWM_Start(&htim3, TIM_CHANNEL_3);
```

```
// TODO: Start TIM2 in Output Compare (OC) mode on channel 1
```

```
HAL_TIM_OC_Start(&htim2, TIM_CHANNEL_1);
```

```
// TODO: Start DMA in IT mode on TIM2->CH1. Source is LUT and Dest is TIM3->CCR3;  
start with Sine LUT
```

```
HAL_DMA_Start_IT(&hdma_tim2_ch1, (uint32_t)Sin_LUT, DestAddress, NS);
```

```
// TODO: Write current waveform to LCD(Sine is the first waveform)
```

```
lcd_command(CLEAR);
```

```
lcd_putstring("Sine");
```

```

// TODO: Enable DMA (start transfer from LUT to CCR)
__HAL_TIM_ENABLE_DMA(&htim2, TIM_DMA_CC1);
/* USER CODE END 2 */

/* Infinite loop */
/* USER CODE BEGIN WHILE */
while (1)
{
    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
}
/* USER CODE END 3 */
}

/**
 * @brief System Clock Configuration
 * @retval None
 */
void SystemClock_Config(void)
{
    RCC_OscInitTypeDef RCC_OscInitStruct = {0};
    RCC_ClkInitTypeDef RCC_ClkInitStruct = {0};

    /** Configure the main internal regulator output voltage
     */
    __HAL_RCC_PWR_CLK_ENABLE();
    __HAL_PWR_VOLTAGESCALING_CONFIG(PWR_REGULATOR_VOLTAGE_SCALE3);

```

```

/** Initializes the RCC Oscillators according to the specified parameters
 * in the RCC_OscInitTypeDef structure.
 */

RCC_OscInitStruct.OscillatorType = RCC_OSCILLATORTYPE_HSI;
RCC_OscInitStruct.HSIState = RCC_HSI_ON;
RCC_OscInitStruct.HSICalibrationValue = RCC_HSICALIBRATION_DEFAULT;
RCC_OscInitStruct.PLL.PLLState = RCC_PLL_NONE;
if (HAL_RCC_OscConfig(&RCC_OscInitStruct) != HAL_OK)
{
    Error_Handler();
}

/** Initializes the CPU, AHB and APB buses clocks
 */

RCC_ClkInitStruct.ClockType = RCC_CLOCKTYPE_HCLK|RCC_CLOCKTYPE_SYSCLK
                               |RCC_CLOCKTYPE_PCLK1|RCC_CLOCKTYPE_PCLK2;
RCC_ClkInitStruct.SYSCLKSource = RCC_SYSCLKSOURCE_HSI;
RCC_ClkInitStruct.AHBCLKDivider = RCC_SYSCLK_DIV1;
RCC_ClkInitStruct.APB1CLKDivider = RCC_HCLK_DIV1;
RCC_ClkInitStruct.APB2CLKDivider = RCC_HCLK_DIV1;

if (HAL_RCC_ClockConfig(&RCC_ClkInitStruct, FLASH_LATENCY_0) != HAL_OK)
{
    Error_Handler();
}
}

```

```

/**
 * @brief TIM2 Initialization Function
 * @param None
 * @retval None
 */
static void MX_TIM2_Init(void)
{

    /* USER CODE BEGIN TIM2_Init 0 */

    /* USER CODE END TIM2_Init 0 */

    TIM_ClockConfigTypeDef sClockSourceConfig = {0};
    TIM_MasterConfigTypeDef sMasterConfig = {0};
    TIM_OC_InitTypeDef sConfigOC = {0};

    /* USER CODE BEGIN TIM2_Init 1 */

    /* USER CODE END TIM2_Init 1 */
    htim2.Instance = TIM2;
    htim2.Init.Prescaler = 0;
    htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
    htim2.Init.Period = TIM2_Ticks - 1;
    htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
    htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
    if (HAL_TIM_Base_Init(&htim2) != HAL_OK)
    {
        Error_Handler();
    }

```

```

}

sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;

if (HAL_TIM_ConfigClockSource(&htim2, &sClockSourceConfig) != HAL_OK)
{
    Error_Handler();
}

if (HAL_TIM_OC_Init(&htim2) != HAL_OK)
{
    Error_Handler();
}

sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;

if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
{
    Error_Handler();
}

sConfigOC.OCMode = TIM_OCMODE_TIMING;
sConfigOC.Pulse = 0;
sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;

if (HAL_TIM_OC_ConfigChannel(&htim2, &sConfigOC, TIM_CHANNEL_1) != HAL_OK)
{
    Error_Handler();
}

/* USER CODE BEGIN TIM2_Init 2 */

/* TIM2_CH1 DMA Init */

__HAL_RCC_DMA1_CLK_ENABLE();

```

```

hdma_tim2_ch1.Instance = DMA1_Stream5;

hdma_tim2_ch1.Init.Channel = DMA_CHANNEL_3;    // TIM2_CH1 is on channel 3

hdma_tim2_ch1.Init.Direction = DMA_MEMORY_TO_PERIPH; // Memory -> TIM3-
>CCR3

hdma_tim2_ch1.Init.PeriphInc = DMA_PINC_DISABLE; // Peripheral address fixed

hdma_tim2_ch1.Init.MemInc = DMA_MINC_ENABLE;    // Memory address
increments

hdma_tim2_ch1.Init.PeriphDataAlignment = DMA_PDATAALIGN_WORD;

hdma_tim2_ch1.Init.MemDataAlignment = DMA_MDATAALIGN_WORD;

hdma_tim2_ch1.Init.Mode = DMA_CIRCULAR;        // Repeat LUT automatically

hdma_tim2_ch1.Init.Priority = DMA_PRIORITY_HIGH;

hdma_tim2_ch1.Init.FIFOMode = DMA_FIFOMODE_DISABLE;


if (HAL_DMA_Init(&hdma_tim2_ch1) != HAL_OK)
{
    Error_Handler();
}


/* Link DMA handle to TIM2 handle */
__HAL_LINKDMA(&htim2, hdma[TIM_DMA_ID_CC1], hdma_tim2_ch1);

/* USER CODE END TIM2_Init 2 */

}


/**
 * @brief TIM3 Initialization Function
 * @param None
 * @retval None
 */

```



```

static void MX_TIM3_Init(void)
{

    /* USER CODE BEGIN TIM3_Init 0 */

    /* USER CODE END TIM3_Init 0 */

    TIM_ClockConfigTypeDef sClockSourceConfig = {0};
    TIM_MasterConfigTypeDef sMasterConfig = {0};
    TIM_OC_InitTypeDef sConfigOC = {0};

    /* USER CODE BEGIN TIM3_Init 1 */

    /* USER CODE END TIM3_Init 1 */

    htim3.Instance = TIM3;
    htim3.Init.Prescaler = 0;
    htim3.Init.CounterMode = TIM_COUNTERMODE_UP;
    htim3.Init.Period = 4095;
    htim3.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
    htim3.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
    if (HAL_TIM_Base_Init(&htim3) != HAL_OK)
    {
        Error_Handler();
    }

    sClockSourceConfig.ClockSource = TIM_CLOCKSOURCE_INTERNAL;
    if (HAL_TIM_ConfigClockSource(&htim3, &sClockSourceConfig) != HAL_OK)
    {
        Error_Handler();
    }

```

```

}

if (HAL_TIM_PWM_Init(&htim3) != HAL_OK)
{
    Error_Handler();
}

sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
if (HAL_TIMEx_MasterConfigSynchronization(&htim3, &sMasterConfig) != HAL_OK)
{
    Error_Handler();
}

sConfigOC.OCMode = TIM_OCMODE_PWM1;
sConfigOC.Pulse = 0;
sConfigOC.OCpolarity = TIM_OCPOLARITY_HIGH;
sConfigOC.OCFastMode = TIM_OCFAST_DISABLE;
if (HAL_TIM_PWM_ConfigChannel(&htim3, &sConfigOC, TIM_CHANNEL_3) !=
HAL_OK)
{
    Error_Handler();
}

/* USER CODE BEGIN TIM3_Init 2 */

/* USER CODE END TIM3_Init 2 */

HAL_TIM_MspPostInit(&htim3);

}

/**

```

```

* Enable DMA controller clock
*/

static void MX_DMA_Init(void)
{

    /* DMA controller clock enable */
    __HAL_RCC_DMA1_CLK_ENABLE();

    /* DMA interrupt init */
    /* DMA1_Stream5_IRQn interrupt configuration */
    HAL_NVIC_SetPriority(DMA1_Stream5_IRQn, 0, 0);
    HAL_NVIC_EnableIRQ(DMA1_Stream5_IRQn);

}

/**
 * @brief GPIO Initialization Function
 * @param None
 * @retval None
 */
static void MX_GPIO_Init(void)
{
    GPIO_InitTypeDef GPIO_InitStruct = {0};
    /* USER CODE BEGIN MX_GPIO_Init_1 */

    /* USER CODE END MX_GPIO_Init_1 */

    /* GPIO Ports Clock Enable */

```

```
__HAL_RCC_GPIOH_CLK_ENABLE();
__HAL_RCC_GPIOA_CLK_ENABLE();
__HAL_RCC_GPIOB_CLK_ENABLE();

// -----
// LCD pins configuration
// -----

// Configure PC14 (RS) and PC15 (E) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_14 | GPIO_PIN_15;
GPIO_InitStruct.Mode = GPIO_MODE_OUTPUT_PP;
GPIO_InitStruct.Pull = GPIO_NOPULL;
GPIO_InitStruct.Speed = GPIO_SPEED_FREQ_LOW;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct);

// Configure PB8 (D4) and PB9 (D5) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_8 | GPIO_PIN_9;
HAL_GPIO_Init(GPIOB, &GPIO_InitStruct);

// Configure PA12 (D6) and PA15 (D7) as output push-pull
GPIO_InitStruct.Pin = GPIO_PIN_12 | GPIO_PIN_15;
HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);

// Set all LCD pins LOW initially
HAL_GPIO_WritePin(GPIOC, GPIO_PIN_14 | GPIO_PIN_15, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOB, GPIO_PIN_8 | GPIO_PIN_9, GPIO_PIN_RESET);
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_12 | GPIO_PIN_15, GPIO_PIN_RESET);
```

```

// -----
// Button0 configuration (PA0)
// -----

GPIO_InitStruct.Pin = Button0_Pin;

GPIO_InitStruct.Mode = GPIO_MODE_IT_RISING; // Interrupt on rising edge

GPIO_InitStruct.Pull = GPIO_PULLUP;    // Use pull-up resistor

HAL_GPIO_Init(GPIOA, &GPIO_InitStruct);


// Enable and set EXTI line 0 interrupt priority

HAL_NVIC_SetPriority(EXTI0_IRQn, 2, 0);

HAL_NVIC_EnableIRQ(EXTI0_IRQn);


/* USER CODE END MX_GPIO_Init_2 */
}


/* USER CODE BEGIN 4 */

void EXTI0_IRQHandler(void){

    // TODO: Debounce using HAL_GetTick()

    current_tick = HAL_GetTick();

    if (current_tick - last_button_press < 1000){

        HAL_GPIO_EXTI_IRQHandler(Button0_Pin); //clear interrupt flags

        return;

    }

    last_button_press = current_tick;

```

// **TODO:** Disable DMA transfer and abort IT, then start DMA in IT mode with new LUT and re-enable transfer

// HINT: Consider using C's "switch" function to handle LUT changes

```
__HAL_TIM_DISABLE_DMA(&htim2, TIM_DMA_CC1);
```

```
HAL_DMA_Abort_IT(&hdma_tim2_ch1);
```

```
current_waveform = (current_waveform + 1) % 6;
```

```
switch(current_waveform){
```

```
    case 0:
```

```
        selected_lut = Sin_LUT;
```

```
        waveform_name = "Sine";
```

```
        frequency = F_SIGNAL;
```

```
        ns_to_use = NS_WAVE;
```

```
        break;
```

```
    case 1:
```

```
        selected_lut = Saw_LUT;
```

```
        waveform_name = "Sawtooth";
```

```
        frequency = F_SIGNAL;
```

```
        ns_to_use = NS;
```

```
        break;
```

```
    case 2:
```

```
        selected_lut = Triangle_LUT;
```

```
        waveform_name = "Triangle";
```

```
        frequency = F_SIGNAL;
```

```
        ns_to_use = NS;
```

break;

case 3:

```
selected_lut = Piano_LUT;  
waveform_name = "Piano";  
frequency = F_SIGNAL;  
ns_to_use = NS_WAVE;  
break;
```

case 4:

```
selected_lut = Guitar_LUT;  
waveform_name = "Guitar";  
frequency = F_SIGNAL/10;  
ns_to_use = NS_WAVE;  
break;
```

case 5:

```
selected_lut = Drum_LUT;  
waveform_name = "Drums";  
frequency = F_SIGNAL/10;  
ns_to_use = NS_WAVE;  
break;
```

default:

```
selected_lut = Sin_LUT;  
waveform_name = "Sine";  
frequency = F_SIGNAL/10;  
ns_to_use = NS;
```

```
}
```

```
TIM2_Ticks = TIM2CLK / (ns_to_use * frequency);
```

```
htim2.Init.Period = TIM2_Ticks - 1;
```

```
HAL_TIM_Base_Init(&htim2);
```

```
lcd_command(CLEAR);
```

```
lcd_putstring(waveform_name);
```

```
HAL_DMA_Start_IT(&hdma_tim2_ch1, (uint32_t)selected_lut, DestAddress, NS);
```

```
__HAL_TIM_ENABLE_DMA(&htim2,TIM_DMA_CC1);
```

```
HAL_GPIO_EXTI_IRQHandler(Button0_Pin);
```

```
}
```

```
/* USER CODE END 4 */
```

```
/**
```

```
 * @brief This function is executed in case of error occurrence.
```

```
 * @retval None
```

```
 */
```

```
void Error_Handler(void)
```

```
{
```

```
/* USER CODE BEGIN Error_Handler_Debug */
```

```
/* User can add his own implementation to report the HAL error return state */
```

```
__disable_irq();
```

```
while (1)
```

```
{
```



```

}

/* USER CODE END Error_Handler_Debug */
}

#ifdef USE_FULL_ASSERT

/**
   * @brief Reports the name of the source file and the source line number
   *        where the assert_param error has occurred.
   * @param file: pointer to the source file name
   * @param line: assert_param error line source number
   * @retval None
   */
void assert_failed(uint8_t *file, uint32_t line)
{
    /* USER CODE BEGIN 6 */
    /* User can add his own implementation to report the file name and line number,
    ex: printf("Wrong parameters value: file %s on line %d\r\n", file, line) */
    /* USER CODE END 6 */
}

#endif /* USE_FULL_ASSERT */

```